GIGCCL Official Website

(Enhancing Academic Excellence and Communication)

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GROUP ID: G-14

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FROM

DEPARTMENT OF COMPUTER SCIENCE,

GOVT ISLAMIA GRADUATE COLLEGE, CIVIL LINES, LAHORE

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CERTIFICATE

This is to certify that Muhammad Taha (126/062973), Ghulam Raza (181/063007), and Abdullah Shahid (190/063022), members of Group-14, have successfully designed and developed the software project titled "GIGCCL Official Website." This project was carried out at Government Islamia Graduate College, Civil Lines, Lahore, affiliated with Punjab University, Lahore, as part of the requirements for the degree of BS Information Technology.

Under my supervision and guidance, this project has been meticulously evaluated and is found to be up-to-date, technically robust, and fully aligned with academic standards and program requirements. The efforts and contributions behind this project are truly commendable, reflecting a strong commitment to excellence and innovation.

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All praise and gratitude belong to Almighty ALLAH, the Most Gracious and Most Merciful, who blessed us with the strength, determination, and wisdom to successfully complete this project, titled “GIGCCL Official Website.” We also express our deepest reverence to The Holy Prophet MUHAMMAD (PBUH), whose teachings continue to illuminate our paths and guide us toward righteousness.

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Finally, we hope that this project serves as a valuable contribution to the academic community and prospective students, fostering enhanced engagement and innovation through the GIGCCL official website. We dedicate this work to all those who believe in the power of education and technology to transform lives.

Group Members

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ABSTRACT

The GIGCCL Official Website project aims to develop a modern, user-centric platform to address the growing needs of the GIGCCL community. This initiative focuses on enhancing communication, accessibility, and engagement among students, faculty, alumni, and prospective applicants, offering a seamless digital experience that aligns with the institution's mission to promote academic excellence and community involvement. The project addresses the challenge of fragmented information dissemination and manual administrative processes. The lack of a centralized platform often leads to inefficiencies in accessing academic resources, program details, and campus life updates, creating barriers to effective communication and user satisfaction. To solve this, we implemented a comprehensive and dynamic website featuring a robust admin panel, a detailed program information module, a career portal, and sections dedicated to admissions, student resources, and campus life. Additionally, external links to critical systems like the Registrar System, LMS, and Sports System are consolidated for user convenience. The design emphasizes responsiveness and accessibility to ensure compatibility across devices and inclusivity for diverse users. The results include an efficient, visually appealing, and user-friendly platform that provides timely updates, facilitates engagement, and enhances operational efficiency for the GIGCCL community. In conclusion, the GIGCCL Official Website project transforms the institution's online presence, offering a scalable and impactful digital solution. Future plans include the development of an AI-powered chatbot integration to further enhance functionality and user experience.

LIST OF ABBREVATION

Sr #	Abbreviation	Description
1	UI	User Interface
2	UX	User Experience
3	RAM	Random Access Memory
5	WCAG	Web Content Accessibility Guidelines
6	HTML	Hyper Text Markup Language
7	CSS	Cascading Style Sheet
8	JS	Java Script
9	WBS	Work Break Structure
10	uWSGI	Universal Web Server Gateway Interface

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1 Introduction

Government Islamia Graduate College, Civil Lines (GIGCCL), Lahore, is a distinguished institution with a rich legacy of academic excellence and community engagement. Established in 1886 as Dayanand Ayurvedic School under the Arya Samaj movement, the institution evolved over the years, offering intermediate classes in 1888 and graduate courses by 1892. Following Pakistan's independence in 1947, it was renamed Islamia College and managed by Anjuman-e-Himayat-i-Islam in 1955. By 1958, the college had gained its distinct identity and became affiliated with Punjab University, continuing its long-standing tradition of academic distinction.

At the intermediate level, GIGCCL offers various programs, including Pre-Medical, Pre-Engineering, I.C.S (Intermediate in Computer Science), I.Com (Intermediate in Commerce), and Arts (Humanities), providing students with solid foundations for future academic pursuits. At the graduate level, the college offers a comprehensive 16-year education program across more than 24 departments, offering 18 programs, including disciplines such as science, arts, humanities, and computer science. This broad range of academic offerings allows students to specialize in their chosen fields, equipping them for professional success. With a strong reputation among students and alumni, GIGCCL maintains an overall rating of 3.5 out of 5 on educational platforms.

In addition to its educational achievements, GIGCCL has a proud history of sporting excellence. The college has produced numerous athletes who have represented Pakistan in national and international events, particularly in cricket, hockey, and various other sports. Its commitment to fostering athletic talent has significantly contributed to the nation's sports achievements. Over the years, GIGCCL has provided its students with the opportunity to excel in sports, offering state-of-the-art facilities and coaching. This commitment ensures that students not only thrive academically but also grow as athletes, benefiting from a well-rounded educational experience.

The current website of Government Islamia Graduate College, Civil Lines, Lahore, lacks a streamlined and user-friendly design. While it offers basic functionality, it falls short in key areas such as content management, as it does not feature an admin panel for efficient updates. Additionally, the user interface is poorly organized, with inconsistent colors and a lack of a cohesive theme, making it difficult to navigate and requiring manual coding for even minor changes. These shortcomings significantly impact the website’s ability to serve its users effectively and professionally.

A college website is a vital online resource created to give important details about the institution to a wide range of audiences, such as faculty, staff, alumni, current and potential students, and more. Its main goal is to provide comprehensive details on the college's academic offerings, admissions procedure, campus culture, and resources. The website acts as an essential communication tool that guarantees all stakeholders can remain informed and involved by offering simple access to information on classes, instructors, student activities, and events. A thoughtfully designed website not only simplifies administrative tasks but also fosters effective communication among stakeholders while enhancing the institution’s online presence and reputation.

A college website plays an important role because it serves as the organization's online presence and aids in creating and promoting its identity and image. It acts as an information hub for potential students to research the college's programs and make well-informed educational decisions. The website is also essential for keeping lines of communication open and providing timely information about announcements, events, and academic timetables.

A college website has several advantages. By reducing the need for direct requests and facilitating continuously access to essential data, it offers convenience. Furthermore, a well-designed website enhances the institution's professionalism and credibility by demonstrating its dedication to modern, effective communication. Moreover, by simplifying procedures like online applications and event registrations, it improves operational efficiency. In the end, a college website supports both administrative duties and student participation, improving the experience for everyone in the college community.

Despite its benefits, the existing website faces several challenges that affect its effectiveness. Its obsolete design reduces its overall appeal and falls short of modern requirements. The navigation is also not user-friendly, which makes it challenging for users to efficiently access important information. Moreover, the website lacks of crucial administrative tools, such an efficient content management system, which causes updates and modifications to be difficult and time-consuming. These problems make it more difficult for the website to successfully serve the campus community.

Our project, the GIGCCL Official Website, modernizes and enhances the current platform, transforming it into a comprehensive and efficient online resource. The website has an intuitive user interface, a unified and attractive design, and a powerful admin panel for smooth administration. Complete information regarding academic programs, campus life, admissions processes, extracurricular activities, and employment opportunities are provided via key aspects. The platform also improves communication and participation, providing a seamless and rewarding experience for every member of the GIGCCL community. The website is now a vital resource for staff, instructors, and students because of to these developments.

1.1 Project Context and Background

The Government Islamia Graduate College Civil Lines (GIGCCL) is well known for its lively community of faculty, students, and alumni as well as its academic brilliance. However, the college’s current website falls short of modern standards in design, functionality, and user experience.

The current website's inconsistent and unbranded color scheme is one of its main problems. Incompatible color schemes give the website a less polished and attractive look, which undermines its ability to represent the college's brand and reduces user confidence.

Moreover, the website does not have a distinct admin panel, which makes it challenging to effectively manage updates and content. Manual procedures are used for tasks like posting notifications and updating academic information, which results in delays and inefficiencies. The GIGCCL community is neglected since key components including established academic resources, career tools, and sections for campus events are missing.

In order to solve these problems, the GIGCCL Official Website project introduces a cohesive, attractive layout with a recognizable color scheme. To ensure timely updates and streamline content management, it comes with a robust admin panel. The project also adds crucial features such as a program information section, career portal, and specialized areas for campus life and student resources, creating a user-friendly and comprehensive platform.

The purpose of this project is to modernize the college's online appearance to a revolutionary, smooth, and easily navigable platform that promotes GIGCCL's principles of community involvement and academic success.

1.2 Problem Statement

The current GIGCCL website fails to fully address the diverse needs of its students, staff, and visitors. It lacks essential features that would provide a seamless experience for all stakeholders, such as comprehensive academic program details, clear admission processes, real-time updates on campus activities, and centralized access to vital resources. As a result, students often struggle to find relevant academic, co-curricular, and job-related information, while staff face challenges in managing and updating content efficiently. Furthermore, the outdated and unintuitive design hinders user engagement, making navigation and access to essential information difficult. For students, obtaining up-to-date details about academic programs, admission deadlines, and career opportunities remains a persistent challenge due to the fragmented nature of the current platform. The absence of centralized communication also complicates their ability to stay informed about key events, job openings, and administrative updates. On the administrative side, the manual process of maintaining the website and managing frequent updates increases the workload, contributing to delays in providing accurate and timely information. This limits the staff's ability to focus on other important tasks, with updates often not being made in real time. Additionally, the website's outdated interface fails to provide the modern, user-friendly experience necessary to foster student engagement. A streamlined and intuitive design is crucial for encouraging students to interact with the available resources, from academic content to co-curricular activities and job opportunities. To address these challenges, there is a pressing need for a modern, comprehensive, and interactive platform that enhances communication, improves user experience, and streamlines administrative tasks. The GIGCCL Official Website project seeks to provide an integrated, user-centric platform that meets the diverse needs of students, faculty, staff, and visitors. This platform will feature a robust admin panel, detailed academic program information, clear admission guidelines, easy access to student resources, and real-time updates, significantly improving communication and reducing manual workloads. The website will become a useful tool for the GIGCCL community by encouraging user engagement through a unified and eye-catching design, which will facilitate productivity, teamwork, and easy access to crucial information.

1.3 Project Title

The title of our final year project is **"GIGCCL Official Website Using Web Technology"**.

1.4 Project Overview

Our "GIGCCL Official Website" project is dedicated to creating a comprehensive, dynamic, and user-centric online platform that serves the academic, administrative, and communication needs of students, faculty, staff, and visitors at Government Islamia Graduate College, Civil Lines, Lahore (GIGCCL). This platform seeks to modernize campus interactions and provide a smooth, interesting experience for all users by incorporating innovative features and tools. The project highlights the institution's rich legacy of academic excellence, diverse extracurricular achievements, and commitment to administrative transparency. It makes crucial resources easily accessible, streamlining procedures and improving usability for both present and potential stakeholders. The website's user-friendly interface ensures effortless navigation, while its robust backend infrastructure guarantees reliability, scalability, and efficiency. In addition to being a source of information, the platform aims to improve the college's digital presence by encouraging active participation from applicants, instructors, students, and the wider community. By embracing innovation and user-focused functionality, the GIGCCL Official Website not only reflects the institution's values but also sets a benchmark for modern educational platforms, fostering a vibrant and connected academic environment.

1.5 Project Goals

The goal of the **GIGCCL Official Website** project aims to provide a modern, user-friendly platform that improves communication, engagement, and accessibility across the whole college community. The platform aims to simplify administrative processes, provide accurate and up-to-date information, and encourage academic excellence by providing thorough resources regarding programs and opportunities. It also aims to build a sense of community by promoting events, campus life, departments, alumni, and job prospects while maintaining inclusion through accessible and adaptable design, resulting in a valuable and efficient resource for all stakeholders.

1.6 Project Objectives

- **Develop a Robust Admin Panel**

To design and implement an admin panel that is easy to use and forms the foundation of website management, allowing administrators to easily manage various administrative tasks, update website information, and handle academic content. By centralizing these tasks, the admin panel will guarantee smooth operation, minimize manual work, and streamline content maintenance, freeing up staff to concentrate on providing a seamless user experience.

- **Design a Program Information Module**

To develop a comprehensive program information module that lists the academic programs offered by the institute. This module gives potential students a concise, easily navigable summary of the programs that are offered. By presenting this information effectively, the module helps students explore their options and identify programs aligned with their academic interests and career aspirations.

- **Integrate Student Resource Tools**

To enhance the student experience by integrating essential tools for accessing academic and extracurricular resources. These tools including timetables, exam schedules, and updates on extracurricular activities, all accessible through a unified and intuitive interface. By consolidating these resources into a single platform, the project aims to make it easier for students to stay organized and informed about their academic and campus life.

- **Incorporate a Career Portal**

To create a dedicated career portal that supports students and alumni in their professional growth. Through this site, visitors can access the college's career development services, internships, and employment possibilities. It helps students and alumni prepare by informing them about job openings and giving them crucial information about the documents needed for interviews. The career portal is an essential tool for making sure students and alumni remain educated and prepared for job prospects by bridging the gap between school and employment.

- **Provide Easy Access to Key External Links**

To include a section that consolidates important external links for the convenience of the GIGCCL community. This section offers easy access to vital platforms connected to the college, including the Learning Management System (LMS), Sports management System, Registrar management System, and others. By providing a single location for these links, it guarantees that customers can quickly and easily access essential services, simplifying their experience overall and eliminating the need to visit several websites.

- **Promote Campus Life and Engagement**

To create a dedicated section that highlights the vibrant campus life at GIGCCL. Information about student organizations, occasions, campus events, and extracurricular activities that promote student and community involvement is included in this section. It strengthens ties within the GIGCCL community by encouraging students, teachers, and alumni to actively participate in college life through the promotion of a sense of belonging.

- **Introduce GIGCCL Through a Dedicated Information Hub**

To create a dedicated page that offers essential information about GIGCCL in a welcoming and professional manner. In order to make it simple for users to get in touch with the college for questions or help, this page includes the campus address, phone number, and email address. The page improves the institution's transparency and builds a closer relationship with stakeholders, parents, and potential students by providing this information in an understandable and efficient manner.

1.7 Project Scope

1.7.1 Core Features and Functionalities

- **Admin Panel**
  A robust, user-friendly backend for managing website content, academic updates, and administrative tasks.
- **Department Information Module**
  A dedicated section listing all departments and programs offered by the institute, targeting prospective students.
- **Admissions Section**
  A portal linking to admissions resources, including eligibility criteria, deadlines, and procedures.
- **Career Portal**
  A resource that helps alumni and others to discover employment that fit their needs by providing job descriptions and qualifications.
- **Student Resource Tools**
  Integration of timetables, exam schedules, and extracurricular activity updates in one centralized platform.
- **Campus Life Section**

A vibrant space showcasing campus events and co-curricular activities to highlight campus life.

1.7.2 Other Functionalities

- **Notifications and Alerts System**
  A notification system for students to receive alerts about important deadlines (e.g., admission deadlines, exam schedules).
- **Search Functionality**
  The admin panel has a search bar with filters for effective content management that provides easy access to events, activities, and resources.
- **Dynamic Content Management**
  Allow administrators to update news, announcements, and event information dynamically without requiring technical knowledge.
- **Progressive Web App (PWA)**
  Transforming the website into a PWA to provide offline functionality, faster load times, and a more app-like user experience.

1.7.3 Exclusions

- **Real-Time Communication Tools**
  Features like real-time chat support or video conferencing for student or faculty interactions are not part of the current project.
- **Advanced Career Services**
  Career features such as AI-based career guidance, or automated job matching are outside the current scope.
- **Multilanguage's Support**
  The current scope does not include support for multiple languages; all content will be available in a single language (e.g., English).

1.7.4 Future Work

- **AI-Powered Chatbot**
  Introducing an intelligent chatbot capable of answering user queries and providing information about programs, admissions, events, and campus life.

- **Alumni Portal**

  Giving alumni a place to connect, exchange job openings, and sign up for events will guarantee a continued connection with the educational institution.

1.8 Assumptions

- The users of the GIGCCL Official Website will have basic internet access and knowledge to navigate the website.
- The bulk of consumers will access the website through modern web browsers on PCs, tablets, or smartphones.
- The website administrators will have received training or have a basic understanding of the admin panel interface.
- This organization will deliver proper information and timely updates for the website's stuff.
- Sensitive information on the platform would be protected by security features like firewalls and encrypted connections.
- Future scalability is taken into consideration, and without significant redesigns, the system can handle more features or more user traffic.
- Users will need less technical expertise because of the website's easy-to-use and intuitive design.
- Regular maintenance will be performed on the website to make sure reliability and continuous performance.
- The vendors of any third-party tools or APIs (such as those used for external connections) will be responsible for maintaining and updating them.

1.9 Risks

1.9.1 Delayed Information from Stakeholders

Risk: Important information or updates from the organization might not arrive on schedule.

Mitigation Strategy: Define exact schedules for information submission from all parties involved, and follow up to ensure on-time delivery.

1.9.2 Limited Technical Expertise

Risk: Complex designs or advanced features like integrations may provide challenges for the development team.

Mitigation Strategy: Set up time for training and research, and when needed, seek advice from mentors or outside experts.

1.9.3 Scope Creep

Risk: During development, new requirements could appear, which may lead the project to be delayed.

Mitigation Strategy: Determine the project scope early on, freeze it, and handle modifications with official approval procedures.

1.9.4 Compatibility Issues

Risk: The website might not work properly on every browser or device.

Mitigation Strategy: Throughout development, test for compatibility regularly to make sure web standards are being followed.

1.9.5 Security Vulnerabilities

Risk: Sensitive information on the website might be compromised by cyberattacks.

Mitigation Strategy: Put strong security measures in place, like firewalls, SSL, and frequent vulnerability analyses.

1.9.6 Resource Constraints

Risk: Project delivery may be impacted by time, money, or human resource constraints.

Mitigation Strategy: To stay on track, prioritize important features, allocate resources as efficiently as possible, and keep updated on developments.

1.9.7 Integration Failure with External Systems

Risk: It is possible that external systems like Registrar Systems or Sports Portal will not interact seamlessly.

Mitigation Strategy: To fix compatibility concerns, communicate with external system vendors and carry out extensive testing.

1.9.8 Insufficient Testing Time

Risk: Minimal testing time could result in bugs going unnoticed.

Mitigation Strategy: Give testing stages enough time, and perform continuous parallel testing as the project is being developed.

1.9.9 Data Loss or Corruption

Risk: Unexpected mistakes may cause website data to be lost or corrupted.
Mitigation Strategy: Utilize reliable database management systems and perform routine data backups.

1.10 Success Criteria

User Registration and Authentication	Admin panel with secure login and management tools.
Core System Functionalities	Department Information Module Student Resources Campus Life Section Career Portal
Data Management	Dynamic content updates through the admin panel. Proper storage and retrieval of user data.
Speed and Responsiveness	Fast page load times. Seamless navigation across the website.
Scalability	Capacity to handle increased traffic or feature expansion.
Data Privacy	Secure handling of user data.
Access Control	Role-based access for admins and users.
User Interface Design	Modern, intuitive, and visually appealing design.
Cross-Platform Compatibility	Responsive design for desktops, tablets, and smartphones.
Error Handling	Clear error messages and proper redirection.

Third-Party Integration	Integration with Sports Portal, Registrar System, and other platforms.
Deployment Readiness	Fully tested and bug-free deployment.

Table 1: 1-1. Succes Criteria for GIGCCL Official Website

1.11 Tools and Technologies

1.11.1 Tools

- **Visual Studio Code**

  For creating the GIGCCL Official Website, Visual Studio Code (VS Code) is the main development environment. Because it supports Python, it is a great option for Django development. Essential features that improve the coding experience, like syntax highlighting, code completion, and debugging tools, are provided by Visual Studio Code. Reliable version control is ensured by the integration with Git, which makes it easier for the development team to collaborate. The development process speeds up with Python and Django-specific extensions, which facilitate efficient testing and deployment. While VS Code's customized interface provides a streamlined and customized development process, the integrated terminal makes it easier to run Django commands, migrate data, and manage the local server.

- **Figma**

  The GIGCCL Official Website's user interface (UI) is designed and prototyped using Figma. It makes it possible to produce interactive prototypes, user flows, and excellent visual designs, facilitating effective real-time collaboration within the design team. By using Figma, different stakeholders can offer input on design components such as layouts, fonts, and color schemes, guaranteeing that the website complements GIGCCL's objectives and identity. Rapid iteration is made possible by the tool's user-friendly interface, which guarantees that the finished design satisfies user expectations and provides an intuitive, aesthetically pleasing experience.

- **Draw.io**

  Draw.io is used to create UML (Unified Modeling Language) diagrams for the GIGCCL Official Website project. It gives the team a clear development plan by enabling them to graphically depict the database models, system architecture, and component relationships. The user-friendly interface of Draw.io makes it simple to create use case diagrams, class diagrams, sequence diagrams, and other crucial components of system architecture. This ensures a common understanding and alignment throughout the development process by assisting stakeholders in recognizing the project's structure, data flow, and essential functionalities.

1.11.2 Technologies

1.11.2.1 Front-End Technologies

- **HTML (Hyper Text Markup Languages)**

  The GIGCCL Official Website relies on HTML, which organizes and displays important content like faculty information, admissions information, academic programs, and updates on campus life. It confirms that every page is formatted appropriately and offers easy navigation between areas such as the contact forms, event pages, program listings, and homepage. The overall structure and layout of the website are supported by HTML, which makes it accessible and easy to use for teachers, students, and potential applications. It is essential for arranging content in a way that improves usability and facilitates smooth website navigation.

- **CSS (Cascading Style Sheet)**

  The GIGCCL Official Website is styled using CSS, which specifies visual components including page layout, color schemes, formatting, and device responsiveness. By using CSS, the website keeps a polished and consistent look that complements the college's identity and branding. Whether users are looking at faculty profiles, educational institutions news, or academic programs, it guarantees a seamless user experience. Certain elements are stylized to produce an attractive and seamless user experience, such as the event pages, program details, and homepage banners.

- **JS (JavaScript)**

  JavaScript improves user engagement on the GIGCCL Official Website by adding dynamic features and interactivity. Smooth page transitions, dynamic dropdown menus, animation transitions, and interactive components like image sliders enhance navigation and provide a responsive, fluid experience across sites. The whole user experience is improved by JavaScript, which makes it simple for users to obtain important information in an interesting and intuitive way, explore academic programs, and keep up with campus events.

- **Bootstrap5**

  To ensure that the GIGCCL Official Website adapts smoothly to various devices, a responsive design is implemented using Bootstrap5. By offering pre-made, adjustable elements like modals, sliders, and navigation bars, it simplifies development and enables a standardized, expert layout across the website. The website dynamically adjusts to different screen sizes due to Bootstrap, providing the best possible user experience across PCs, tablets, and smartphones. By using a grid structure and pre-built styles, the framework also speeds up development while reducing the need for unique coding and preserving a responsive, high-quality design.

1.11.2.2 Back-End Technologies

- **Python**

  The GIGCCL Official Website is developed using Python, the core programming language of the system, providing a solid foundation for back-end development. The implementation of several modules, such as Programs, Events, Jobs, Timetables, Alumni, and Notifications, is streamlined by Python's ease of use and built-in capabilities. Its smooth interaction with the Django framework enables dynamic functionality throughout the website, effective data management, and content rendering. The security features of the language, like safe password hashing and automatic input validation, improve the website's defense against vulnerabilities. Python is the perfect option to meet the digital requirements of GIGCCL because of its scalability, which guarantees that the system can readily handle future updates and growing information needs.

- **Django**

  The back-end development of the GIGCCL Official Website is based on the high-level Python web framework Django. Its Model-View-Template (MVT) design guarantees a clear division of responsibilities, facilitating effective development and simple upkeep. User management and content handling are made easier by Django's built-in capabilities, which include URL routing, user authentication, and an admin interface. The website can adjust to growing traffic and the addition of new features thanks to the framework's scalability. Furthermore, Django's strong security features safeguard private information, maintaining the website's dependability and security while it adapts to the college's changing requirements.

1.11.2.3 Database

- **SQLite3**

  Essential information for the GIGCCL Official Website, such as faculty profiles, event information, and campus life news, is stored and managed using SQLite3. Because it is lightweight and serverless, it can be set up quickly and requires little configuration, which makes it ideal for the scope of the project. Through Django's ORM, SQLite3 and Django easily combine to enable effective database queries. It makes local testing and development simple because it is an embedded database. SQLite3's portability guarantees that the database may be readily moved and maintained in various environments, providing flexibility for upcoming deployments and updates.

1.11.3 Libraries

- **Django ORM**

  The project's database interactions are made simpler using Django ORM (Object-Relational Mapping). Because it converts Python code into SQL queries, developers can use Python syntax instead of raw SQL when working with database objects. As a result, generating, updating, and querying records for events, programs, and user profiles is made easier. By managing the intricacy of database queries, the ORM offers an abstraction layer that improves development efficiency and guarantees database scalability.

- **SQLParse**

  The project uses SQLParse to effectively manage and parse complex SQL queries, especially when accessing dynamic databases. Beyond Django's ORM, this module enables sophisticated query processing, allowing for the creation of unique, adaptable SQL queries for elaborate database operations. SQLParse gives you more control over data retrieval and manipulation in the SQLite3 database by ensuring that queries are well-formed, safe, and efficient.

- **Django Authentication**

  The GIGCCL Official Website's admin panel is secured with Django Authentication. It guarantees that program details, online content, and updates can only be managed by authorized persons. This feature protects sensitive data and preserves the integrity of the website's backend operations by offering administrators secure login, password management, and session handling.

- **Django Filter**

  Django Filter allows administrators to filter and retrieve data according to particular criteria, improving the search capabilities within the admin panel. When used with Django's ORM, this module makes it possible to effectively filter and manage content, including events, academic programs, and campus changes. Content administration is made easier with Django Filter's accurate filter choices, which guarantee administrators can find and update pertinent material fast.

1.12 Agile Methodology

The GIGCCL Official Website is being developed in phases with the help of the Agile methodology, which delivers services like program management and campus updates gradually. This strategy makes sure a user-focused platform, flexibility, and responsiveness to feedback.

Figure 1:1-1. Agile Model of GIGCCL Official Website

- **Requirement Gathering**

  Ensures the team understands user needs, like dynamic content management, program details, and event updates, forming a strong foundation for development.

- **Sprint planning Design Document**

  Breaking development into smaller sprints helps prioritize key features and deliver functionalities like the campus life and student resource tools, aligning with project goals.

- **Development and Coding**

  Implementing and testing each feature incrementally, such as student resources tools and notifications, ensures a smoother development process and reduces errors.

- **Testing**
  Regular testing ensures the GIGCCL Official Website functions effectively, providing reliable performance and user satisfaction.
- **Deployment**
  Deploying functional modules early, like student resource tools and campus life sections, allows for real-time usage and feedback.
- **Feedback and Adaptation**
  By considering user and stakeholder feedback, the website promises to adapt to their needs, increasing engagement and usability.

1.13 Gantt Chart

- **Requirement Gathering**
  The project starts with a 10-day requirement-gathering phase. The team gathers all the data required for the design and development of the GIGCCL Official Website during this time, including user requirements, stakeholder input, and technical specifications. This phase is crucial to determining specific targets for the development process and comprehending the project's scope.
- **Proposal**
  After gathering requirements, the team spends 20 days writing a comprehensive project proposal. This proposal describes the goals, budget, schedule, and scope of the GIGCCL Official Website. It ensures that all parties involved are in agreement with the project's objectives and deliverables by acting as the formal plan and agreement.
- **Documentation**
  The team spends forty days writing comprehensive project documentation when the proposal is accepted. This covers design plans, diagrams, system requirements, and any other paperwork required to direct the development process. The documentation guarantees that the development team gets precise instructions for creating the website in accordance with the specified specifications.
- **UI Designing**
  The design team spends ten days to the UI design phase, which aims to create a user-friendly and intuitive interface for the website. The layout, visual components, and navigation system are designed at this phase to make the website aesthetically pleasing

and user-friendly for all target groups, including students, professors, alumni, and potential applicants.

- **Front-End Development**

  The project enters the 40-day front-end development phase following the completion of the documentation and user interface design. The group works on developing client-side capabilities, like integrating dynamic functionality with UI design. This stage ensures that the website's design is accessible and responsive across all platforms, including PCs, tablets, and smartphones.

- **Back-End Development**

  The project moves onto back-end development after the 40-day front-end development phase. In order to make sure the system runs effectively and safely, the development team works on the server-side elements of the website during this phase, such as database setup, API creation, and server-side logic.

- **Unit Testing**

  The team spends 15 days working unit testing following the coding stages. This step involves examining the website's various elements or components to make sure they all work as expected. Unit testing ensures the quality of the code by identifying and fixing problems early in the development process.

- **Integration Testing**

  The team spends 14 days on integration testing after unit testing. Testing the overall functionality of the website's many components is part of this phase. The database, APIs, and user interface are a few examples of the systems that are tested for seamless interaction.

- **Bug Fixing**

  A specific 10-day bug-fixing phase is provided because problems are usually found during the testing phases. The team concentrates on fixing any bugs, errors or issues identified during unit and integration testing at this phase. Improving the website's general stability and functionality is the aim.

- **Final Testing**

  The project then enters the 13-day final testing phase after all issues have been fixed. To make sure that every element of the website is functioning as intended, this phase includes comprehensive testing. Making sure the GIGCCL Official Website is reliable, operational, and prepared for launch is the last stage before to deployment.

1.13.1 Project Timelines

Task	Start Date	End Date	Duration
Requirement Gathering	October 24, 2024	November 3, 2024	10
Proposal	November 4, 2024	November 24, 2024	20
Documentation	November 25, 2024	January 4, 2025	40
UI Designing	January 5, 2025	January 15, 2025	10
Front-End Development	January 16, 2025	February 25, 2025	40
Back-End Development	February 26, 2025	April 7, 2025	40
Unit Testing	April 8, 2025	April 23, 2025	15
Integration Testing	April 24, 2025	May 8, 2025	14
Bug Fixing	May 9, 2025	May 19, 2025	10
Final Testing	May 20, 2025	June 2, 2025	13

Table 2:1-2. Project Timeline for GIGCCL Official Website

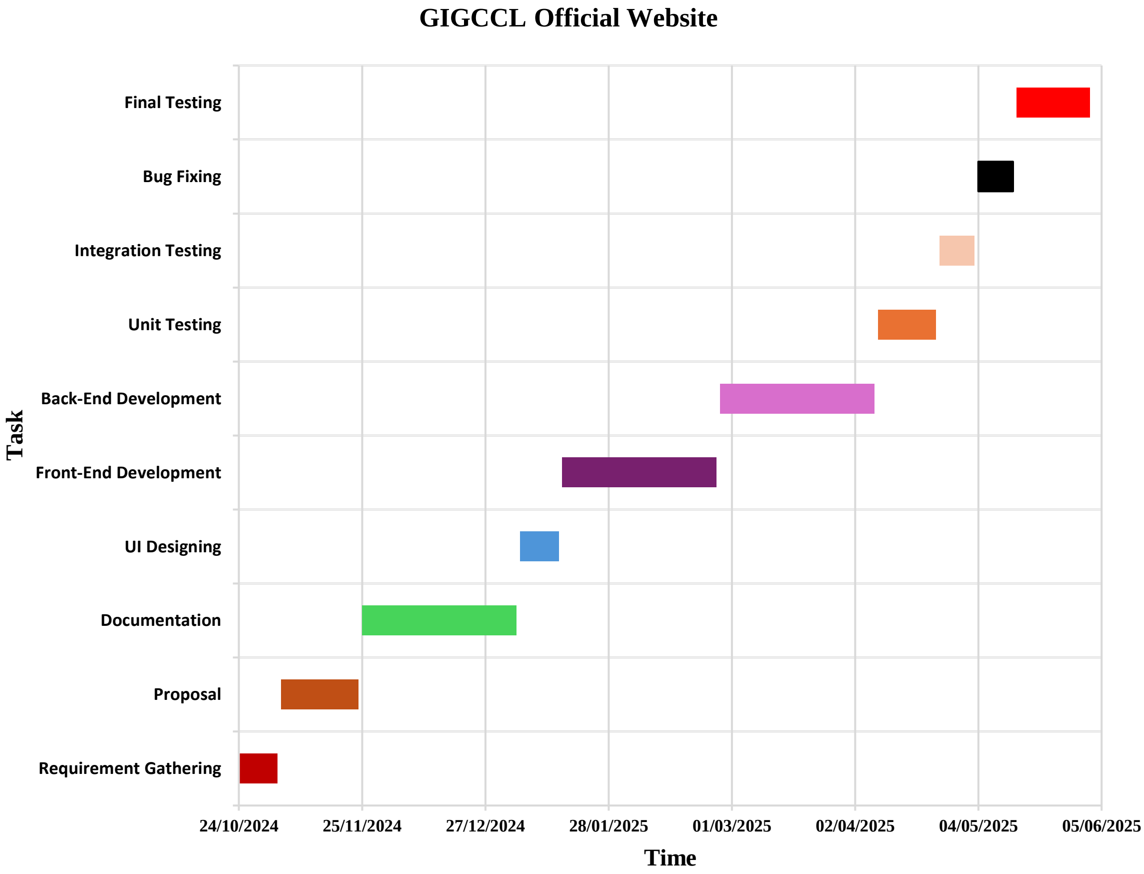


Figure 2:1-2. Timeline for GIGCCL Official Website

1.14 Cost Estimation

Activities	Start Date	End Date	Effort	Hours	Cost
Requirements	October 24, 2024	November 24, 2024	10%	50	10000
Documentation	November 25, 2024	January 4, 2025	20%	100	25000
UI Design	January 5, 2025	January 15, 2025	10%	50	20000
Coding	January 16, 2025	April 7, 2025	50%	1250	100000
Bugs Fix & Testing	April 8, 2025	June 2, 2025	10%	50	15000
Total			**100%**	**1500**	**170000**

Table 3:1-3. Cost Estimation of College Official Website

2 Literature Review

University websites have become increasingly important, especially since the COVID-19 pandemic shifted much of education online. These websites serve as critical tools for attracting new students and providing essential information about courses and campus life. Essential features include degree programs, staff profiles, campus highlights, and student testimonials. Best practices from top university websites highlight the importance of using stunning imagery, dedicated subdomains for academic contributions, and prominent search bars for easy navigation. Tools like Astra's WordPress templates simplify the creation process, emphasizing mobile optimization and quick loading times. By incorporating these features and strategies, universities can create engaging online platforms that effectively communicate their value and boost enrollments. (Abhijeet Kaldate, 2024)

A website is a collection of web pages and content identified by a domain name and hosted on at least one web server. Websites serve various purposes, such as news, education, commerce, entertainment, or social media, with navigation starting from a home page. All public websites form the World Wide Web, while private websites are accessible only on internal networks. Created by Tim Berners-Lee in 1989, the Web became free to use in 1993, leading to its rapid growth. Websites can be static, displaying the same content to all users and requiring manual updates, or dynamic, changing content based on user interactions and using databases. Early websites had text and images, later adding multimedia and interactive features with the help of plugins and technologies like JavaScript and HTML5. Websites are categorized as static or interactive, with many now combining both types. They serve various business models, including advertising, e-commerce, and subscription services. (wikipedia, 2024)

Push notifications can greatly improve communication and engagement in educational institutions by sending quick, personalized messages to students, teachers, and staff. These notifications help share important updates, remind students of deadlines, invite them to events, provide academic support, and enhance safety during emergencies. They also aid in recruitment and keep alumni connected. By integrating push notifications with existing school systems, institutions can ensure timely information delivery, improve event attendance, and streamline administrative tasks. Overall, push notifications make communication more efficient, support student success, and enhance the overall experience in schools and universities. (Ankur, 2024)

The Virtual University of Pakistan's website is highly comprehensive, catering to a large student population with detailed sections on academic programs, admissions, examinations, and student services. It includes interactive features like a learning management system (LMS) and e-learning resources. While robust in functionality, the user interface could benefit from modernization to improve aesthetics and accessibility. (VU, 2024)

Massachusetts Institute of Technology website exemplifies its reputation as a global leader in technology and innovation. It provides detailed information on academic programs, research, and campus activities. The homepage highlights groundbreaking research and innovation through multimedia content. The website also includes resources for entrepreneurship and community outreach, aligning with MIT's mission to advance knowledge and innovation for a better world. (MIT, 2024)

The effectiveness of search functionality on higher education websites is crucial for improving user experience and satisfaction, especially for prospective students. As online platforms grow, effective search capabilities are essential, much like major search engines. Good search features enhance student engagement and satisfaction, and universities must serve to various user groups, each with unique needs. Key strategies for improving search functionality include making the search bar easily visible, adding advanced filtering options, and using effective metadata. Continuous optimization and regular updates based on user feedback and analytics are necessary to keep search features effective. High-quality search functionality directly impacts user engagement, satisfaction, and enrollment rates, making it a vital aspect of university websites. (Stella Power, 2024)

Web accessibility is crucial for higher education institutions as digital technologies become integral to education. It ensures that online resources are usable by everyone, including those with disabilities, fulfilling both legal and ethical obligations. Following the principles of being perceivable, operable, understandable. Common barriers include inaccessible content and complex navigation. Strategies to improve accessibility include creating user-friendly interfaces, providing alternative text for multimedia, and using assistive technologies. Regular audits and user feedback are essential. Ensuring accessibility not only complies with laws like the ADA and WCAG but also enhances overall user experience, benefiting all students and promoting an inclusive educational environment. Additionally, ongoing training and awareness programs for staff and developers are vital to keep up with evolving standards and technologies. By prioritizing web accessibility, institutions can support diverse user needs, improve engagement, and foster a more inclusive digital landscape for all. (Rayed Chaudhry, 2024)

As educational institutions increasingly rely on technology, they face significant cyber security risks. Schools, colleges, and universities store sensitive data, making them targets for cyber attacks. There has been a 55% increase in reported cyber incidents from 2022 to 2023, highlighting the need for strong security measures. Key threats include data breaches, ransomware, phishing, and insider threats. To mitigate these risks, institutions should follow guidelines from the National Cyber Security Centre, such as updating software, limiting data access, and using multi-factor authentication. Best practices include segmenting networks, encrypting sensitive data, conducting regular audits, and securing cloud services. By implementing these measures, educational institutions can protect sensitive information, ensure smooth operations, and maintain trust with students and staff. Investing in cyber security is crucial for safeguarding the institution's data and reputation while ensuring the continuity of educational services. (Guy Lawrence, 2024)

Schools have transitioned from analog to digital learning, using technology for tasks like assignments and communication. While this increases efficiency and provides vast information, it also poses cybersecurity risks. To stay safe online, schools should educate students and staff on cybersecurity best practices, such as creating strong passwords, protecting sensitive data, and avoiding suspicious emails. They must secure student data by storing it safely and limiting access. Implementing strong password policies and two-factor authentication adds security. Ensuring Wi-Fi networks are secure and regularly updating systems to fix vulnerabilities is essential. Setting up 24/7 system monitoring helps catch cyber threats early, and having a data breach response plan is crucial. By following these steps, schools can create a safer online environment for everyone. (Cybercivics, 2024)

University websites showcase their identity and image by publishing academic offerings, mission, vision, objectives, achievements, regulations, news, and other university activities. Despite their importance, previous evaluations of their accessibility haven't been summarized for an overall view. This research conducts a systematic literature review (SLR) to consolidate, analyze, synthesize, and interpret accessibility results from selected papers. Using Kitchenham's methodology, the SLR involves planning, conducting, and reporting the review. It evaluates universities in various countries, assessing web pages, YouTube videos, and PDF documents through manual methods, automated tools, and their combination. The evaluations mainly used ISO/IEC 40500:2012 and Section 508 standards, with common accessibility issues in adaptability, compatibility, distinguishability, input assistance, keyboard accessibility, navigation, predictability, readability, and text alternatives. The SLR reveals significant accessibility problems across these websites and documents, consolidating findings from the studies to highlight global accessibility trends in university websites. (Sergio Luján-Mora, 2021)

Ensuring the safety of students, faculty, and staff is a top priority for educational institutions. To mitigate potential threats, it's important to take proactive measures. This involves conducting thorough risk assessments to identify and address vulnerabilities, developing comprehensive emergency response plans, and enhancing physical security with surveillance cameras, access control systems, and alarms. Leveraging new technologies like visitor management systems and biometric authentication can further bolster security. Strengthening cybersecurity with firewalls, antivirus software, and regular audits is also essential. Fostering a culture of awareness and vigilance through safety campaigns and training sessions helps everyone recognize and report suspicious activities. Collaborating with law enforcement and local authorities ensures a coordinated response to emergencies. By prioritizing these security measures, institutions can create a safer environment conducive to learning and growth. (Kalpa Kalhara Sampath, 2023)

The increasing reliance on mobile and desktop devices has heightened user expectations for website performance, making speed critical for retaining users and enhancing conversion rates. High-performing websites improve user experience, engagement, and business success, with faster sites correlating to higher conversion rates. Delays in loading can cause significant stress and dissatisfaction, impacting user retention. Optimizing web performance is essential, especially with affordable mobile data plans, as it ensures accessibility, improves SEO, and saves users money. Investing in web performance leads to substantial benefits, including better engagement, retention, and revenue. (Mayank Joshi, 2023)

Many websites are like that, causing low user engagement. College and university websites may have fewer visitors, but those visitors are viewing more pages per visit, indicating that students are doing their research. To engage potential students, schools need to provide the information they're seeking. This involves understanding student personas, creating career-focused content, and optimizing website content to be concise and relevant. Additionally, offering multiple communication channels and clear calls to action can help guide students through their decision-making process. By focusing on user experience and providing the right information at the right time, schools can keep students engaged and encourage them to take the next step toward enrollment. (Taylor Precure, 2023)

In today's digital age, a university's website is crucial as it serves as the digital front door, impacting prospective students, families, faculty, and alumni. Effective higher education websites cater to diverse stakeholders, from prospective students to alumni, requiring personalized digital experiences. The main website and specialized sites for admissions, academics, and student life must provide clear, valuable information. Designing user-friendly websites with SEO and UI/UX best practices ensures easy navigation and responsiveness across devices. Continual data-driven improvements enhance user engagement. Crafting a clear Statement of Work (SOW) or Request for Proposal (RFP) is key for successful website projects. This includes defining objectives, scope, audience, competitor analysis, content strategy, design, technical requirements, SEO, budget, timeline, vendor selection, and post-project maintenance. A well-planned website project aligns institutional goals with user needs, fostering a strong digital presence. (Gideon Harrington, 2023)

SQLite is a server-less relational database widely used in both desktop and mobile applications due to its lightweight, fast, reliable, portable, and accessible nature. It is easily embedded into applications without requiring installation or configuration and works with all programming languages. SQLite efficiently loads only necessary data, continuously updates content to minimize data loss from crashes, and is available on multiple platforms like Windows, MacOS, Linux, iOS, and Android. Additionally, it can be accessed through many third-party tools. However, SQLite might not be suitable for handling high traffic, complex queries, highly concurrent users, or large-scale applications. For such needs, databases like PostgreSQL or MongoDB may be better. In conclusion, SQLite is ideal for small to medium-sized apps needing a lightweight, low-maintenance database, but other databases might be preferable for high traffic or complex queries. (Andy Shen, 2023)

Digital transformation is essential for higher education institutions (HEIs) in today's technology-driven world. However, many challenges hinder the successful implementation of digital transformation initiatives. Through a systematic review of the literature, I identified twenty key barriers that prevent progress in this area. These barriers are grouped into six main categories: environmental, strategic, organizational, technological, people-related, and cultural. Understanding these challenges is crucial for developing effective strategies to overcome them. This analysis aims to provide valuable insights for HEIs, policymakers, and stakeholders involved in driving digital transformation forward. (Gkrimpizi, T, Peristeras, V, & Magnisalis, I, 2023)

An organization's website is its gateway to information, products, and services, and should meet the needs of its clients. However, website design often focuses on technology, organizational structure, or business goals rather than user needs. With higher education institutions using their websites to recruit students, effective website design has become essential in attracting them. Website usability is vital for an institution's survival in the digital world. The COVID-19 pandemic has further highlighted the importance of websites, especially for universities. Both users and designers consider usability a key factor in website development. If a website is difficult to use, visitors may turn to other institutions' websites instead. (Ersin Caglar, 2022)

To start building your education website, list all the pages you need by writing them on post-it notes and arranging them. Essential pages include the homepage, about, faculty & staff, admissions, academics, and contact. Consistent branding and messaging are crucial; use a dedicated color palette and limited fonts. Content should be concise, engaging, and easy to scan. Invest in professional photography if possible. Ensure the website loads quickly by optimizing images and using quality website builders to avoid bloated code. Make the site mobile-responsive to cater to users on different devices, and keep it search engine friendly by incorporating keywords and filling out meta-tags. Use built-in analytics to track performance and invest in strategy before spending on expensive tools. Your website should provide a great user experience, highlight your school's culture, and act as the hub for all your marketing activities. (Chuck Bankoff, 2022)

A social-academic platform has been identified as crucial for promoting collaborative learning and boosting the use of academic web portals. However, the lack of social and academic features in the platforms studied has led to usability issues and reduced interactions between tutors and students. The COVID-19 pandemic has highlighted the importance of web portals for learning outside of physical classrooms. This study evaluated a platform using the System Usability Scale (SUS) to assess usability and validate the social learning and knowledge-sharing aspects. The results showed a perceived learnability of 70.9% and a usability score of 83.9%, indicating excellent usability. The findings suggest that during the pandemic, the social academic web portal is adaptable, user-friendly, and can enhance interactions between lecturers and students, thereby increasing portal usage. (IJEDICT, 2021)

This study examines the relationship between user experience design, usability, and the click-through rate of an internationally-focused university website. These relationships are identified based on online patterns and user behavior. The top pages of two studies were analyzed: high-converting pages focused on broad information, and low-converting pages focused on detailed information. These pages formed the basis of the user experience design and usability analysis. Relevant and irrelevant relationships were identified. The main relevant relationship is the internal navigability of a website, linking user experience design, usability, and click-through rate. Usability supports graphic design and navigability. Differences in click-through rates stem from poor usability and less user-friendly design of high-converting pages in the less-converting study. The analysis provides valuable information to improve these pages. An integrated model for future use is also presented. (Jeroen G. Mombarg, 2021)

Online scheduling systems offer numerous benefits. These systems digitize organization and planning for teacher schedules and streamline student registration for classes and events. Students can book and manage meetings and activities anytime, providing convenience and independence. Teachers can manage their schedules easily with notifications for bookings. Additional features include seamless integration with existing IT systems, automatic notifications, and data security. This system enhances communication, reduces costs, and increases attendance by providing a streamlined, efficient experience for students, teachers, and administrators. By implementing an online scheduling system, educational institutions, students, and teachers benefit from digitized scheduling and registration processes. It enhances organizational efficiency and enables teachers to manage their schedules effectively. These advantages lead to improved scheduling control, increased student engagement, and significant time savings for educators, making it a valuable tool for modern educational environments. (TIMIFY, 2020)

The effectiveness of visual elements and aesthetics in university websites for recruiting prospective students is significant. College and university websites serve as the gateway to the institution and the public face for both academics and athletics. According to Ruffalo Noel Levitz’s study on consumer behavior in 2014, these websites are crucial for providing information to prospective students (both new and transfer), current students, parents, and alumni. University recruiters aim to maximize the utility and depth of information on their websites while maintaining appealing and impactful aesthetics to attract potential students and donors. This entirely qualitative study examines the role of aesthetics in three key categories: web design, website functionality, and universities’ online recruitment strategies. The website functionality category is further divided into subcategories such as website usability, accessibility, and credibility. (Fatemeh Bordbar, 2016)

Higher education institutions need to have a reliable, effective, and attractive online presence because technology is becoming an important part of education. Higher education institutions play a essential role in society's development, and their websites have many responsibilities. They need to provide information for potential students, current students, faculty, and alumni. They often have to present a lot of information in an easy-to-navigate way, which is quite challenging. The goal of this research is to highlight the importance of higher education websites. We've reviewed a lot of literature to focus on the topic and conducted a survey to assess the usability of two websites. This paper will be helpful for higher authorities and web developers in understanding the significance and usability of higher education institute websites. (Mirfa Manzoor, 2012)

The relationship between college website impact and the impression formed by students is significant. Previous research has demonstrated that websites influence organizational impressions through aspects such as presentation, navigation structure, information quality, trust, perceived risk, and internet self-efficacy. This study extends these findings to the educational sector, using a survey methodology to reveal that college websites indeed affect the impressions prospective students form about the institution. Key influencing factors include website presentation, navigation structure, and the quality of posted information. (Thiagarajan Ramakrishnan, 2011)

School websites are crucial for transforming and enhancing the educational experience, offering platforms for innovation in teaching, learning, administration, and marketing. However, the importance placed on these websites and the resources allocated for their development vary widely. I reviewed how 30 South Australian schools utilized their websites, focusing on three key questions: (1) what variables determine website usage, (2) what makes a website engaging and useful, and (3) how a development framework can help improve website effectiveness. The study evaluated website design, purpose, content, and technology integration, identifying key indicators of progressive school websites. Based on these findings, a development framework was proposed to help schools create websites that accurately represent their identity and meet stakeholder needs. This research provides valuable insights for developing a holistic and effective school website for my project. (Beck, S. E. 2009)

Having an inviting and easy-to-navigate website that provides necessary information is crucial for any organization. People have high expectations whether they're online for work or personal reasons. In the academic world, many colleges and universities use the Internet to recruit students. These institutions know that parents and high school students use the Internet to find the best college. Their goal should be to present their academic programs and opportunities on their websites effectively. A key concern should be whether their websites are presenting their schools effectively to their audience. This study focuses on determining the features and elements that the target audience looks for in a university website when browsing online. (Marzie Astani, 2003)

An institution's website acts as its digital front door and is often the first-place students and their families visit to learn about the institution. If the website isn't welcoming, students may not explore further. A positive first impression can influence a student's decision to attend. Removing frustrating elements like unnecessary clicks and outdated information is vital, as students expect answers and easy navigation 24/7. Effective website design is key, especially during the COVID-19 pandemic, and must meet student expectations, be flexible, and accessible to all. Challenges include limited resources, rapidly changing digital landscapes, and maintaining consistent messaging across platforms. Involving students in the website creation process ensures it meets their needs. A high-quality website enhances an institution's brand, provides vital information, and can increase student enrollment and retention. Ultimately, a great website can turn prospective students into lifelong supporters and proud alumni. (Kenneth Mashinchi, 2024)

An education portal is essential for seamless information exchange between tutors and learners. It streamlines administrative tasks, manages resources, and centralizes student data. Teachers benefit from easy access to student information, attendance tracking, and class management. Students can view their performance, fee details, schedules, and leave requests. A well-designed portal connects administrators, teachers, parents, and students, offering user-friendly navigation, quick access to information, and integration with existing systems. It also provides dashboards for relevant data and a mobile app for on-the-go access. Secure portals ensure the privacy of student information. Online learning portals allow for virtual classes and interactions between teachers and students. Overall, a comprehensive school education portal enhances the efficiency of school operations and improves the learning experience. (SkoolBeep, 2024)

Effective website design is crucial for universities to attract and engage students. The article highlights ten tips for improving university websites, such as using storytelling to connect with students, incorporating images and videos to make content engaging, and ensuring easy navigation for a smooth user journey. Clear, honest design builds trust, while visual hierarchy directs attention to key information. Microinteractions and accessible, mobile-friendly design enhance user experience. Personalizing content based on user behavior increases relevance, and clear calls to action guide users through the website. Integrating social media buttons boosts online presence and community engagement. Examples from various universities demonstrate these principles in action. By focusing on these elements, universities can create websites that are informative, engaging, and user-friendly, meeting the needs of current and prospective students. (Manno Notermans, 2024)

The Government Graduate College of Science Lahore website offers basic information about the institution, including academic programs, admissions, and contact details. However, its design is simplistic, relying on the Google Sites platform. The lack of advanced features such as dynamic content, AI-powered tools, or mobile responsiveness limits user engagement and accessibility. It provides static content, which may not fully cater to the needs of a diverse audience like students, parents, and faculty. (GGCS, 2024)

The GCU Lahore website stands out for its professional design, user-friendly navigation, and detailed information on academic programs, admissions, and research activities. The website integrates multimedia content and provides updates on events, notifications, and academic calendars. However, GCU’s site also has strong branding and a responsive design, making it accessible across devices. (GCU, 2024)

Harvard University’s website is a well-designed and content-rich portal providing detailed information on academic programs, research initiatives, admissions, and campus life. The homepage emphasizes its global reputation through impactful visuals and concise updates. The website also includes sections dedicated to news, events, and alumni, reflecting its commitment to community engagement. Navigation is intuitive, with links tailored for prospective students, current students, and faculty. (Harvard University, 2024)

Harvey Mudd College’s website highlights the institution’s focus on STEM education through a clean design and informative content. The site includes sections for admissions, academics, campus life, and events, with a special emphasis on innovation and interdisciplinary learning. It also features detailed profiles of faculty and their research interests. While functional, the website could benefit from more dynamic interactive elements to enhance user engagement. (Harvey Mudd College, 2024)

The University of Michigan’s website reflects its status as a leading public research university. It provides comprehensive information on academic programs, admissions, and research opportunities. The site features sections for news, events, and university rankings, showcasing its global impact. Its responsive design ensures accessibility across devices, and interactive elements like a campus map and virtual tour enhance user engagement. (University of Michigan, 2024)

Stanford University’s website combines minimalist design with rich functionality. It offers detailed sections on academics, research, campus life, and innovation. The homepage features news, events, and multimedia content, effectively engaging users. The site also includes dedicated resources for prospective students, faculty, and alumni, reflecting Stanford’s diverse community. Its design prioritizes usability and accessibility, with an intuitive navigation structure. (Stanford University, 2024)

In today's digital age, school websites are essential for communication, education, and community engagement. They provide a centralized hub for students, parents, and staff to access important information like schedules, events, and contact details, improving transparency and communication. Websites help schools share announcements, updates, and policies, reducing misunderstandings. They also facilitate parental involvement by offering resources and keeping parents informed about their child's progress. Schools can showcase achievements and attract new students by highlighting their programs and values. Websites enhance learning by hosting educational resources and supporting teacher-student communication. They also build community by connecting students, parents, alumni, and community members. Overall, investing in a well-designed, regularly updated website strengthens relationships and improves education. (iPistis, 2024)

Nielsen’s ten usability heuristics provide essential guidelines for creating user-friendly and intuitive website by ensuring visibility of system status, matching the system with the real world, granting user control and freedom, maintaining consistency and standards, preventing errors, supporting recognition over recall, enhancing flexibility and efficiency, adopting aesthetic and minimalist design, helping users recognize, diagnose, and recover from errors, and providing help and documentation. These principles help designers and stakeholders stay aligned, identify usability issues early, and ensure products meet user needs effectively. (Jakob Nielsen, 2024)

The Web Content Accessibility Guidelines (WCAG) 2, developed through the W3C process with global cooperation, aim to provide a unified standard for web content accessibility, catering to the needs of individuals, organizations, and governments. These guidelines detail making web content—text, images, sounds, and code—accessible to people with disabilities. Targeted primarily at web content developers, authoring tool developers, accessibility evaluation tool developers, and anyone needing a web accessibility standard, WCAG serves as a technical guide rather than an introductory resource to accessibility. (W3C, 2024)

The evolution of higher education websites demonstrates how technology and user expectations have transformed over time. Initially, in the 1990s, college websites were simple, static pages. Over the years, they have become dynamic and interactive platforms designed to meet the diverse needs of users. A successful college website requires a deep understanding of its audience, organized content architecture, and effective design principles like intuitive navigation and mobile responsiveness. It's also essential to maintain the website with ongoing management and regular updates to keep it relevant and user-friendly. By focusing on these key areas, institutions can create websites that engage users, provide valuable information, and support their strategic goals. This involves analyzing user data, aligning content with user needs, and employing high-quality visuals and multimedia to enhance the user experience. The goal is to create an engaging, informative, and accessible website that effectively serves prospective students, current students, faculty, and the broader community. (Jenelle Kruse, 2024)

Agile methodology is a favored approach in software development due to its flexibility and customer focus. It emphasizes iterative development, frequent customer involvement, and collaborative teamwork, which lead to high-quality products and better project control. Agile allows for faster launches, continuous delivery, transparency, and risk reduction. It differs from traditional methods like Waterfall, which are more rigid and linear, by promoting ongoing feedback and real-time adjustments. Research shows 71% of companies use Agile, with a 28% higher success rate in projects. Agile's emphasis on communication and collaboration enhances project outcomes and makes it particularly effective for complex, changing environments. This methodology ensures that projects remain aligned with customer needs and can adapt to changes efficiently, making it a popular choice across industries. (Gurpreet Kaur, 2024)

Django is a powerful, open-source web development framework built with Python, designed for quick and secure web application development. It is popular due to its simplicity, flexibility, and robust features, making it suitable for various projects like content management systems and eCommerce platforms. Django follows the Model-View-Template (MVT) architecture, making it easy to manage application logic, user interfaces, and data. It offers built-in tools like an Object-Relational Mapper (ORM), dynamic CRUD interface, and testing framework, which speed up development and ensure security against common threats like SQL injection and cross-site scripting. Django's scalability makes it suitable for both small and large applications, and it supports multiple operating systems and databases. The framework's active community and extensive documentation provide valuable support to developers. Companies like Instagram, Spotify, The Washington Post, and Dropbox use Django for its efficiency and reliability. Overall, Django's rapid development capabilities, strong security measures, and versatile features make it a top choice for developers. (Julia Korsun, 2024)

This comprehensive examination focuses on optimizing college website infrastructure by implementing personalized databases and enhancing UI/UX design. Utilizing the MERN stack, known for its scalability, flexibility, and real-time data processing, we aimed to fortify website security, streamline operations, and improve user engagement metrics. The research addressed challenges like security vulnerabilities and scalability concerns due to increasing server traffic. By systematically identifying and mitigating vulnerabilities, we ensured a robust security framework. Personalized databases for students, teachers, and administrators were meticulously crafted to facilitate streamlined access to relevant information and services tailored to individual roles and preferences. Additionally, significant efforts were made to enhance the website's UI/UX design, fostering intuitive navigation, accessibility, and aesthetic appeal. Integrating interactive elements, optimizing page load times, and implementing responsive design principles significantly improved user satisfaction metrics, resulting in a more immersive and rewarding digital experience for all stakeholders. (Rekha Sahare, 2024)

Effective website design is crucial for universities to attract and engage students. The article highlights ten tips for improving university websites, such as using storytelling to connect with students, incorporating images and videos to make content engaging, and ensuring easy navigation for a smooth user journey. Clear, honest design builds trust, while visual hierarchy directs attention to key information. Micro interactions and accessible, mobile-friendly design enhance user experience. Personalizing content based on user behavior increases relevance, and clear calls to action guide users through the website. Integrating social media buttons boosts online presence and community engagement. Examples from various universities demonstrate these principles in action. By focusing on these elements, universities can create websites that are informative, engaging, and user-friendly, meeting the needs of current and prospective students. (Manno Notermans, 2024)

When creating a new website, it's important to decide between a static and dynamic design. Static websites display the same content to everyone and are made with HTML, CSS, and JavaScript. They're easy to create, secure, and perform well, but updating content is time-consuming, and they can't offer personalized experiences. Dynamic websites generate content based on user interactions, using databases to tailor information. They simplify content management, enhance user experience, and allow personalization, but they're more complex and costly to develop. Most modern websites use a hybrid approach, combining static pages for stable content and dynamic pages for frequently updated information. This combination allows flexibility in managing different types of content, catering to various user needs. The choice between static and dynamic depends on the website's goals and the experience you want to offer your audience. (Nick Babich, 2023)

Mobile-first design prioritizes the needs of mobile users in web development due to Google's mobile-first indexing, which favors mobile-friendly sites in search rankings. This approach ensures websites work well across all devices, including Android and iOS. Essential elements for a mobile-first design include good UX/UI, responsive web design, and simple navigation. The benefits of mobile-first design include improved user experience, faster loading times, better SEO, increased conversions, cost-effectiveness, and future-proofing. Best practices involve keeping designs simple, prioritizing content, optimizing for touch, ensuring responsive design, and testing on various devices. However, challenges include keeping up with new technologies, balancing design priorities, ensuring accessibility, optimizing performance, and adjusting for different screen sizes. Embracing mobile-first design is crucial in today's digital landscape to enhance user experience and stay competitive. (Loveneet singh, 2023)

Student recruitment involves strategies to attract and convert leads into enrolled students. To do this, educational institutions need to understand their potential students' profiles, fears, and dreams, offering them the best solutions. Effective educational marketing can attract and retain students, transforming them into brand advocates. Eight tips to attract students include understanding the student's profile, investing in content marketing, making the website attractive, forming partnerships with companies, promoting community events, participating in educational programs, investing in sponsored links, and being active on social networks. Each step involves tailored actions like creating personas to understand students, producing relevant content, designing user-friendly websites, and engaging with the community both online and offline. By implementing these strategies, schools can enhance their recruitment efforts and create lasting connections with potential students. (Gustavo Goncalves, 2023)

Educational institutions handle large volumes of data related to students, instructors, and administration, requiring efficient systems for management. The College Management System (CMS) addresses this need by providing a comprehensive database management solution. This system, built using JavaScript, and CSS, helps reduce administrative burdens, improve communication, and enhance student engagement. It allows administrators to manage and update student and instructor data through an intuitive interface, making tasks more efficient. By implementing CMS, institutions benefit from centralized data storage, real-time updates, improved communication, and better student participation. These advantages lead to data-driven decision-making and institutional improvements, offering enhanced educational experiences. The CMS is therefore a valuable tool for streamlining operations, addressing administrative challenges, and fostering a more effective educational environment. (Varma, A., & Sharma, A., 2022)

Technology has revolutionized education, making functional websites essential for universities. These sites provide critical information to parents, students, and education stakeholders and need key elements to ensure high conversion rates. Important features include seamless navigation for easy page transitions, mobile responsiveness to cater to smartphone users, and high-quality, authentic images that reflect the institution's diversity. Valuable content that meets user needs, visible information on programs and majors, and details on job placements after graduation are crucial. Additionally, clear admissions requirements, a strategically located search box, and accurate location and contact details enhance user experience. Moreover, these websites should offer valuable, regularly updated content to keep users informed and engaged. Highlighting programs and majors prominently helps prospective students find relevant information quickly. Sharing success stories of alumni can inspire new students. Clear admissions guidelines and strategically placed search functionality improve usability. Accurate location and contact information facilitates campus visits. (Webdesign, 2021)

Figma is a popular design tool known for its real-time collaboration, browser-based access, and easy-to-use prototyping features. It allows multiple users to work together on designs simultaneously, making feedback and communication efficient. Accessible from any device with internet, Figma offers flexibility and is platform-independent. It features a user-friendly interface for creating animations and interactions, making it suitable for both beginners and experienced designers. The Inspect tool helps simplify the developer handoff process by providing necessary code. Figma’s component system ensures consistent design updates across projects. It also supports plugins, offers extensive community resources, and provides regular updates, making it a cost-effective and comprehensive solution for modern design needs. Figma's real-time collaboration, strong community support, and continuous improvements make it a versatile choice for individual designers and large teams alike. (Daniel Danielyan, 2021)

Building an appealing web design is a front-end developer’s dream, but it often involves lots of repetitive coding. Bootstrap, a free and powerful front-end framework for HTML, CSS, and JavaScript, has revolutionized this process. Created by Mark Otto and Jacob Thornton for Twitter, Bootstrap is now a popular tool for web development. It saves time with pre-made design themes and templates, making it easy to use even for beginners. Bootstrap’s responsive grid system ensures websites are mobile-friendly, and it allows for customization to suit specific project needs. Compatible with modern browsers, it ensures consistency across different devices. As an open-source framework with strong community support, Bootstrap provides tutorials, updates, and user-generated resources. Its user-friendly features, like cross-browser compatibility and easy content management, make it an essential tool for developers looking to improve work efficiency without spending money. Bootstrap can streamline your web design process and boost productivity. (Simon Dwight Keller, 2019)

Websites have become a popular way to access information, but many users still struggle with navigation and finding relevant data. This study assessed the accessibility of 30 Jordanian university websites against WCAG 2.0 guidelines. It also tested the usability of Al-Zaytoonah University’s website by interviewing six students. The students were asked to perform 14 tasks, such as navigating the homepage and accessing e-learning sites, without assistance. The study found that most university websites were not easily accessible and had usability issues, including poor web link organization and a lack of tags and information. Additionally, the students' feedback highlighted that the user interface needed improvements, such as better layout design and clearer navigation paths. These findings emphasize the need for universities to enhance website accessibility and user experience. (Mahafzah, A., 2017)

Educational institutions are increasingly targeted by cybercriminals due to the large amounts of sensitive data they manage. They face challenges like limited cybersecurity resources and frequent staff turnover, making them vulnerable to threats like phishing, ransomware, and insider attacks. Recent data breaches at places like the Los Angeles Unified School District and the University of Michigan highlight the need for better security measures. Common issues include outdated systems and insufficient training. It's crucial for schools to invest in robust cybersecurity, provide consistent training, and create awareness to protect sensitive information and maintain safe educational operations in the digital age. (Sara Jelen, 2024)

The Government Shalimar College website is more structured and offers features like academic information, faculty details, and admissions guidance. It still lacks advanced interactive tools. Additionally, the site does not provide personalized user experiences, which could enhance engagement for prospective students. (GSC, 2024)

An admin panel is a user-friendly control center for managing various aspects of a web application, like content, user accounts, and analytics, which is especially crucial for educational institution websites. These sites need to provide essential information to parents, students, and stakeholders, and an admin panel makes this management much easier. It allows administrators to update and maintain the site without extensive coding, enabling both admins and developers to work more efficiently. With centralized information, businesses and institutions can make better decisions and enjoy simplified reporting. Admin panels save time and money by allowing quick content management, providing real-time data insights, and ensuring security. They help prevent performance issues and automate tasks, freeing up resources for other activities. For educational institutions, an admin panel is vital to keep content relevant and accessible, enhancing the overall user experience. (Karan Datwani, 2023)

HTML is essential in web development, providing the structure and content for web pages. It enables browsers to display information in a human-readable format. HTML has evolved from a simple markup language to a versatile tool that meets the needs of web developers and users. Mastering HTML, following best practices, and using complementary tools allow developers to create accessible and efficient web experiences. As the digital landscape evolves, HTML remains core to building the future web, forming the foundation for both beginners and experienced developers. (joswellahwasike, 2024)

CSS (Cascading Style Sheets) defines the presentation and design of web pages, including colors, fonts, and layouts. It separates content from presentation, ensuring compatibility across different devices. Maintained by the W3C, CSS can be used with any XML-based markup language. Its rule sets consist of selectors and declaration blocks. CSS is crucial for styling web interfaces, making it a valuable tool for achieving project goals based on specific needs and business requirements. (Priya Pedamkar, 2023)

JavaScript, introduced by Netscape in 1995, is a versatile programming language used for both client-side and server-side web development. It enhances interactivity, manipulates the Document Object Model (DOM), supports asynchronous programming, and provides access to various Web APIs. With a rich ecosystem of frameworks and libraries like React and Angular, JavaScript simplifies complex web development tasks. Its expansion into server-side development with Node.js allows developers to build scalable, high-performance web applications using a single language, making it a crucial tool in modern web development. (Codeworks, 2024)

Visual Studio Code (VS Code) is a free, open-source code editor developed by Microsoft for Windows, Linux, and macOS. Announced in 2015, it quickly became the most popular developer tool by 2019. Written in TypeScript, JavaScript, HTML, and CSS, VS Code offers features like debugging, syntax highlighting, intelligent code completion (IntelliSense), snippets, code refactoring, and embedded Git. The key reasons to choose VS Code include user-friendliness, cross-platform support, lightweight design, robust architecture, IntelliSense, and being free. Additionally, it supports web technologies like HTML, CSS, and JSON, making it an excellent choice for developers. (Shruti Sharma, 2021)

Figma, a web-based design application launched in 2016, revolutionizes UI/UX design with its intuitive interface and real-time collaboration features, making it more accessible than traditional tools like Photoshop. Ideal for creating wireframes, prototypes, and mockups, Figma supports cross-platform use and enhances productivity, despite requiring an internet connection and having limited offline capabilities. Its versatility extends to product design, marketing materials, and UX enhancements, making it a preferred choice for designers of all levels seeking to improve their workflow and design quality. (Fita, 2024)

Universities, colleges, and schools critically need robust network security to protect their data and systems. These institutions face challenges in balancing open access to information with strong security measures. Implementing effective network access security, controlling careless user behavior, and promoting best practices through alerts and notifications for students, staff, and faculty are essential. UserLock, for example, helps secure logins, prevent simultaneous sessions, and make users accountable for their actions. It controls access based on roles, devices, and locations, allowing remote session management. Such measures ensure safe use of personal devices and help enforce security policies, reducing unauthorized access risks. (Chris Bunn, 2018)

3 Software Requirement

A crucial stage of the software development lifecycle is requirement analysis, during which the goals of the system, user requirements, and limitations are carefully considered and noted. This stage of the **GIGCCL Official Website: Enhancing Academic Excellence and Communication** focuses on figuring out the needs of all parties involved and making sure they support the project's objectives of updating the current system into a comprehensive, cutting-edge platform that effectively handles academic and extracurricular data, improves user engagement, and facilitates smooth communication via a user-friendly web interface.

3.1 User Classes and Characteristics

The four main user classes of the **GIGCCL Official Website** are Admin, Students, Faculty members, and Alumni. The unique features, responsibilities, and duties of every user class ensure the system's efficient operation. The main user classes and their attributes are provided in the following table:

User Class	**User Characteristics**
Administrators	Manage and update website content, academic programs, events, announcements, and handle alerts and notifications effectively.
Students	Explore academic programs, admissions information, timetables, exam schedules, campus updates, extracurricular resources, and notifications.
Faculty Members	Manage and update timetable, exam schedules, events, academic as per define roles.
Alumni	Use the career portal to find job opportunities, explore job descriptions and access guidelines for job applications.

Table 4: 3-1. User Classes and Characteristics

3.2 Functional Requirements

The functional requirements for **GIGCCL Official Website** are classified based on the system core features and functionalities.

3.2.1 User Management

- **Department Information Module**
  Create a department information module where user can view all departments and programs offered by institute.
- **Students Resources Tools**
  Provide tools that are accessible through a single interface, including as timetables, exam schedules, and updates on extracurricular activities.
- **Campus Life**
  Create a section which uses to encourage involvement on campus, highlight events, conferences, and extracurricular activities.
- **Job Opportunities**
  Provide jobs description, qualifications and application listed by institution that are relevant to careers.
- **Admissions Section**
  Provide links to the application submission portal for admission which also facilitate eligibility criteria and further procedure.
- **Notifications and Alerts System**
  Provide crucial information about upcoming activities on campus, exam dates, extracurricular activities and admissions deadlines.

3.2.2 Admin Management

- **Admin Dashboard**
  Allow administrators access to a complete dashboard using valid credentials that includes information on events, academic programs, and overall content of website.
- **Content Management**

Offer website administrators the ability to add, modify, and remove information, including announcements, campus activities, and academic program data. Maintain dynamic content changes without the need for technological expertise.

- **User Management**

  Offer administrators the ability to create, modify, and deactivate admin accounts in order to control access to the admin panel. Make sure that all admin users have safe login and password management.

- **Search and Filtering Tools**

  Provide the admin panel powerful search and filter capabilities so that it can quickly find and handle particular content, such event entries or listed jobs.

- **Notification and Alert System**

  Offer administrators the resources they need to plan and create alerts or notifications about crucial dates, such admissions and exam dates, to make sure quick interaction with the college community.

3.3 Non-Functional Requirements

Non-functional requirements (NFRs) make sure the **GIGCCL Official Website's** security, reliability, and operation. Rather than focusing on particular actions, these requirements illustrate the system's working features.

- **Performance**

  Manage at least 500 users at once without experiencing any reduction in performance. Under typical traffic situations, the website should load pages in two to three seconds.

- **Scalability**

  The system should support future development of the system such as more features or higher user traffic, should be supported. Use horizontal scaling techniques to add more servers when traffic increases.

- **Usability**

  All user categories must be able to navigate the website with ease because of its intuitive and user-friendly interface. Make sure that the platform's design components are consistent using CSS framework like Bootstrap5.

- **Responsiveness**
  The website has to be completely responsive so that it works on mobile devices, tablets, and PCs. By using CSS media queries which provides a mobile-first design approach to make sure proper rendering on all screen sizes.
- **Security**
  To stop unwanted access, use secure authentication for the admin panel and website. Manage data integrity and defend against security flaws like cross-site scripting (XSS) and SQL injection by using Django middleware.
- **Accessibility**
  Develop the website using accessibility guidelines (like WCAG 2.1 Level AA) in mind to make it usable by people with difficulties.
- **Maintainability**
  Use a version control systems like Git to manage modular code, make changes easily and also maintained detailed documentation to simplify future updates and troubleshooting.
- **Reliability**
  Make sure minimal downtime while performing server maintenance or updates. The website should function 99.9% of the time with no major disruptions. To prevent single points of failure, use redundant server configurations.
- **Portability**
  The system should be easily deployable across various hosting environments.
- **Backup and Recovery**
  To make sure data recovery in the case of a system breakdown, set up regular backups or use cloud storage solutions like Azure, AWS to store backup securely.

3.4 External Interface Requirements

3.4.1 Software Interfaces

3.4.1.1 Operating System

- **Development Environment:** Supports Linux, macOS, and Windows 10/11 for development computers.
- **Server Environment:** Windows Server or Linux-based servers (such as Ubuntu Server or CentOS) for the application's hosting.

3.4.1.2 Tools and Framework

- **Python 3.x:** Used with Django as the main web framework for back-end development.
- **Django Libraries:** Includes Pillow for image processing and Django Filter for more sophisticated search capabilities.

3.4.1.3 Database Management System:

- **SqLite3:** An embedded database that is lightweight and used to store notifications, event details, and program details.

3.4.1.4 Development Tools

- **Code Editor:** Use Visual Studio code for writing, testing and debugging code.
- **Version Control:** Git with repositories hosted on GitHub or GitLab for source code management.

3.4.1.5 Web Server and Application Server

- **Web Server**: For hosting and handling HTTP requests, use Apache or Nginx.
- **Application Server:** Gunicorn or uWSGI for serving the Django application in production.

3.4.1.6 Security Tools

- **SSL/TLS Protocols:** Data security in transmission is ensured for safe HTTPS communication.
- **Firewall and Security Plugins:** To defend against cyberthreats and illegal access.

3.4.1.7 Browser Compatibility

- Fully compatible with modern web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

3.4.1.8 Backup and Storage Solutions

- Data recovery in the event of a system failure is ensured by routine backups to cloud storage services like AWS S3, Microsoft Azure, Google Cloud Storage, or comparable ones.

3.4.2 Hardware Interfaces

3.4.2.1 Development Machine

- At least 8GB RAM
- Intel i5 or equivalent processor
- 256GB SSD for storage and local development

3.4.2.2 Server Requirement

- At least 4GB RAM and 2 CPU cores
- At least 50GB storage for hosting the website
- Hosted on a cloud-based Platform or physical server to make sure scalability and reliability

3.4.2.3 User Devices

- The website is accessible on desktops, tablets, and smartphones, requiring standard hardware and an internet connection.

3.4.3 Communication Interfaces

3.4.3.1 Web Protocols

- Use HTTP/HTTPS to make sure secure communication between server and users.

3.4.3.2 Web Browsers Interface

- Standard web browsers are used by users to access the website, guaranteeing cross-platform usability.

3.4.3.3 Internet Connectivity

- Reliable high-speed internet is required to ensure seamless access to the website.

3.4.3.4 Network Protocol

- Functioning over TCP/IP to handle data transmission between clients and servers.

3.5 Use Case Analysis

3.5.1 Use Case # 1: Admin/Faculty Login (UC1)

UC Identifier	UC-1
UC Name	Admin/Faculty Login
Requirements Traceability	Related to functional requirement for Admin authentication.
Purpose	Allow admin and faculty to securely log into the system and access their features.
Priority	High
Preconditions	Admin/Faculty has valid login credentials.
Post conditions	Admin/Faculty gains access to the admin dashboard.
Actors	Admin, Faculty
Extends	None
Main Success Scenario	Admin/Faculty opens the login page. Admin/Faculty enters credentials and submits. System validates credentials and redirects to the admin dashboard.
Alternate Flows	Admin/Faculty forgets credentials and resets them.
Exceptions	Invalid credentials result in an error message.
Includes	None

Table 5: 3-2. Admin/Faculty Login (UC1)

3.5.2 Use Case # 2: Update Offered Programs (UC2)

UC Identifier	UC-2
UC Name	Update Offered Programs

Requirements Traceability	Related to functional requirements for academic program management.
Purpose	Allow Admin and Faculty to update the details of academic programs offered.
Priority	High
Preconditions	Admin/Faculty is logged into the system with proper privileges.
Post conditions	Updated programs are reflected on the website
Actors	Admin, Faculty
Extends	None
Main Success Scenario	Admin/Faculty logs into the system. Navigates to the "Programs" section. Updates program details (add/edit/delete). System validates and saves changes in the database.
Alternate Flows	Admin/Faculty cancels the operation.
Exceptions	Validation errors (e.g., missing fields).
Includes	None

Table 6: 3-3. Update Offered Programs (UC2)

3.5.3 Use Case # 3: Update Timetable (UC3)

UC Identifier	UC-3
UC Name	Update Timetable
Requirements Traceability	Related to functional requirements for timetable management.
Purpose	Allow Admin and Faculty to update and manage timetables.
Priority	High

Preconditions	Admin/Faculty is logged into the system.
Post conditions	Updated timetable is displayed to students.
Actors	Admin, Faculty
Extends	None
Main Success Scenario	Admin/Faculty logs in. Navigates to "Timetable" section. Updates timetable (add/edit/delete). Changes are saved and visible to students.
Alternate Flows	Operation cancelled by Admin/Faculty.
Exceptions	Input validation errors.
Includes	None

Table 7: 3-4. Update Timetable (UC3)

3.5.4 Use Case # 4: Update Exam Detail (UC4)

UC Identifier	UC-4
UC Name	Update Exam Detail
Requirements Traceability	Related to functional requirements for managing exam details.
Purpose	Allow Admin and Faculty to add or update exam details.
Priority	High
Preconditions	Faculty is logged in with valid credentials.
Post conditions	Updated exam details are visible to students.
Actors	Admin, Faculty

Extends	None
Main Success Scenario	Admin/Faculty logs in. Navigates to "Exam Details" section. Updates exam details (add/edit/delete). Changes are saved and visible to students.
Alternate Flows	Admin/Faculty cancels the operation.
Exceptions	Input validation errors.
Includes	None

Table 8: 3-5. Update Exam Detail (UC4)

3.5.5 Use Case # 5: Update Event Detail (UC5)

UC Identifier	UC-5
UC Name	Update Event Detail
Requirements Traceability	Related to functional requirements for event management.
Purpose	Allow Admin and Faculty to update event details.
Priority	Medium
Preconditions	Admin/Faculty is logged into the system.
Post conditions	Updated event details are displayed to users.
Actors	Admin, Faculty
Extends	None

Main Success Scenario	Admin/Faculty logs in. Navigates to "Events" section. Updates event details (add/edit/delete). System saves changes and updates the website.
Alternate Flows	Operation cancelled by Admin/Faculty.
Exceptions	Input validation errors.
Includes	None

Table 9: 3-6. Update Event Detail (UC5)

3.5.6 Use Case # 6: List Job Opportunities (UC6)

UC Identifier	UC-6
UC Name	List Job Opportunities
Requirements Traceability	Related to functional requirements for job opportunities management.
Purpose	Allow Admin to add, update, or delete job opportunities.
Priority	Medium
Preconditions	Admin is logged into the system with appropriate privileges.
Post conditions	Updated job opportunities are reflected on the website and visible to alumni.
Actors	Admin
Extends	None
Main Success Scenario	Admin logs in to the system. Navigates to the "Job Opportunities" section. Adds, edits, or deletes job postings. System validates the inputs and updates the database. Changes are reflected on the website for alumni to view.

Alternate Flows	Operation cancelled by Admin.
Exceptions	Validation errors such as missing or incorrect job details.
Includes	None

Table 10: 3-7. List Job Opportunities (UC6)

3.5.7 Use Case # 7: Update Alumni (UC7)

UC Identifier	UC-7
UC Name	Update Alumni (Old Franians)
Requirements Traceability	Related to functional requirements for alumni management.
Purpose	Allow Admin and Faculty to add, update, or remove alumni details (Old Franians).
Priority	Medium
Preconditions	Admin/Faculty is logged into the system with appropriate privileges.
Post conditions	Alumni details are updated in the database and reflected on the website.
Actors	Admin, Faculty
Extends	None
Main Success Scenario	Admin/Faculty logs in to the system. Navigates to the "Alumni Management" section. Adds, edits, or deletes alumni records. System validates the inputs and updates the database. Changes are reflected on the website under the alumni section.

Alternate Flows	Admin/Faculty cancels the operation.
Exceptions	Validation errors such as missing or incorrect alumni data
Includes	None

Table 11: 3-8. Update Alumni (UC7)

3.5.8 Use Case # 8: View Departments (UC8)

UC Identifier	UC-8
UC Name	View Departments
Requirements Traceability	Related to functional requirement for displaying department.
Purpose	Allow students to view departments.
Priority	Medium
Preconditions	Departments are available in the system.
Post conditions	Departments are displayed to the student.
Actors	Student
Extends	None
Main Success Scenario	Student navigates to the department section. Views the list of departments.
Alternate Flows	None
Exceptions	System fails to retrieve department details.
Includes	None

Table 12: 3-9. View Departments (UC8)

3.5.9 Use Case # 9: View Offered Programs (UC9)

UC Identifier	UC-9
UC Name	View Offered Programs
Requirements Traceability	Related to functional requirements for displaying offered programs.
Purpose	Allow Student to view academic programs offered by institute.
Priority	High
Preconditions	Program information must be up-to-date in the database.
Post conditions	Students can access the program information.
Actors	Student
Extends	None
Main Success Scenario	Student navigates to the program information section. Student view the offered programs.
Alternate Flows	None
Exceptions	System fails to retrieve program data.
Includes	None

Table 13: 3-10. View Offered Programs (UC9)

3.5.10 Use Case # 10: View Timetable (UC10)

UC Identifier	UC-10
UC Name	View Timetable
Requirements Traceability	Related to functional requirements for displaying timetables.

Purpose	Allow Students to view class timetables.
Priority	Medium
Preconditions	Timetables are updated in the system.
Post conditions	Timetables are displayed to students.
Actors	Student
Extends	None
Main Success Scenario	Student navigates to "Timetable" section. Views timetable details.
Alternate Flows	None
Exceptions	System fails to retrieve timetable data.
Includes	None

Table 14: 3-11. View Timetable (UC10)

3.5.11 Use Case # 11: View Exam Details (UC11)

UC Identifier	UC-11
UC Name	View Exam Details
Requirements Traceability	Related to functional requirements for displaying exam schedules and details.
Purpose	Allow Students to view exam schedules.
Priority	Medium
Preconditions	Exam details must be updated and available in the system.
Post conditions	Students can view the latest exam details on the website.
Actors	Student

Extends	None
Main Success Scenario	Student navigates to the "Exam Details" section on the website. Exam details are displayed to the student.
Alternate Flows	None
Exceptions	System fails to retrieve exam details due to database or server issues.
Includes	None

Table 15: 3-12. View Exam Details (UC11)

3.5.12 Use Case # 12: View Events (UC12)

UC Identifier	UC-12
UC Name	View Events
Requirements Traceability	Related to functional requirements for displaying campus events.
Purpose	Allow Students and Alumni to view details of campus events.
Priority	Medium
Preconditions	Event details must be updated and available in the system.
Post conditions	Student/Alumni can view the latest event information on the website.
Actors	Student, Alumni
Extends	None

Main Success Scenario	Student/Alumni navigates to the "Events" section on the website. Browses the list of available campus events. Selects a specific event to view more details. The system retrieves and displays event information.
Alternate Flows	None
Exceptions	System fails to retrieve event details due to database or server issues.
Includes	None

Table 16: 3-13. View Events (UC12)

3.5.13 Use Case # 13: View Co-Curricular Activities (UC13)

UC Identifier	UC-13
UC Name	View Co-Curricular Activities
Requirements Traceability	Related to functional requirement for co-curricular activities.
Purpose	Allow students to view co-curricular Activities.
Priority	Medium
Preconditions	Co-curricular activities must be updated and available in the system.
Post conditions	Users can view the latest co-curricular on the website.
Actors	Student
Extends	None

Main Success Scenario	Student navigates to the "co-curricular activities" section on the website. Co-curricular activities are displayed to the student.
Alternate Flows	None
Exceptions	System fails to retrieve co-curricular details due to database or server issues.
Includes	None

Table 17: 3-14. View Co-Curricular Activities (UC13)

3.5.14 Use Case # 14: View Alerts and Notification (UC14)

UC Identifier	UC-14
UC Name	View Alerts and Notification
Requirements Traceability	Related to functional requirement for Alerts and Notifications.
Purpose	Allow Student/Alumni to view alerts and notifications on latest news.
Priority	Medium
Preconditions	Notifications are created and published by administrators.
Post conditions	Student/Alumni can view the latest notifications on the website.
Actors	Students, Alumni
Extends	None
Main Success Scenario	Students, Alumni view the notification on the home screen as a pop-up message.

Alternate Flows	None
Exceptions	No Notification found or notification details are unavailable due to database issues.
Includes	None

Table 18: 3-15. View Alerts and Notification (UC14)sss

3.5.15 Use Case # 15: View Contact Information (UC15)

UC Identifier	UC-15
UC Name	View Contact Information
Requirements Traceability	Related to functional requirements for contact details
Purpose	Allow Student/Alumni to view contact details of the institute.
Priority	Low
Preconditions	Contact details are available in the system.
Post conditions	Contact details are displayed to the Student/Alumni.
Actors	Student, Alumni
Extends	None
Main Success Scenario	Student/Alumni navigates to "Contact Us" section. Views the institute's contact information.
Alternate Flows	None
Exceptions	System fails to retrieve contact information.
Includes	None

Table 19: 3-16. View Contact Information (UC15)

3.5.16 Use Case # 16: View Job Opportunities (UC16)

UC Identifier	UC-17
UC Name	View Job Opportunities
Requirements Traceability	Related to functional requirement for Job Opportunities.
Purpose	Allow Alumni to view job opportunities.
Priority	Medium
Preconditions	Job opportunities are updated in the system.
Post conditions	Alumni can view and apply for jobs.
Actors	Alumni
Extends	None
Main Success Scenario	Alumni navigate to "Job Opportunities" section. Views a list of available job postings.
Alternate Flows	None
Exceptions	System fails to load job opportunities.
Includes	None

Table 20: 3-17. View Job Opportunities (UC16)

3.6 Use Case Diagram

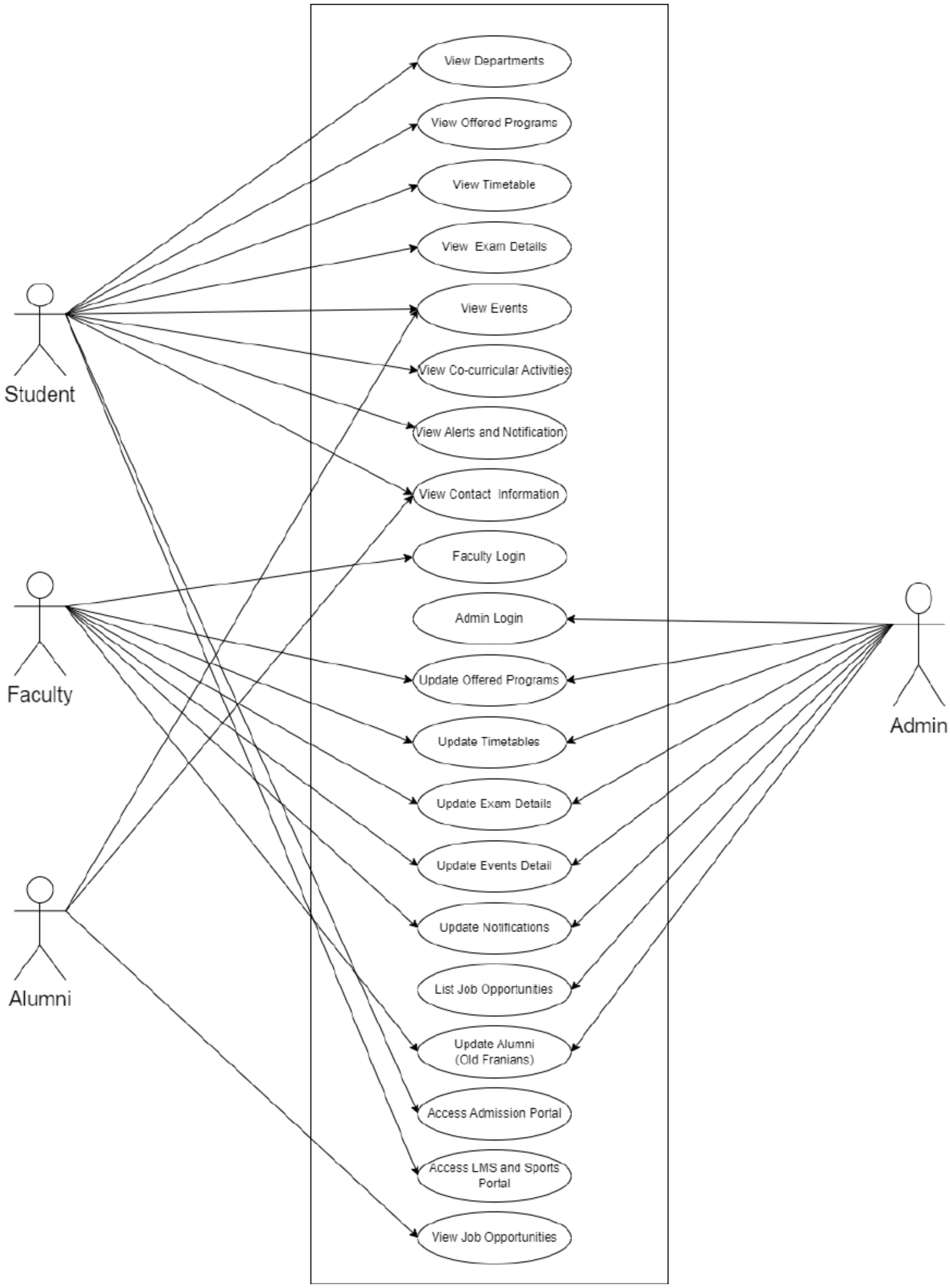


Figure 3:3-1. Use Case Diagram

4 Software Design

The design of the **GIGCCL Official Website** follows a modular and scalable architecture, rooted in the system requirements analysis phase. Through clearly defined components including the Admin Panel, Department Information Module, Student Resources Tools, Career Portal, and Campus Life Section, it effectively manages academic programs, admissions, events, co-curricular activities, Exams schedules, alumni records, career opportunities, and notifications. To provide a clear and well-organized representation of the system's architecture, each component is carefully developed and depicted using UML diagrams. With the help of innovative web technologies and industry best practices, the website offers a smooth user experience, quick access to data, and ease of maintenance, making it a complete and interesting platform for the GIGCCL community.

4.1 Work Breakdown Structure

The Work Breakdown Structure (WBS) for the GIGCCL Official Website arranges the tasks into clear phases to make sure effective execution. It starts with Project Initiation, addressing planning, resource allocation, and risk assessment to build a strong foundation. This is succeeded by System Design, where architectural, database, and user interface designs are developed to determine implementation. Building and integrating front-end and back-end components is part of the development phase, which also includes thorough testing to guarantee functionality. In the Testing and Quality Assurance phase, the system is validated through unit, integration, and system tests, as well as stakeholder acceptance. Finally, the Deployment phase makes sure a seamless launch with production readiness, deployment, and further maintenance. This structured approach makes sure efficient project management and successful system delivery.

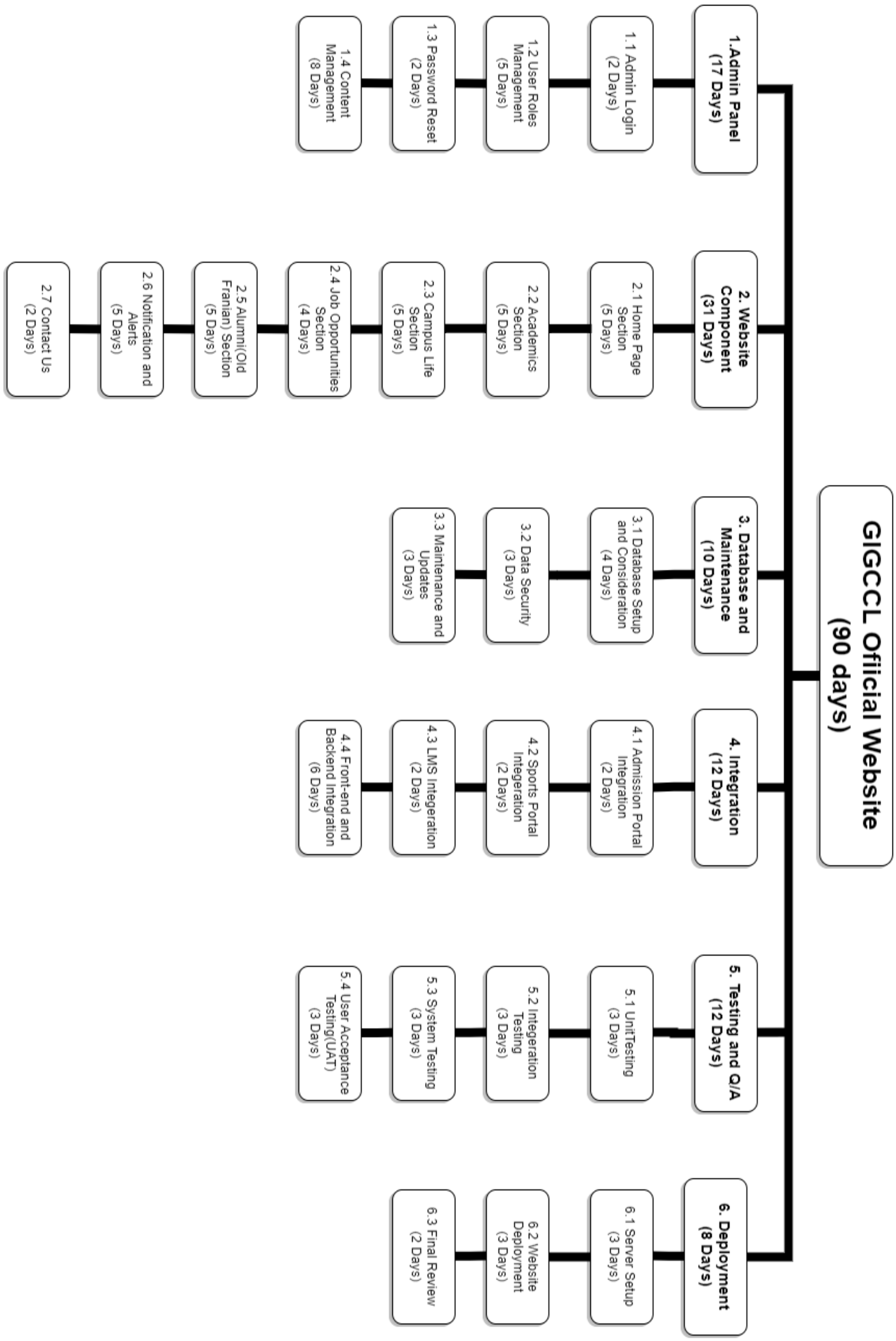


Figure 4: 4-1. Work Breakdown Structure for GIGCCL Official Website

4.2 Design Models

4.2.1 Class Diagram

The primary components of the system and their relationships are shown in the class diagram on the official college website. The main classes and their characteristics or functions are listed below:

- **Admin**

Attributes: adminId, name, email, password

Methods: login(), logout(), resetPassword(), addAdmin(), editAdmin(), deleteAdmin(), viewAdmin(),

Description: The admin class is responsible for administrator related tasks, providing methods to manage users and all system configuration.

- **Faculty**

Attributes: facultyId, name, email, password, facultyRole

Methods: login(), logout(), addFaculty(), editFaculty(), deleteFaculty(), viewFaculty(), assignRole()

Description: The faculty class is responsible for manage task assigned by the admin.

- **Department**

Attributes: deptId, deptPicture, deptName

Methods: addDept(), editDept(), deleteDept(), viewDept()

Description: The department class represents departments, and provide methods for managing them seamlessly.

- **Program**

Attributes: programId, programName, section

Methods: addProgram(), editProgram(), deleteProgram(), viewProgram()

Description: The program class represents programs offered by institute, and provide methods for managing them effectively.

- **Event**

Attributes: eventId, eventPicture, eventTitle, postedDate, description

Methods: addEvent(), editEvent(), deleteEvent(), viewEvent()

Description: The event class represents the event details organized by institutes, and provides methods for managing them efficiently.

- **Job**

Attributes: jobId, jobPicture, jobTitle, postedDate, description

Methods: addJob(), editJob(), deleteJob(), viewJob()

Description: The job opportunity class represents the jobs listed by admin, and also provide methods for managing them smoothly.

- **Timetable**

Attributes: timetableId, timetablePicture, timetableTitle

Methods: addTimetable(), editTimetable(), deleteTimetable(), viewTimetable()

Description: The timetable class represents the timetable for each program, and provide methods for managing them effectively.

- **Exam Schedule**

Attributes: examScheduleId, examSchedulePicture, examTitle, description

Methods: addExamSchedule(), editExamSchedule(), deleteExamSchedule(), viewExamSchedule()

Description: The exam schedule class represents the schedule of examination, and provides methods for managing them efficiently.

- **Co-Curricular Activities**

Attributes: activityId, activityPicture, activityTitle, postedDate, description

Methods: addActivity(), editActivity(), deleteActivity(), viewActivity()

Description: The co-curricular activities class represents the sports related activities, and also provide methods for managing them efficiently.

- **Alumni**

Attributes: alumniId, alumniPicture, description

Methods: addAlumni(), editAlumni(), deleteAlumni(), viewAlumni()

Description: The alumni class represents the alumni details, and provides methods for managing them seamlessly.

- **Notifications**

Attributes: notificationId, message, postedDate, createdBy

Methods: createNotification(), editNotification(), deleteNotification(), viewNotification()

Description: The notification class represents the notification related to campus life events or other information, and provides methods for managing them effectively.

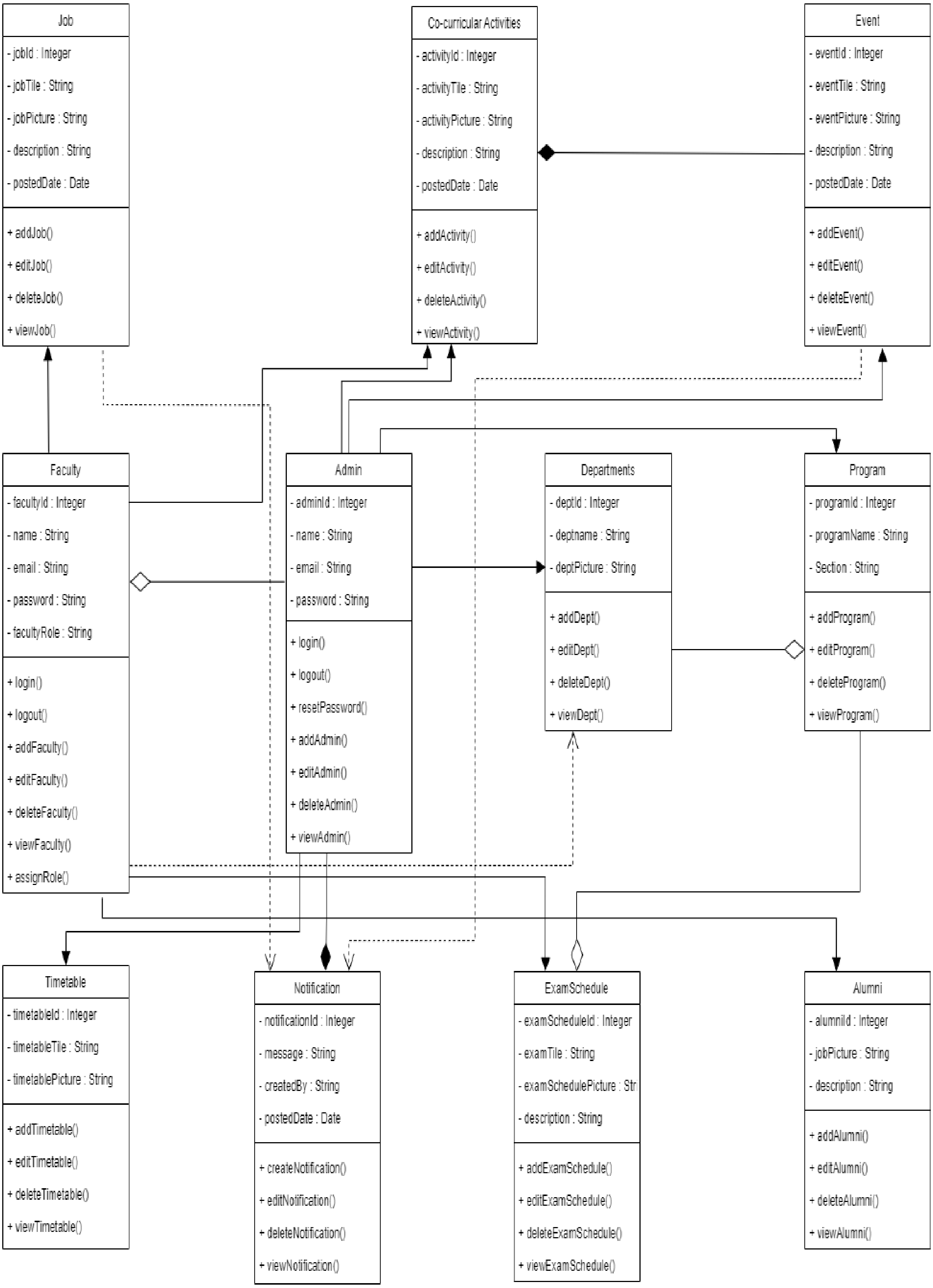


Figure 5: 4-2. Class Diagram for GIGCCL

4.2.2 Object Diagram

The GIGCCL Official Website's object diagram may display a system snapshot during interactions between different modules. It might contain instances of each class such as Admin, Faculty, Department, Program, Event, and Notification together with their present characteristics and connections at a particular moment in time.

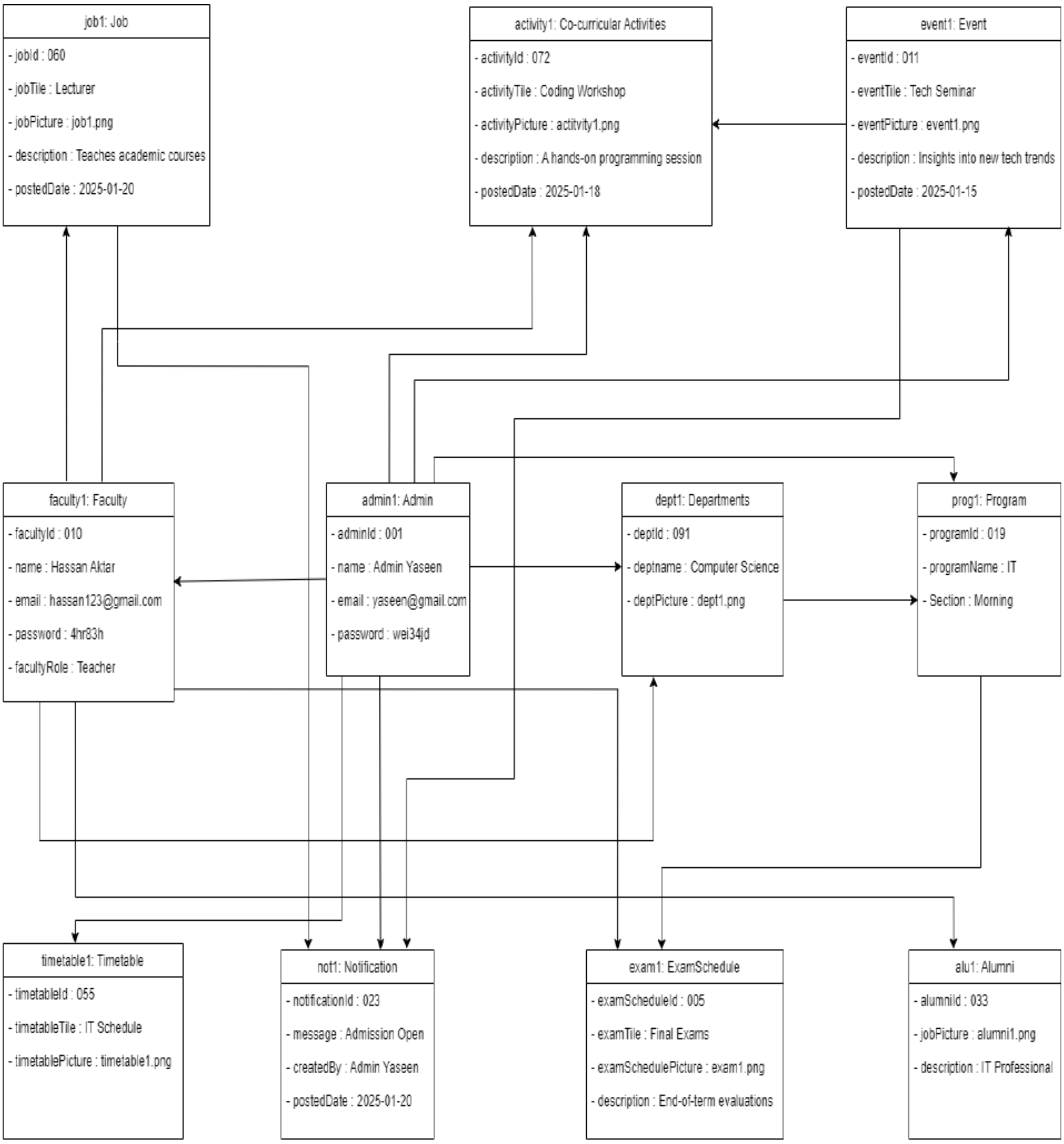


Figure 6: 4-3. Object Diagram for GIGCCL

4.2.3 Sequence Diagram

- **Admin Sequence Diagram**

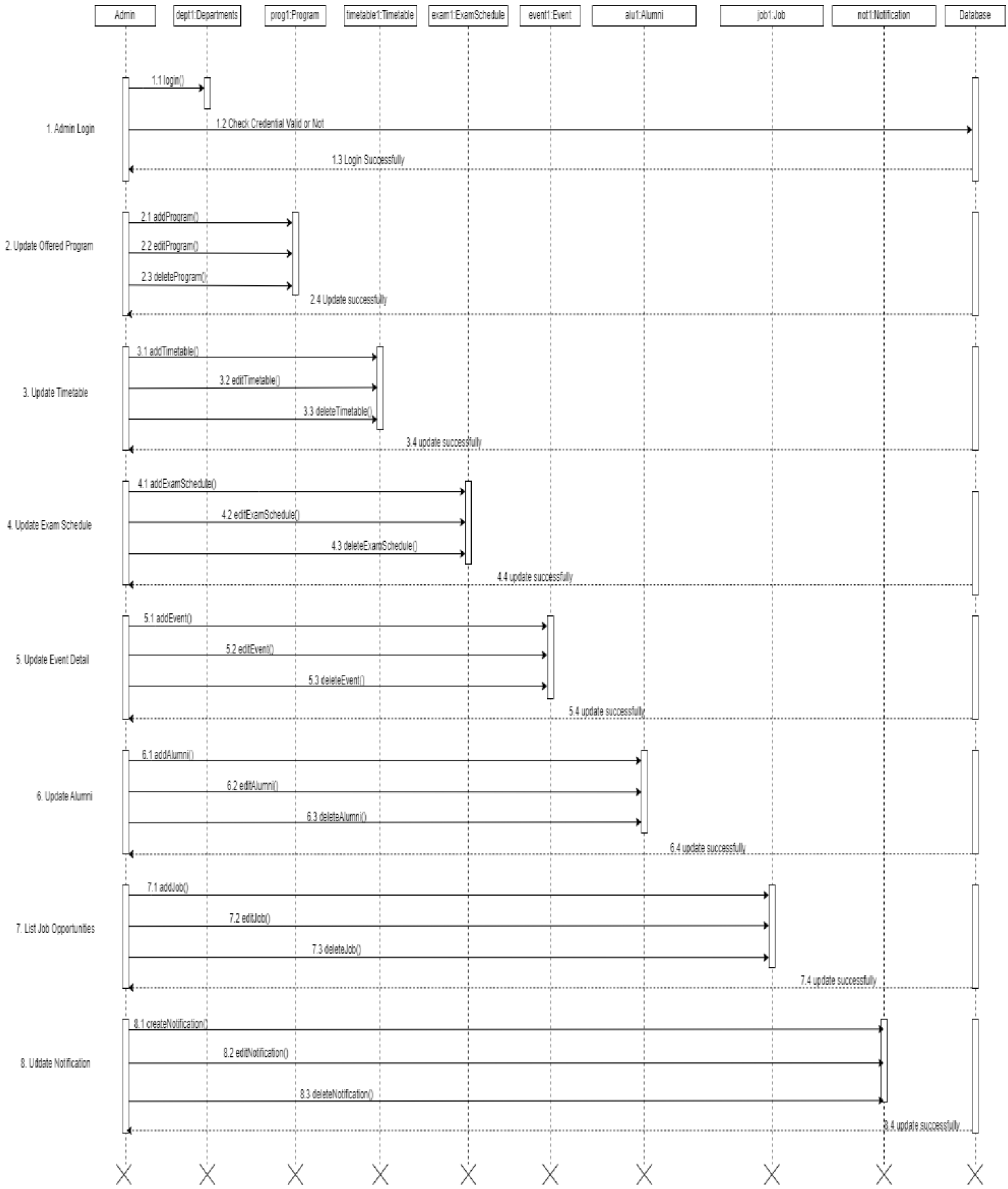


Figure 7: 4-4. Admin Sequence Diagram for GIGCCL

- **Faculty Sequence Diagram**

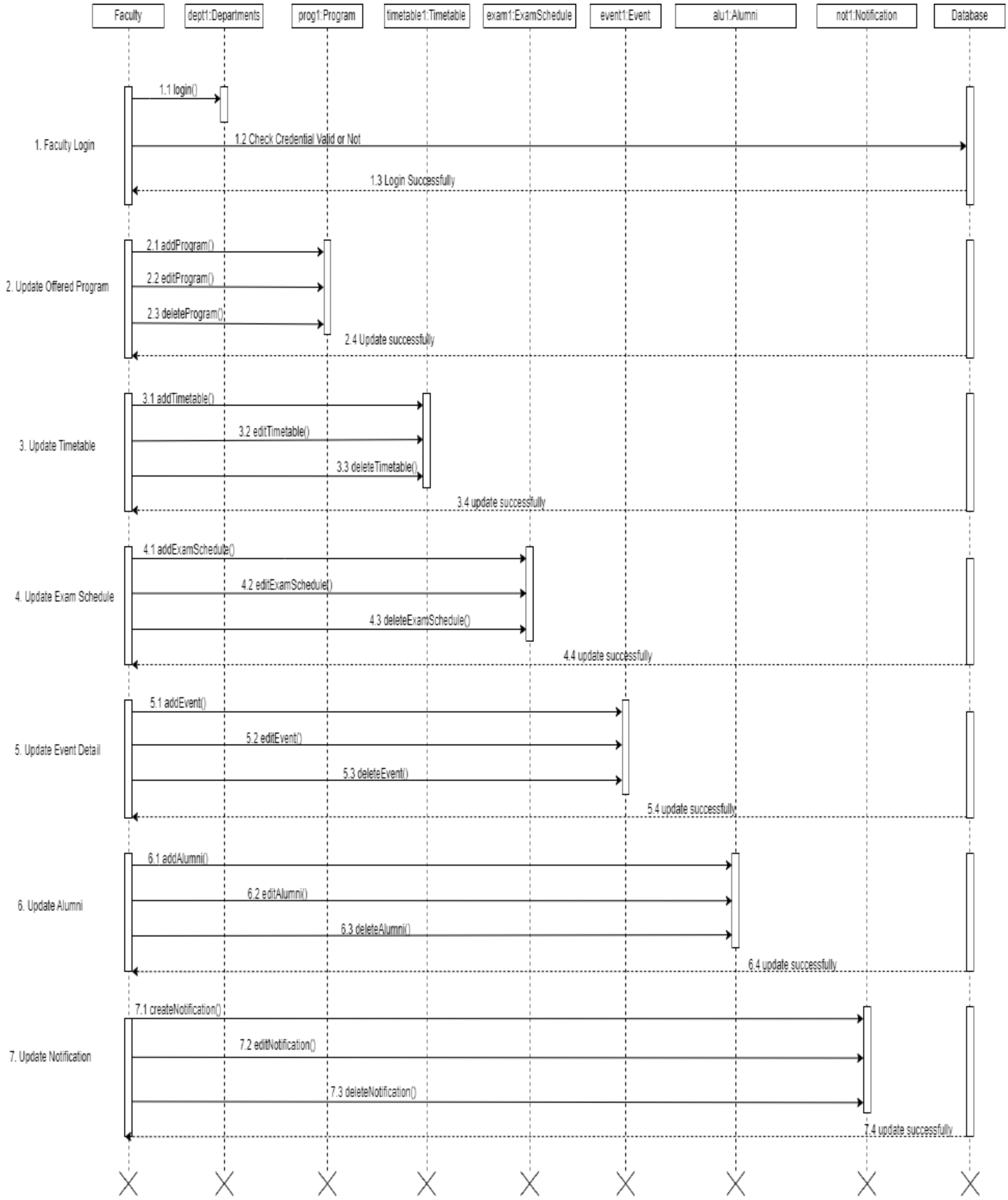


Figure 8: 4-5. Faculty Sequence Diagram for GIGCCL

- **Student Sequence Diagram**

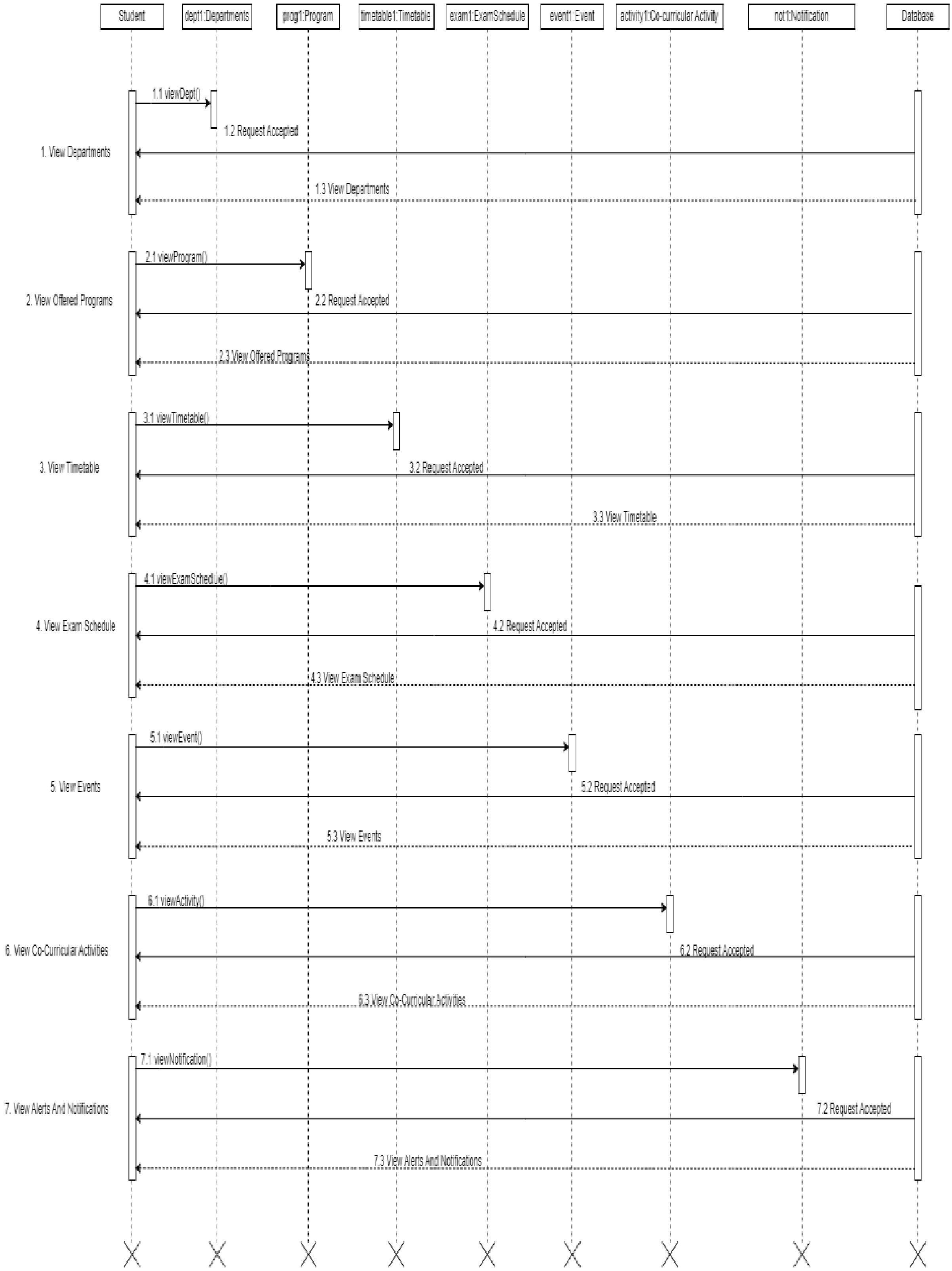


Figure 9: 4-6. Student Sequence Diagram for GIGCCL

- **Alumni Sequence Diagram**

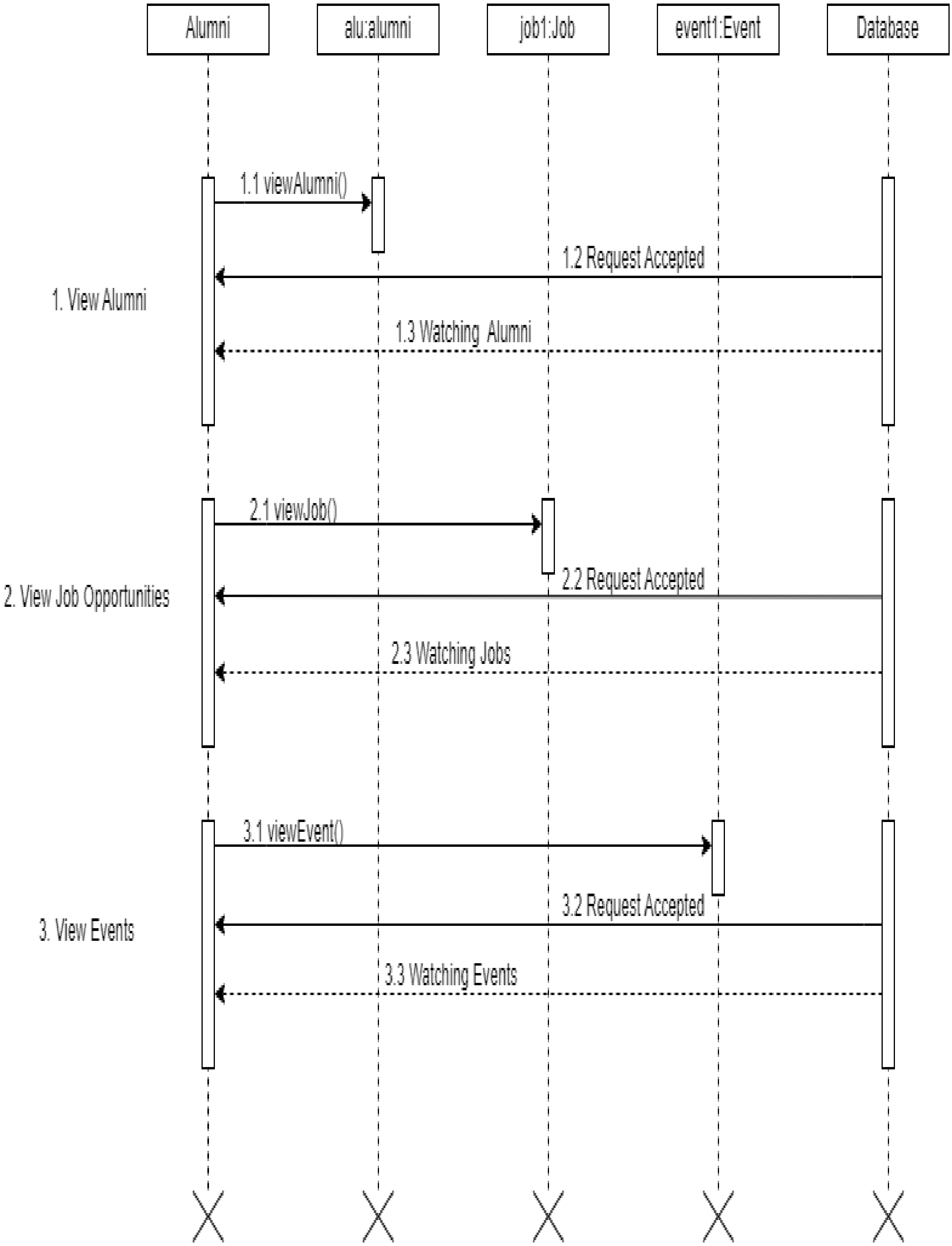


Figure 10: 4-7. Student Alumni Sequence Diagram for GIGCCL

4.2.4 Activity Diagram

- **Admin Activity Diagram**

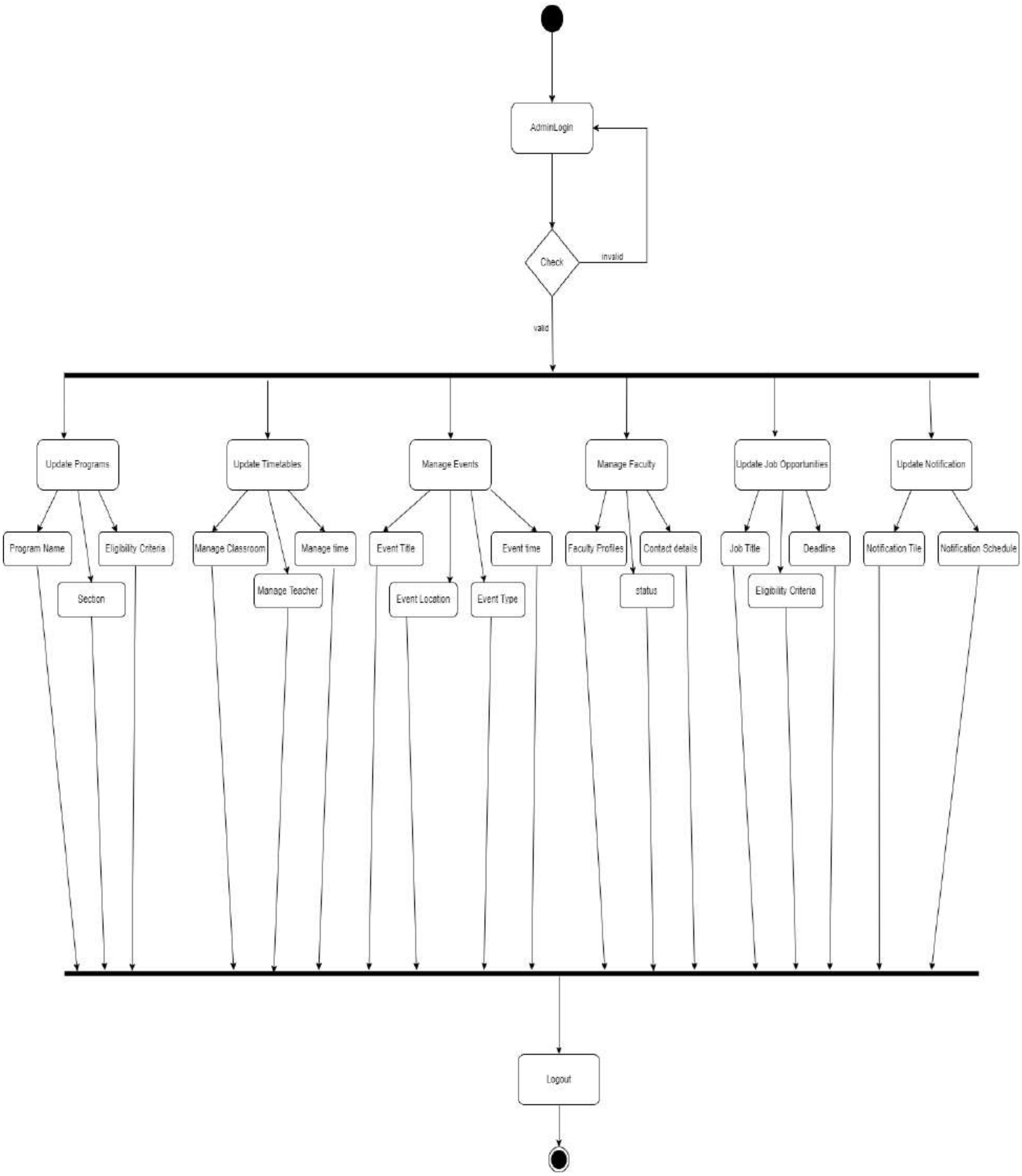


Figure 11: 4-8. Admin Activity Diagram for GIGCCL

- **Faculty Activity Diagram**

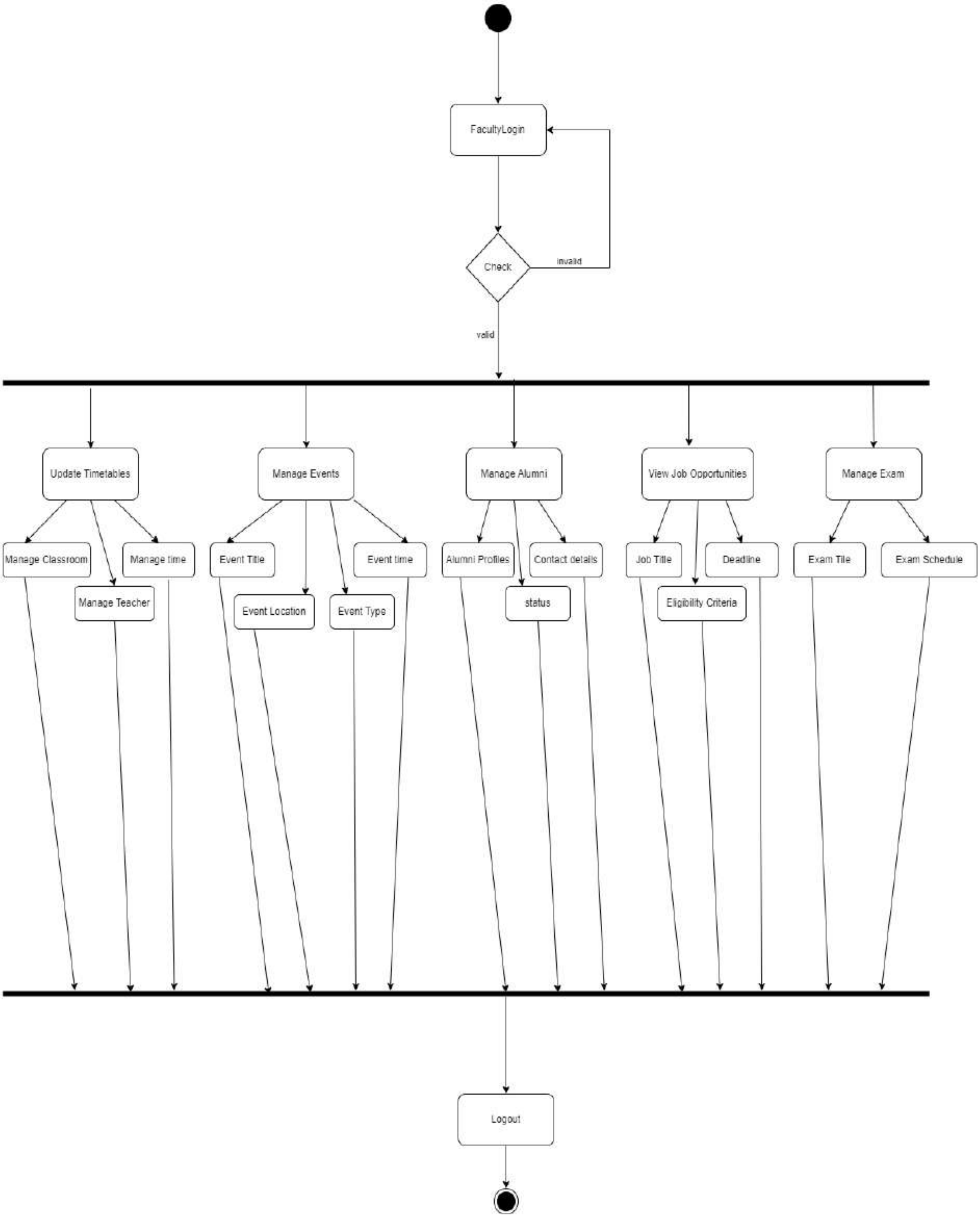


Figure 12: 4-9. Faculty Activity Diagram for GIGCCL

- **Student Activity Diagram**

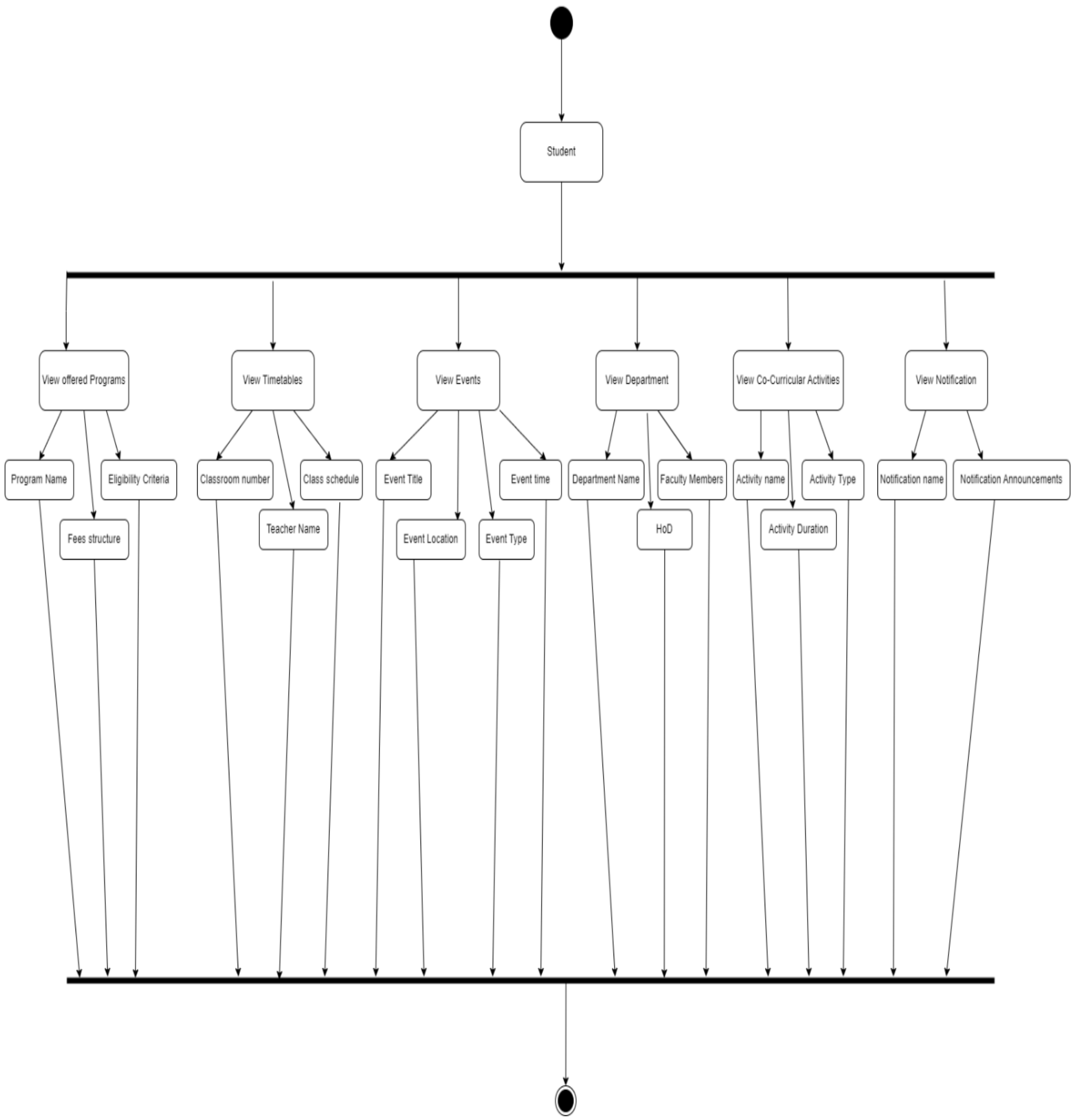


Figure 13: 4-10. Student Activity Diagram for GIGCCL

- **Alumni Activity Diagram**

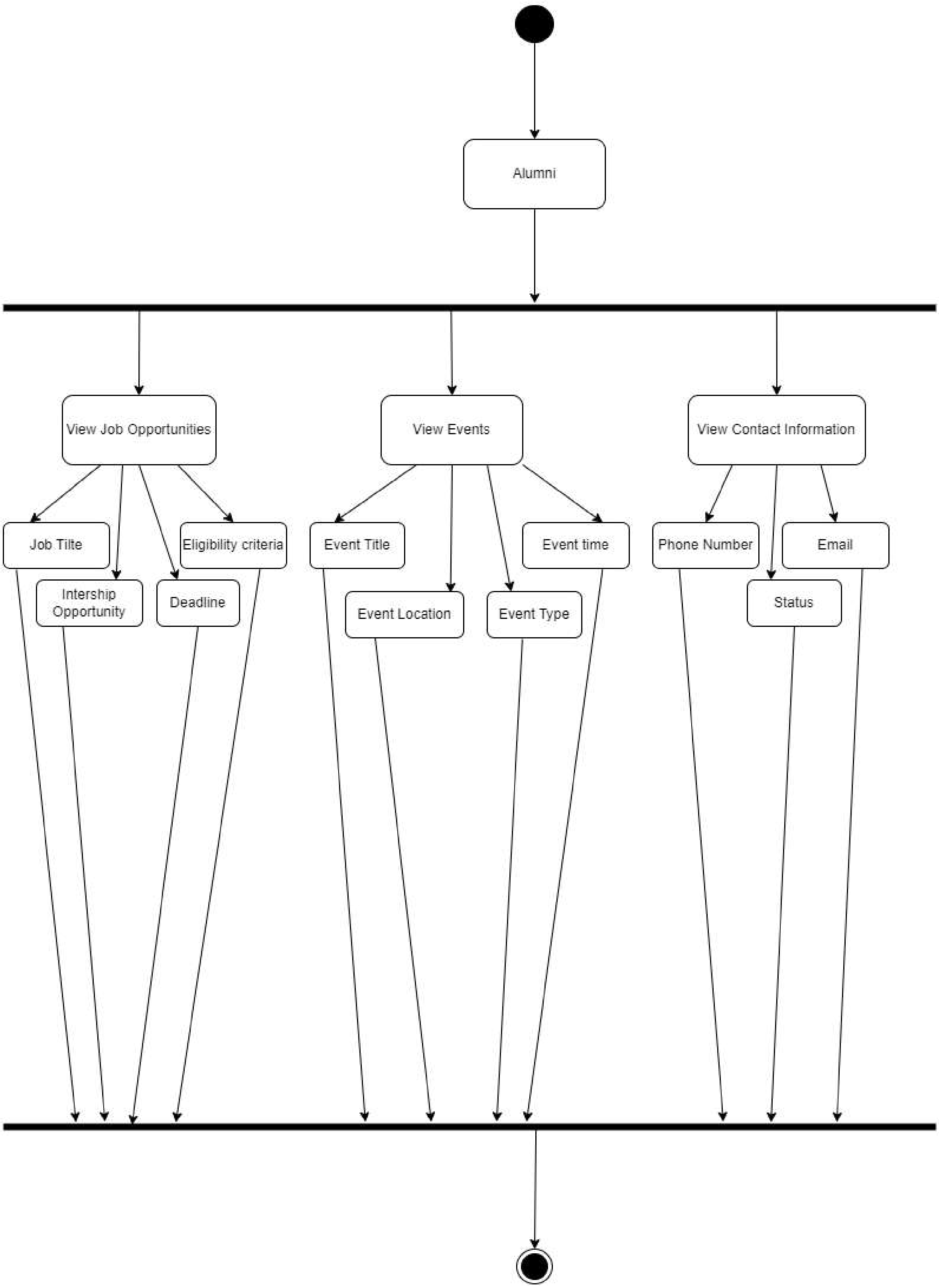


Figure 14: 4-11. Alumni Activity Diagram for GIGCCL

4.2.5 Collaboration Diagram

- **Admin Collaboration Diagram**

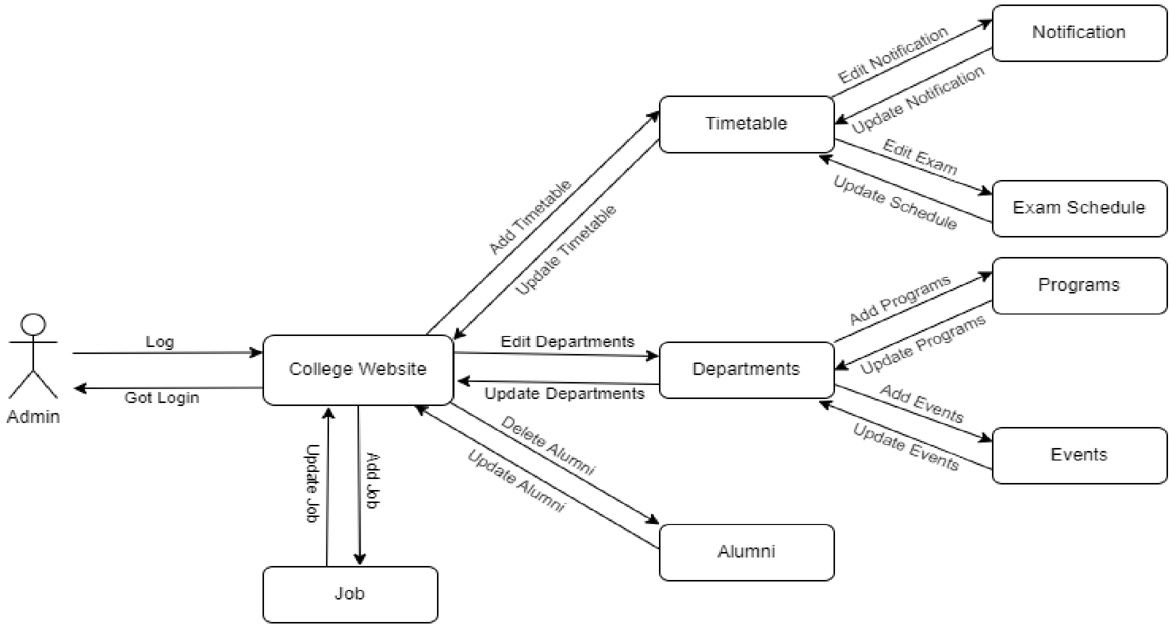


Figure 15: 4-12. Admin Collaboration Diagram for GIGCCL

- **Faculty Collaboration Diagram**

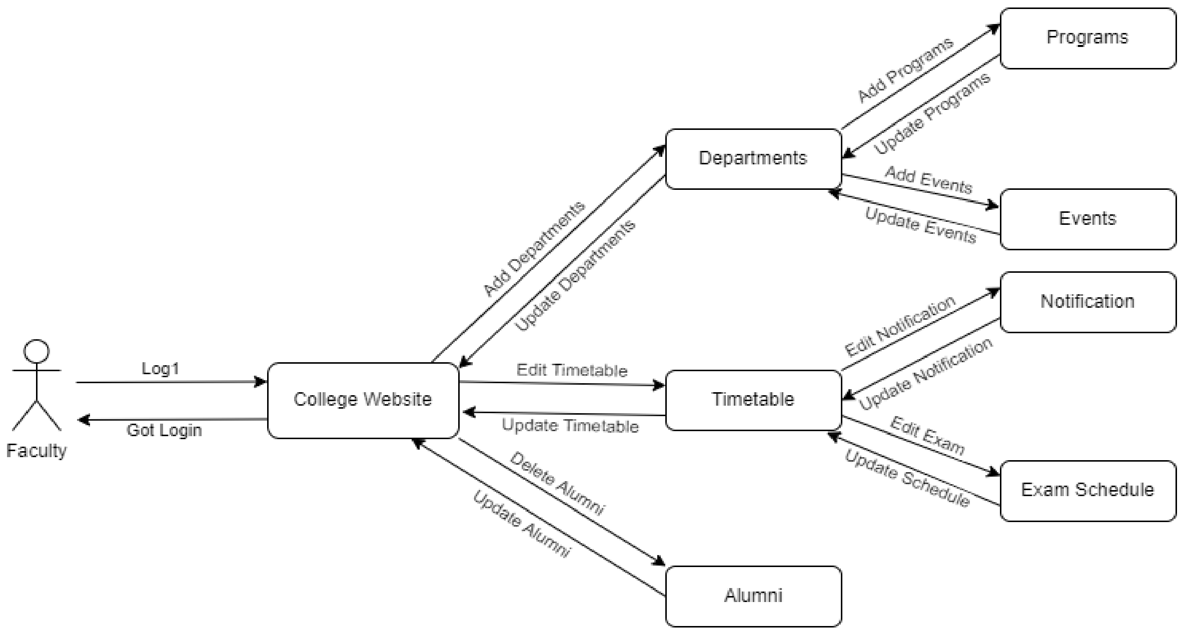


Figure 16: 4-13. Faculty Collaboration Diagram for GIGCCL

- **Student Collaboration Diagram**

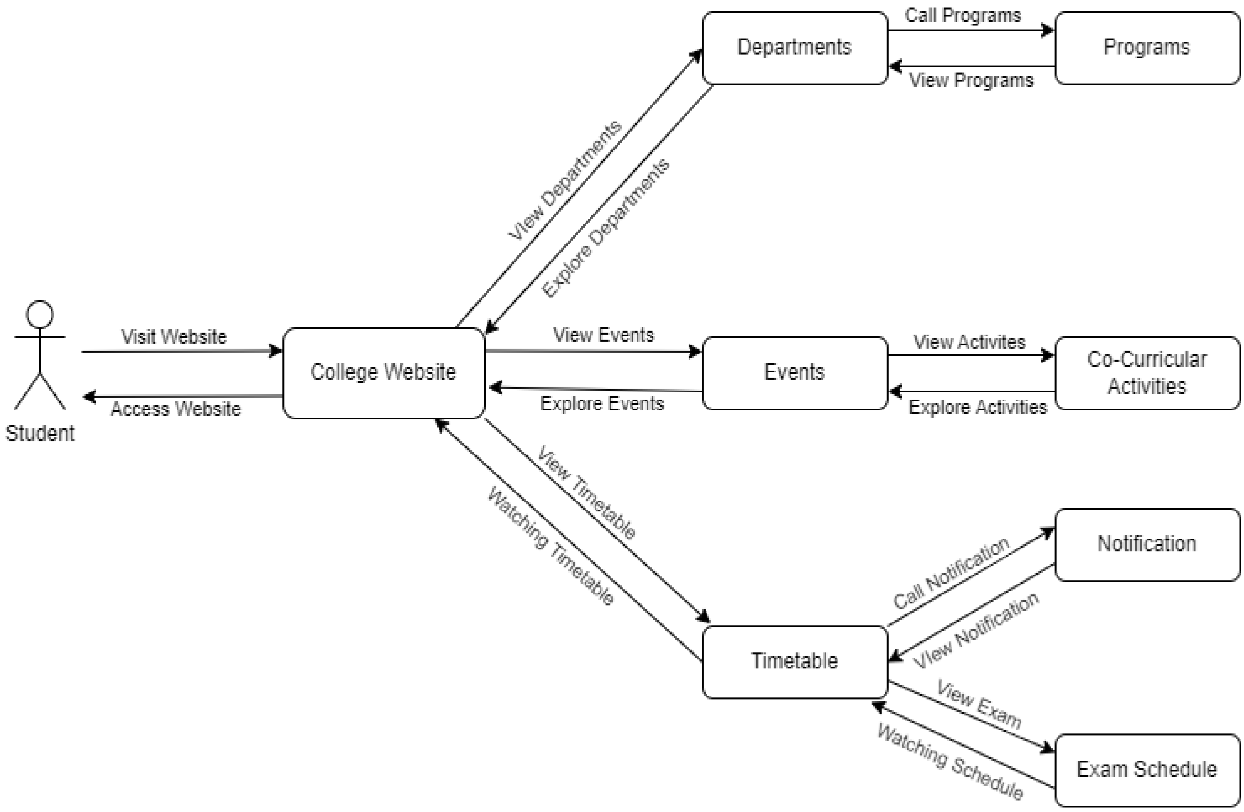


Figure 17: 4-14. Student Collaboration Diagram for GIGCCL

- **Alumni Collaboration Diagram**

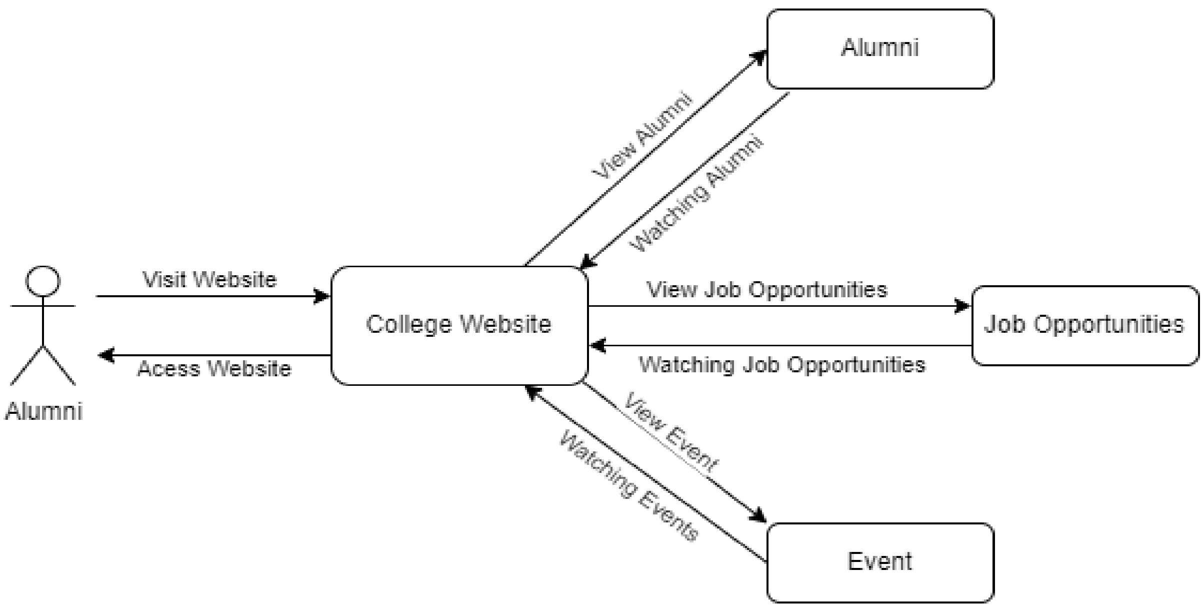


Figure 18: 4-14. Alumni Collaboration Diagram for GIGCCL

4.2.6 State Transition Diagram

- **Admin State Transition Diagram**

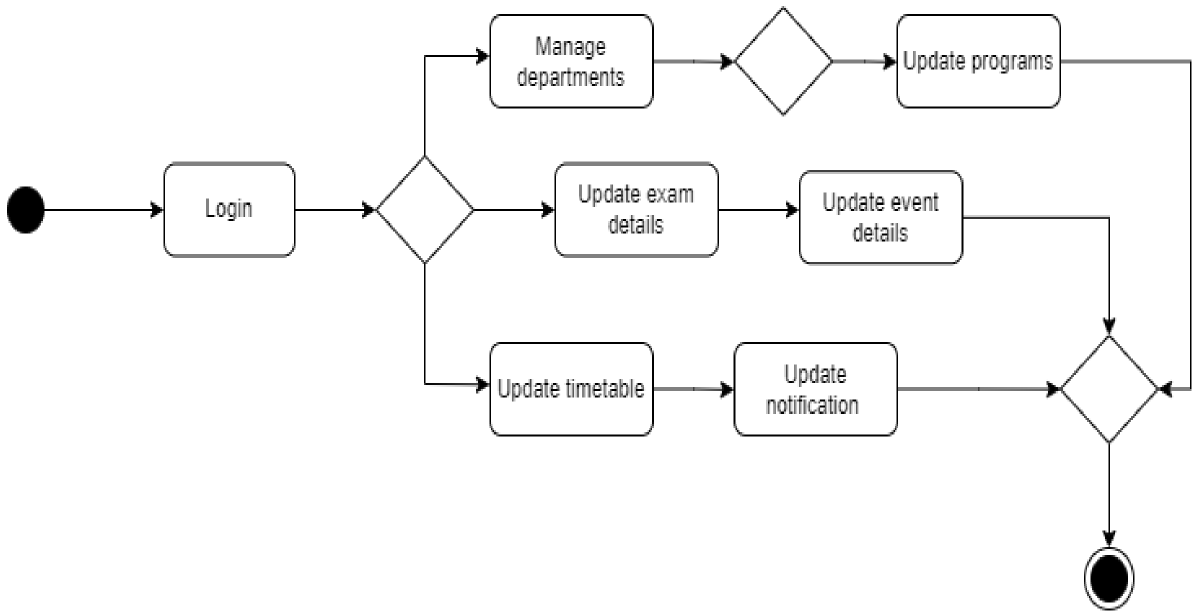


Figure 19: 4-15. Admin State Transition Diagram for GIGCCL

- **Faculty State Transition Diagram**

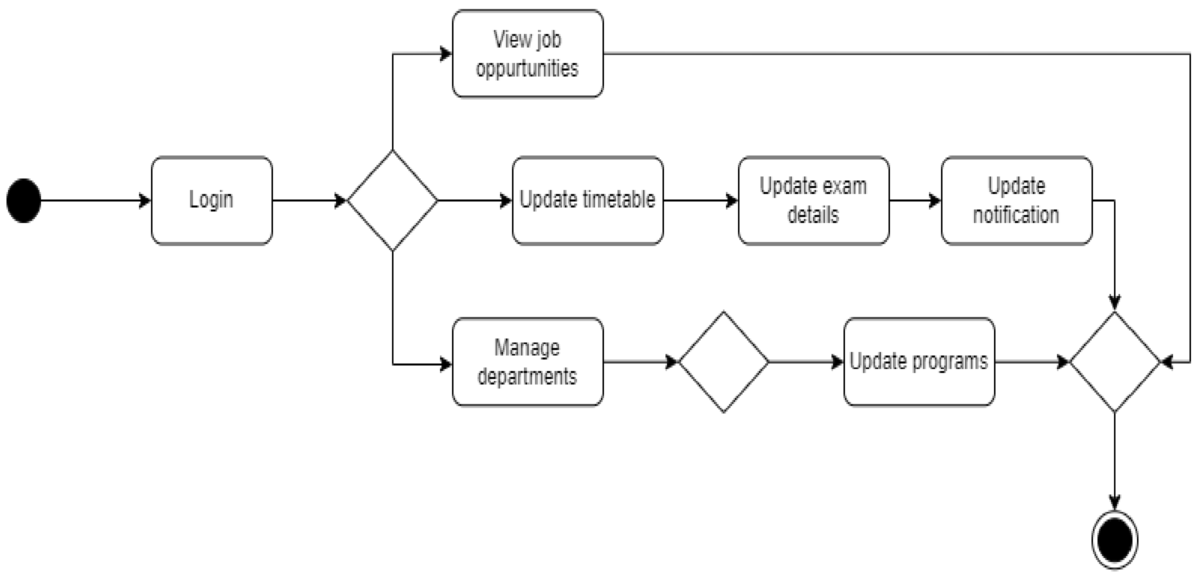


Figure 20: 4-16. Faculty State Transition Diagram for GIGCCL

- **Student State Transition Diagram**

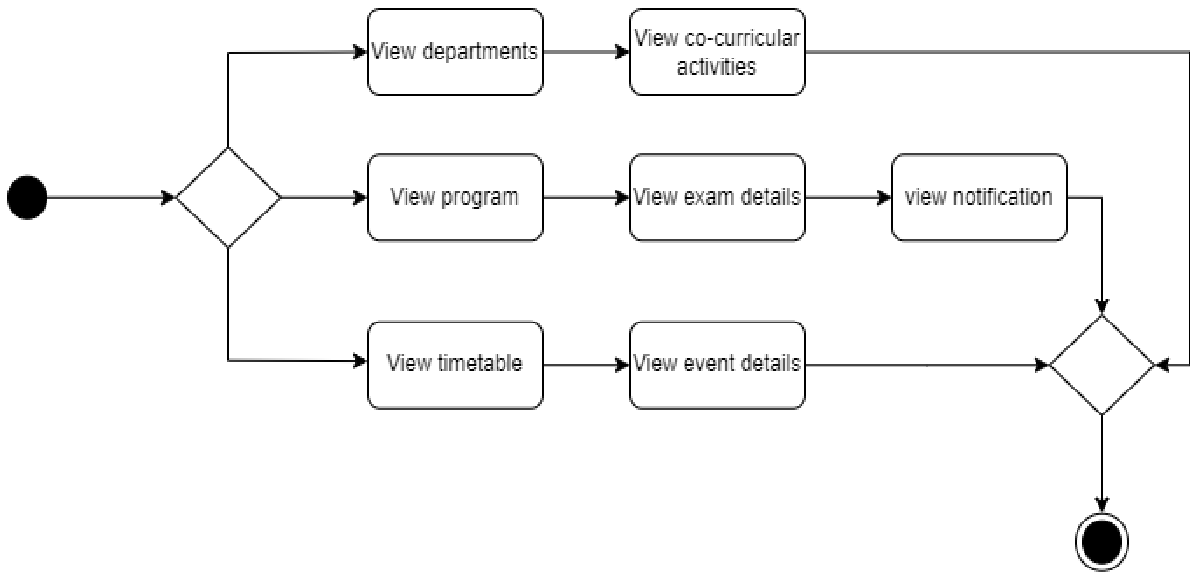


Figure 21: 4-17. Student State Transition Diagram for GIGCCL

- **Alumni State Transition Diagram**

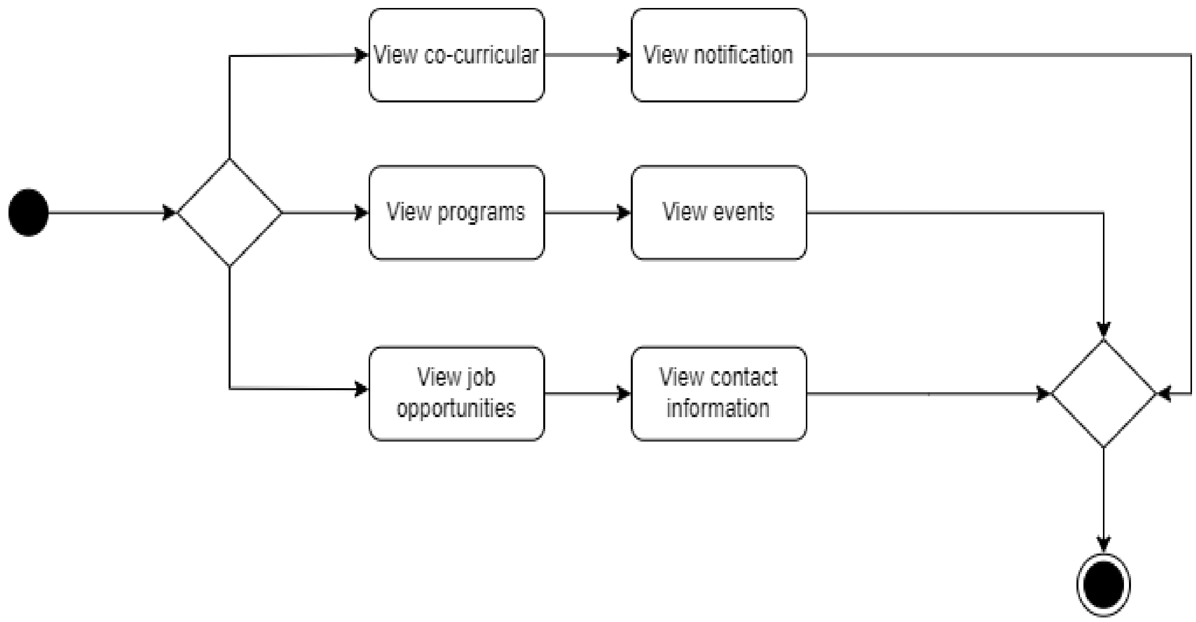


Figure 22: 4-18. Alumni State Transition Diagram for GIGCCL

4.3 Database Design

A key component of building the **GIGCCL Official Website** is creating the organizational structure needed to effectively handle college information, academic programs, departments, events, and resources. A robust and organized database schema designed especially to meet the particular needs of the GIGCCL platform may be found below. This design guarantees smooth administration and maximum information accessibility throughout the system, resulting in a unified and intuitive user experience.

4.3.1 Entities

Admin

- adminId
- name
- email
- password

Faculty

- facultyId
- name
- email
- password
- facultyRole

Department

- deptId
- deptPicture
- deptName

Program

- programId
- programName
- section

Event

- eventId
- eventPicture
- eventTitle
- postedDate
- description

Job

- jobId
- jobPicture
- jobTitle
- postedDate
- description

Timetable

- timetableId
- timetablePicture
- timetableTitle

Exam Schedule

- examScheduleId
- examSchedulePicture
- examTitle
- description

Co-curricular Activities

- activityId
- activityPicture
- activityTitle
- postedDate
- description

Alumni

- alumniId
- alumniPicture
- description

Notification

- notificationId
- message
- postedDate
- createdBy

4.3.2 Relationships

- One-to-Many between Admin and Departments
- One-to-Many between Departments and Programs
- One-to-Many between Programs and Timetables

- One-to-Many between Programs and Exam Schedules
- One-to-Many between Admin and Jobs
- One-to-Many between Admin and Notifications
- One-to-Many between Admin and Events
- One-to-Many between Admin and Co-Curricular Activities
- One-to-Many between Admin and Alumni
- One-to-Many between Faculty and Programs
- One-to-Many between Faculty and Events
- One-to-Many between Faculty and Co-Curricular Activities
- Many-to-Many between Jobs and Alumni
- Many-to-Many between Notifications and Alumni
- Many-to-Many between Notifications and Faculty
- Many-to-Many between Events and Programs
- Many-to-Many between Co-Curricular Activities and Programs

4.3.3 Entity Relationship Diagram

The entities, their characteristics, and the connections between them must be represented in an Entity-Relationship (ER) diagram created for the College Website Portal. The structure of the ER diagram is summarized as follows:

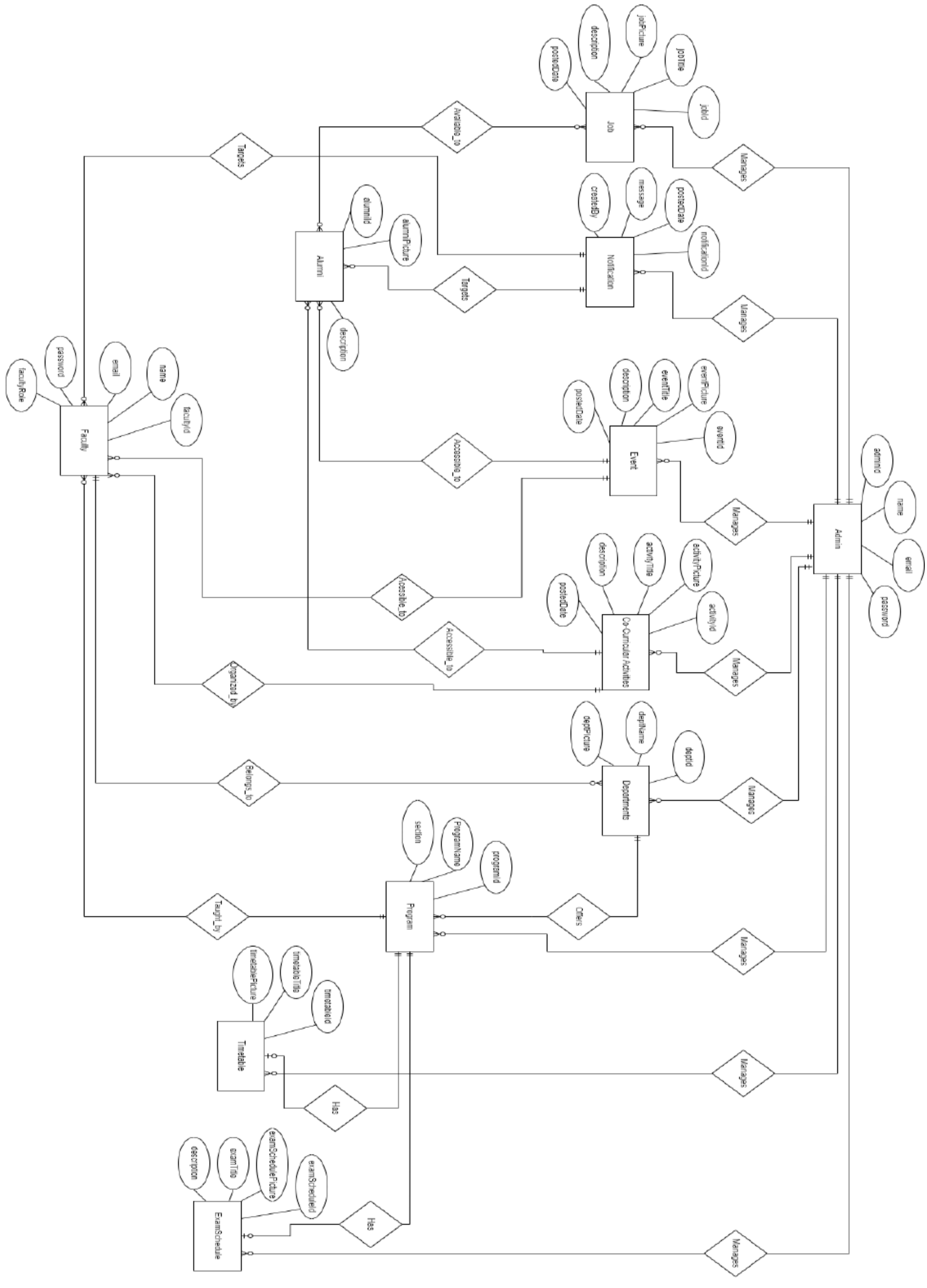


Figure 23: 4-19. ERD Diagram for GIGCCL Website

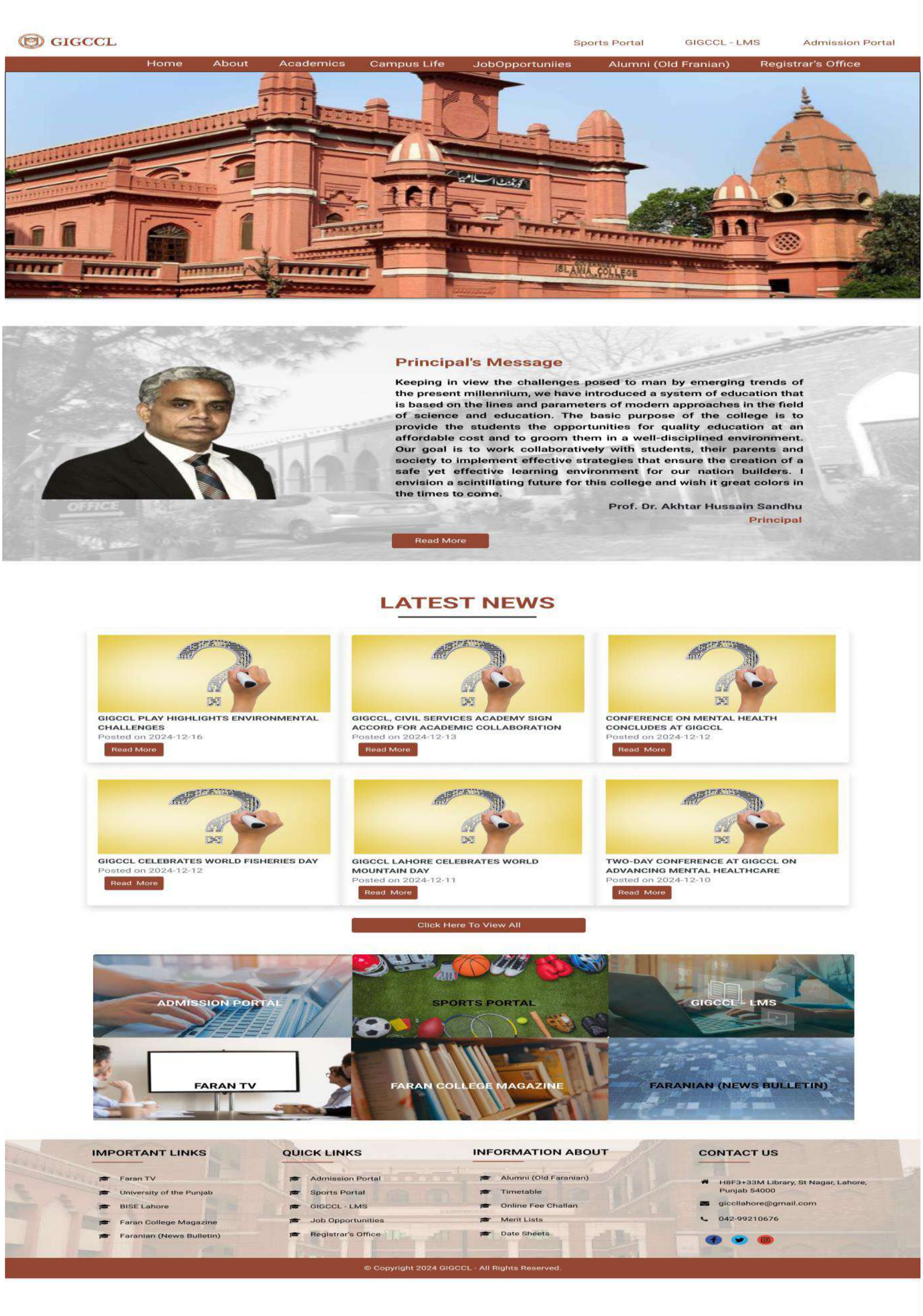
5 Implementation

5.1 User-Interface

The GIGCCL Official Website's User Interface (UI) acts as an intermediary between the user and the technology. Users may easily traverse the platform and obtain important information about academic programs, admissions, events, and career prospects due to the visual and interactive feature. Because a well-designed user interface (UI) makes it simple and intuitive for users to locate what they need, it increases user engagement. The user interface (UI), which serves as the entry point to the GIGCCL community, is designed with functionality and simplicity in mind, giving all users a responsive, innovative, and visually pleasing experience.

5.1.1 Home Page

This is the main page of the official GIGCCL website. It has a modern layout with a navigation bar that makes it simple to obtain information on academics, campus life, job opportunities, and other topics. Portals that improve the user experience include Admissions, Sports, and LMS. The college's goal of offering high-quality, reasonably priced education and encouraging innovation is reflected in the principal's message. Key events are highlighted in the Latest News section along with "Read" buttons for in-depth updates and a link to further explore. The footer ensures easy access to necessary services by providing rapid links to resources such as timetables, contact information, and merit lists.


GIGCCL
Sports Portal
GIGCCL - LMS
Admission Portal
Home
About
Academics
Campus Life
JobOpportuniies
Alumni (Old Franian)
Registrar's Office
Principal's Message
Keeping in view the challenges posed to man by emerging trends of the present millennium, we have introduced a system of education that is based on the lines and parameters of modern approaches in the field of science and education. The basic purpose of the college is to provide the students the opportunities for quality education at an affordable cost and to groom them in a well-disciplined environment. Our goal is to work collaboratively with students, their parents and society to implement effective strategies that ensure the creation of a safe yet effective learning environment for our nation builders. I envision a scintillating future for this college and wish it great colors in the times to come.
Prof. Dr. Akhtar Hussain Sandhu
Principal
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GIGCCL PLAY HIGHLIGHTS ENVIRONMENTAL CHALLENGES
Posted on 2024-12-16
Read More
GIGCCL, CIVIL SERVICES ACADEMY SIGN ACCORD FOR ACADEMIC COLLABORATION
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GIGCCL CELEBRATES WORLD FISHERIES DAY
Posted on 2024-12-12
Read More
GIGCCL LAHORE CELEBRATES WORLD MOUNTAIN DAY
Posted on 2024-12-11
Read More
TWO-DAY CONFERENCE AT GIGCCL ON ADVANCING MENTAL HEALTHCARE
Posted on 2024-12-10
Read More
Click Here To View All
ADMISSION PORTAL
SPORTS PORTAL
GIGCCL - LMS
FARAN TV
FARAN COLLEGE MAGAZINE
FARANIAN (NEWS BULLETIN)
IMPORTANT LINKS
Faran TV
University of the Punjab
BISE Lahore
Faran College Magazine
Faranian (News Bulletin)
QUICK LINKS
Admission Portal
Sports Portal
GIGCCL - LMS
Job Opportunities
Registrar's Office
INFORMATION ABOUT
Alumni (Old Faranian)
Timetable
Online Fee Challan
Merit Lists
Date Sheets
CONTACT US
H8F3+33M Library, St Nagar, Lahore, Punjab 54000
gicclahore@gmail.com
042-99210676
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Figure 24: 5-1. Home Page for GIGCCL Website

5.1.2 Departments Page

This is the departments page of my website where you discover the several academic departments at GIGCCL, all of which are committed to innovation and high standards of instruction. Our departments, which range from the humanities to the sciences, provide courses that prepare students for success in their chosen professions.

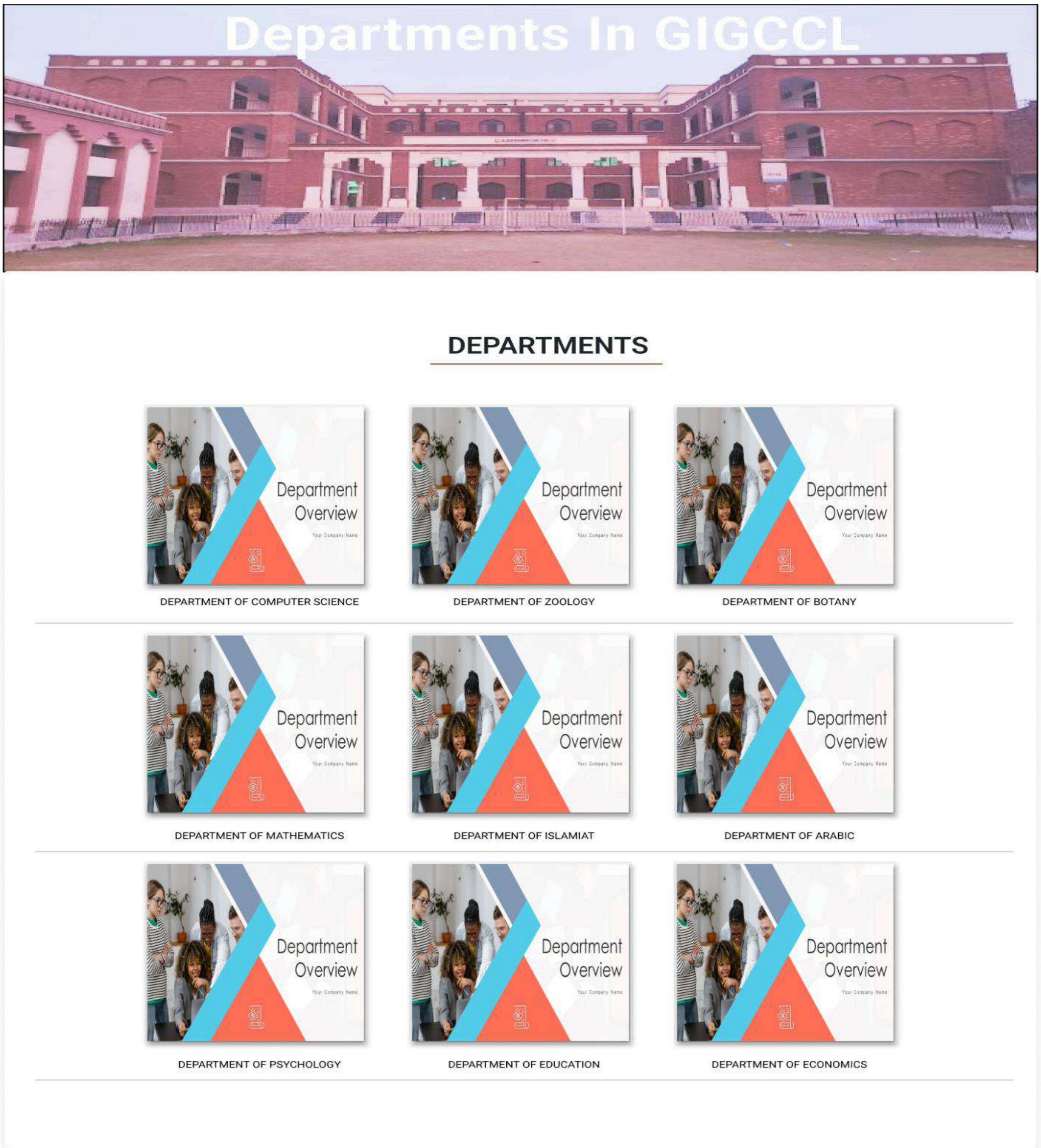

Figure 25: 5-2. Departments Page for GIGCCL Website

5.1.3 Events Page

Explore **GIGCCL** most recent events and activities, where healthy community involvement and academic achievement meet. Our activities, spanning from conferences and workshops to cultural festivities, demonstrate our dedication to promoting innovation and overall growth. To keep up with the exciting life at GIGCCL, check out previous and planned events.

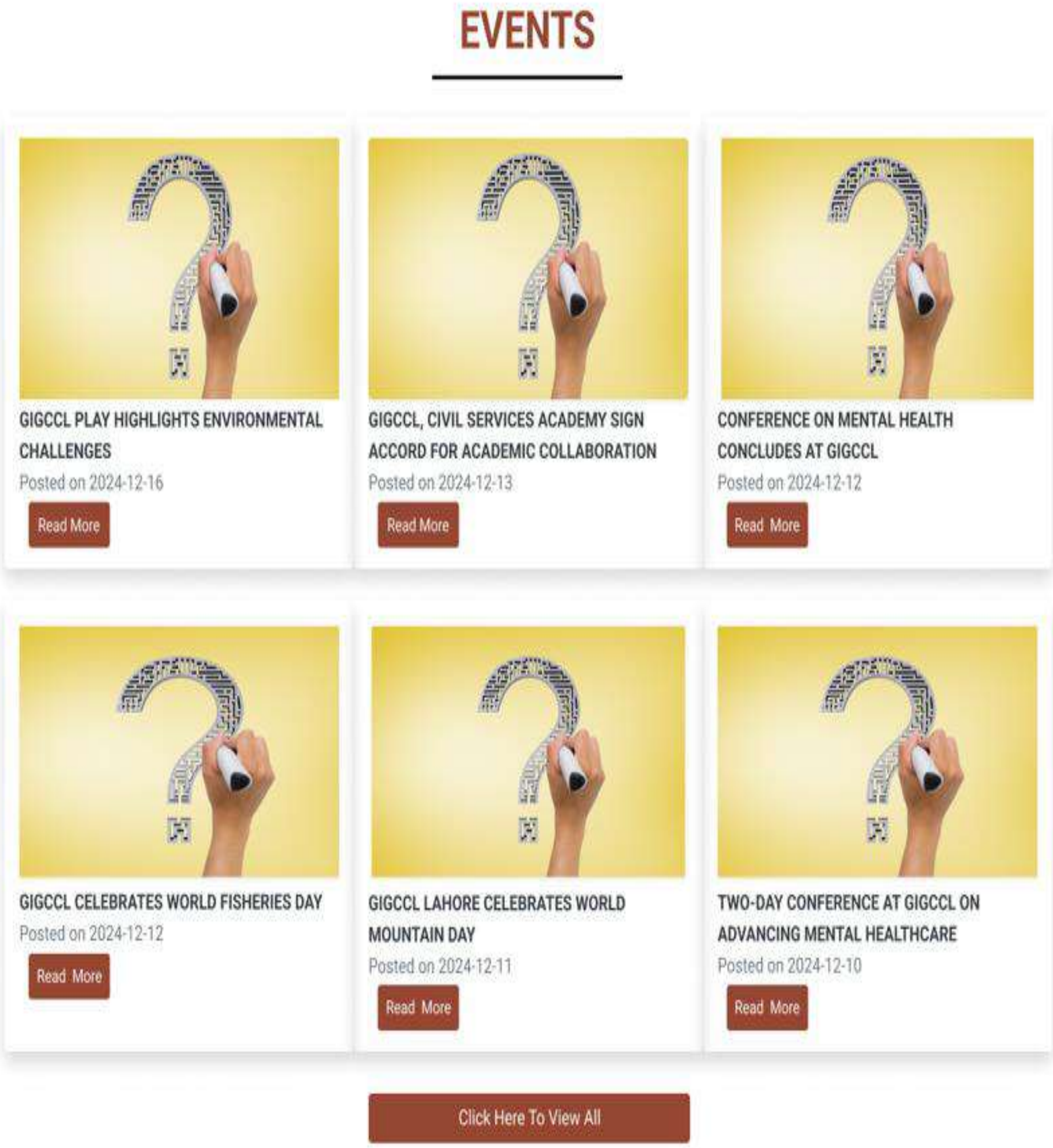

Figure 26: 5-3. Events Page for GIGCCL Website

5.1.4 Alumni (Old Faranian) Page

The GIGCCL Official Website's Alumni Page showcases the successes and efforts of former students, or Faranians, highlighting the institution's rich past. This page provides a forum for highlighting the accomplishments of notable alumni, sharing special moments from alumni gatherings, and fostering a closer relationship between the school and its graduates. Visitors can view photos of alumni events, meetings, and special ceremonies hosted by the Faranian Alumni Fraternity (FAF) in this section. The page fosters a sense of pride and camaraderie among GIGCCL members by honoring the enduring relationship that former students have with the college.

Figure 27: 5-4. Alumni (Old Faranian) Page for GIGCCL Website

5.1.5 Contact Us Page

This is our Contact Us page, which aims to provide you all the information you need to get in touch with GIGCCL. Please contact us by phone or email if you have any questions regarding academic programs, admissions, or general information. The map below makes it simple for you to visit our campus as well.

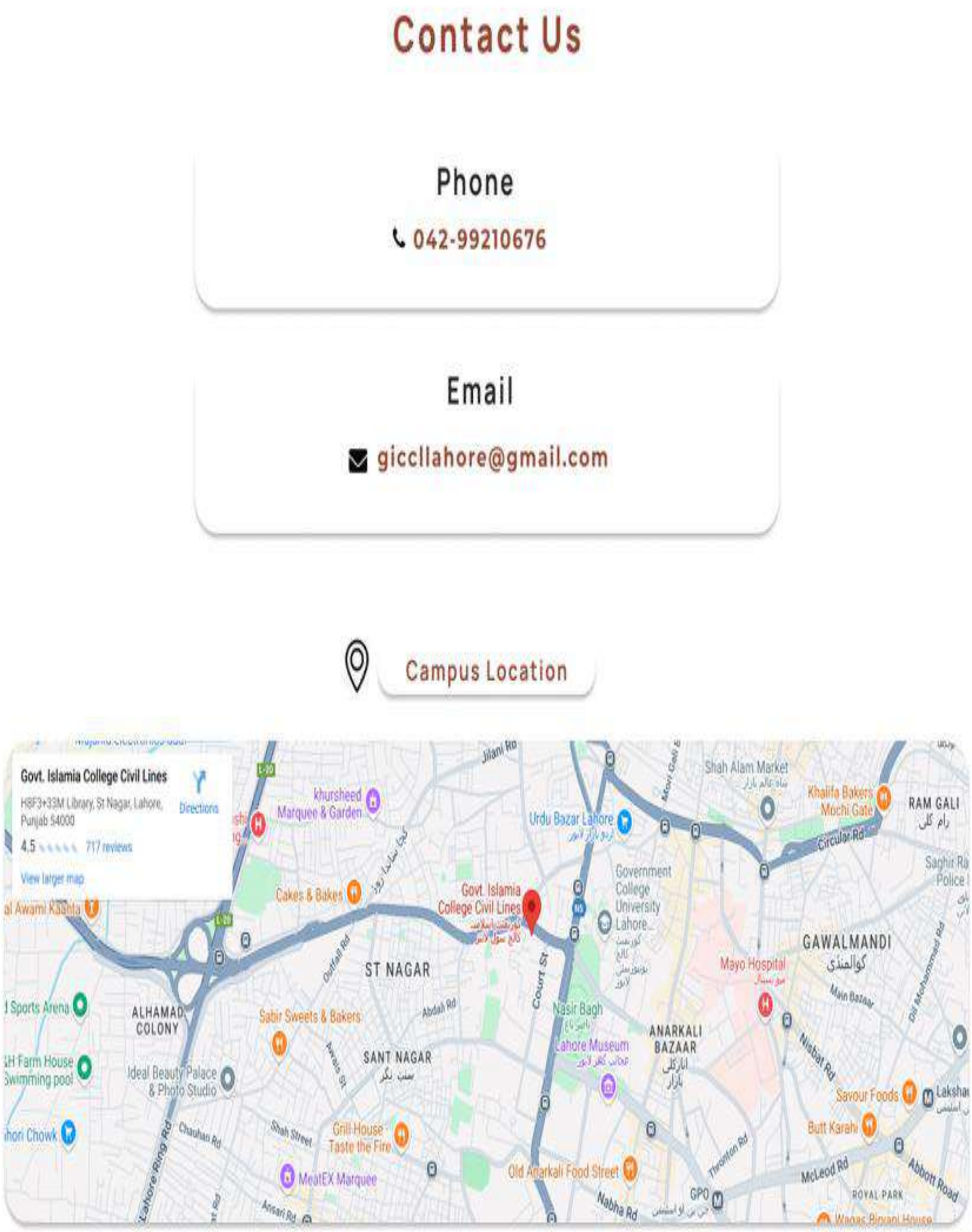

Figure 28: 5-5. Contact Us Page for GIGCCL Website

5.1.6 Admin Login Page

This is the protected login page for the admin panel of the GIGCCL Official Website. By providing their login credentials, only authorized users can access the dashboard. The login page protects administrative functions, enabling administrators to effectively manage content, update website data, and carry out necessary duties.

Figure 29: 5-6. Admin Login Page for GIGCCL Website

5.1.7 Admin Dashboard Page

An effective and safe platform for running the official website is the GIGCCL Admin Dashboard. It gives administrators the ability to manage event material, adjust website themes, and restrict user access. While the content management module makes it simple to update material, the Authentication & Authorization section guarantees proper role management. To improve workflow monitoring, admin activities are tracked via a Recent Actions Panel. The dashboard's user-friendly layout simplifies website management and guarantees a current and orderly online presence.

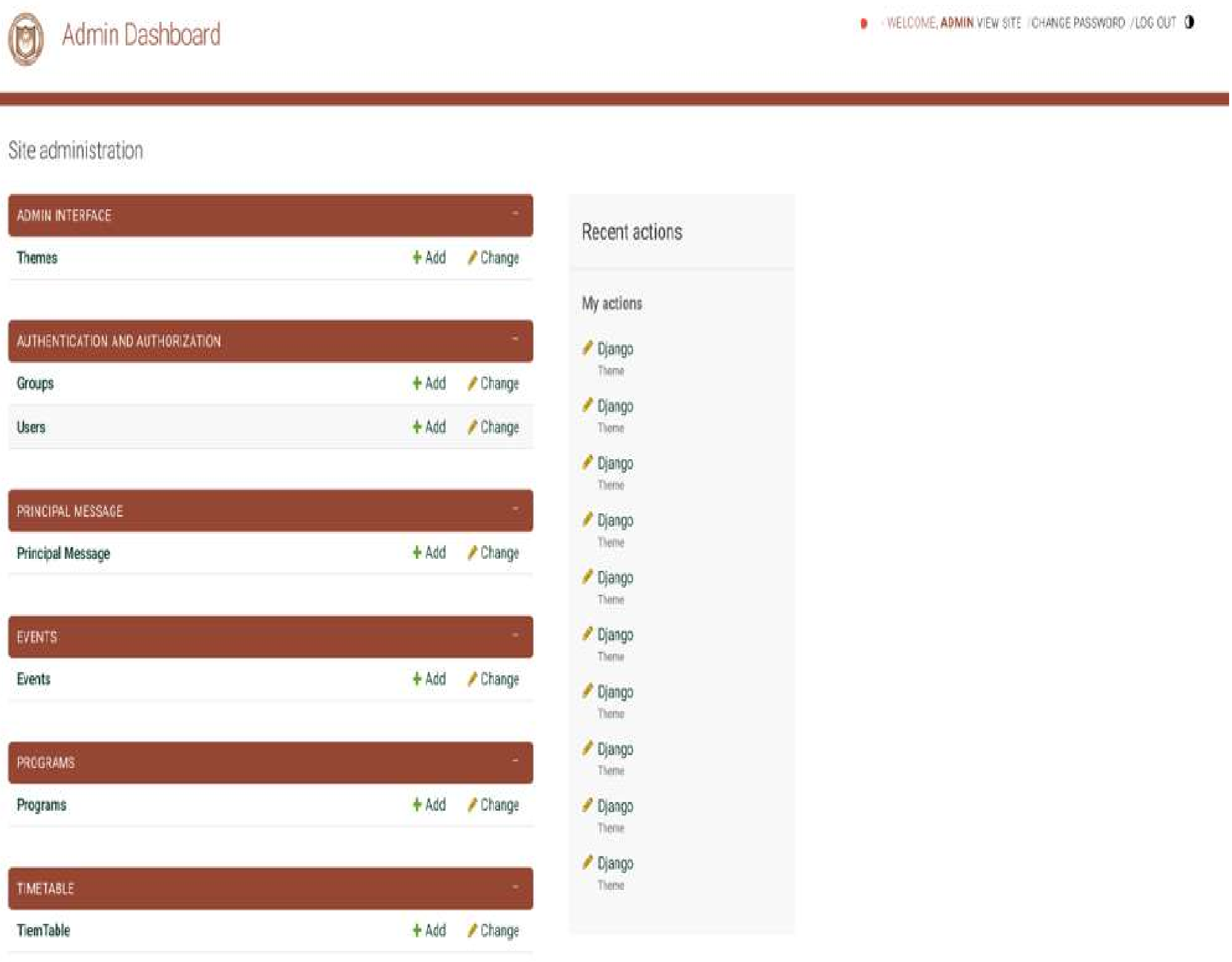

Figure 30: 5-7. Admin Dashboard Page for GIGCCL Website

5.1.8 Change Admin Password Page

Administrators can safely update their login information on the GIGCCL Admin Password Change Page. Users must enter their existing password before creating a new one in order to increase security. Passwords must adhere to certain standards, such as being at least eight characters long and avoiding popular or completely numeric passwords. To make sure correctness and avoid mistakes, the system requires the new password to be typed twice. By limiting access to the admin dashboard to authorized staff, this page is essential to preserving account security.

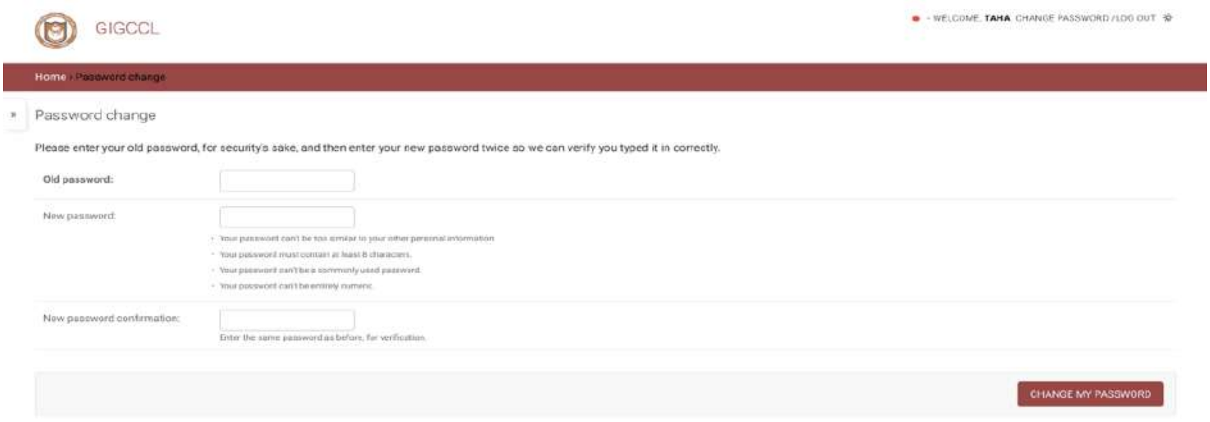

Figure 31: 5-8. Change Admin Password Page for GIGCCL Website

5.1.9 Add User Page

This is the admin add user page where admin can manage user accounts quickly using the GIGCCL Admin Panel. Create new users, specify strong passwords, and enable authentication via password or external means like SSO and LDAP. With integrated password policies and simplified user administration, you can guarantee platform security.

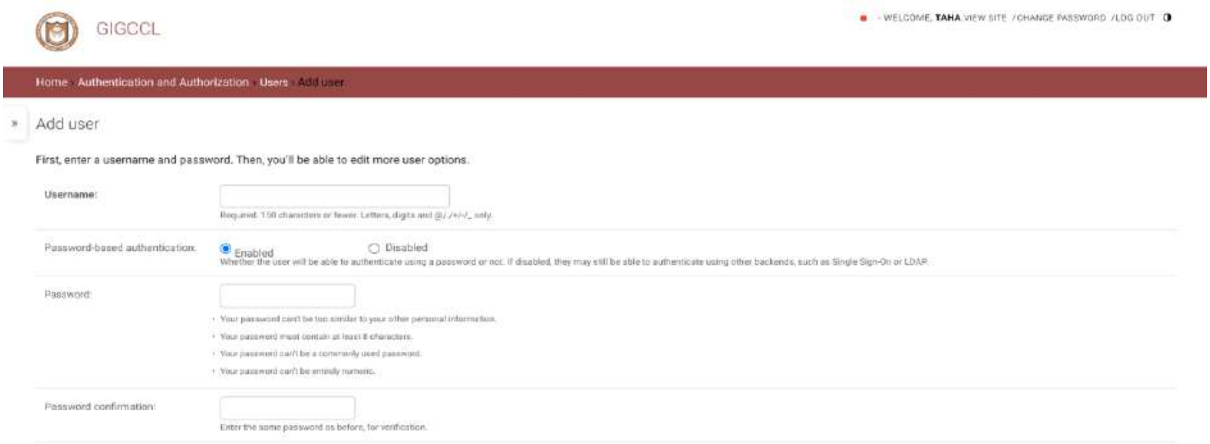

Figure 32: 5-9. Add User Page for GIGCCL Website

5.1.10 Change User Password Page

This is the admin change user password page where admins can safely alter user credentials on the Change Password page. Administrators can create a new password while adhering to security regulations and enable or stop password-based authentication. This improves platform security for all users and guarantees strict access control.

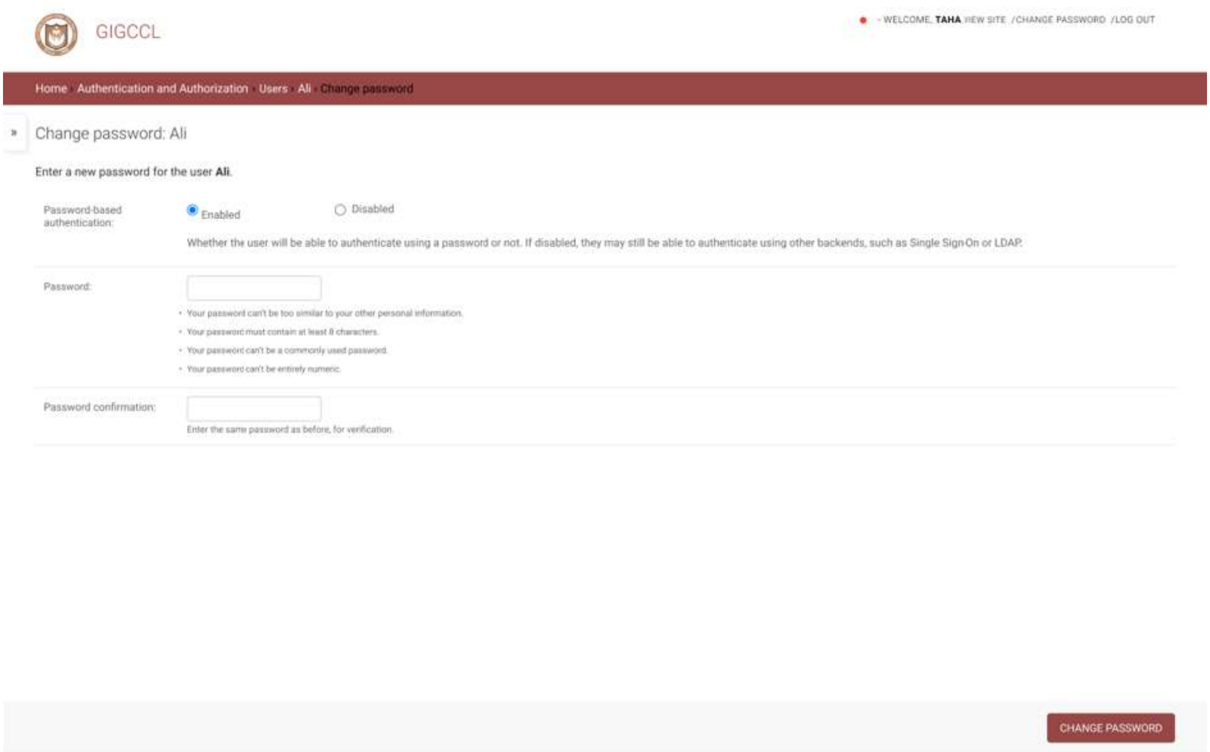

Figure 33: 5-10. Change User Password Page for GIGCCL Website

5.1.11 Assign Faculty Role Page

This is the admin panel page where administrators can give roles and permissions, change user information, and reset passwords via the User Management page. Admins have the ability to manage group memberships, provide staff or superuser privileges, and activate or delete accounts. Precise access control is guaranteed by this technology, enabling effective user privilege management for a safe and well-structured platform.

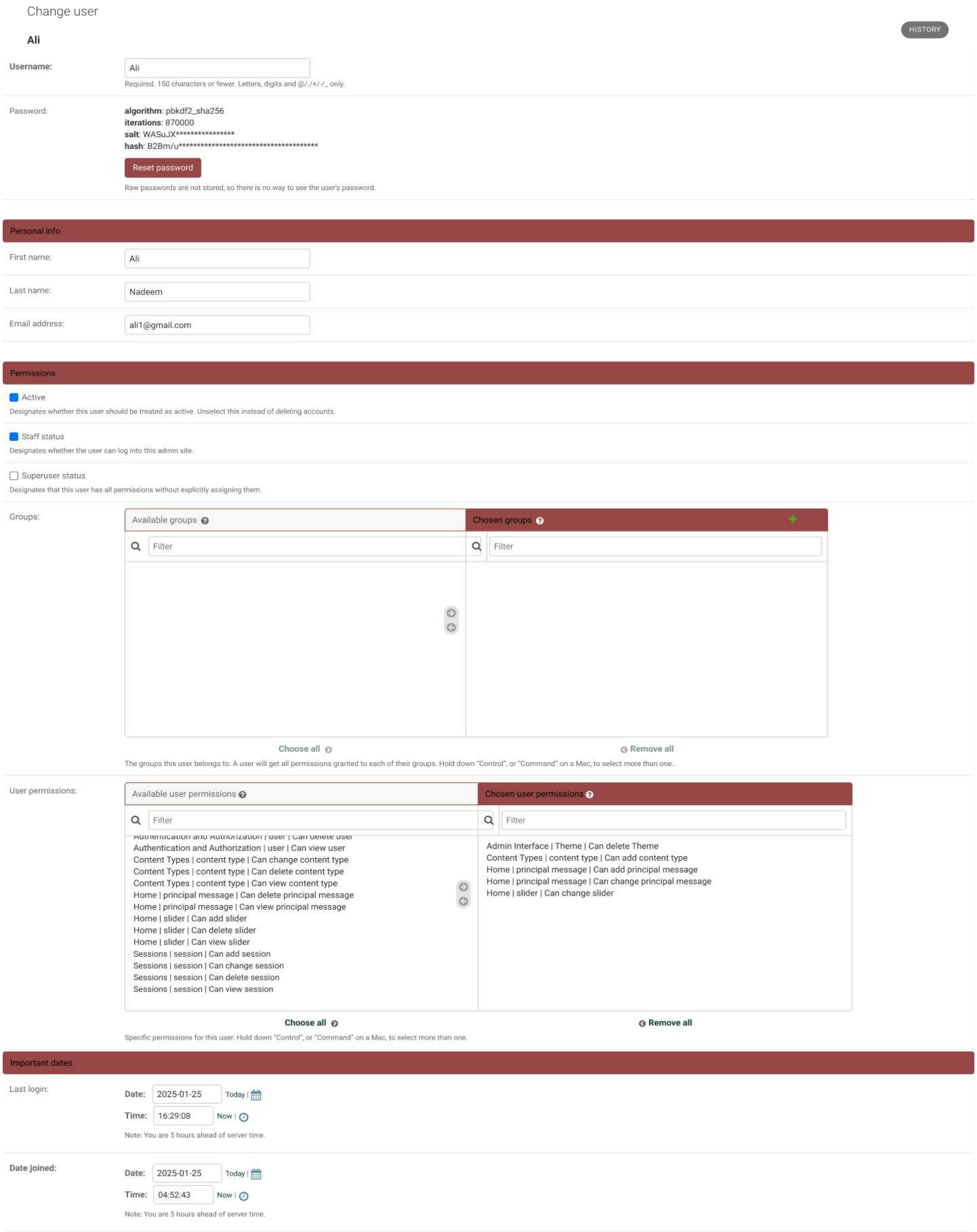

Figure 34: 5-11. Assign Faculty Role Page for GIGCCL Website

6 Testing and Verification

The GIGCCL Official Website development process relies heavily on software testing to make sure the system's efficiency, stability, and functionality meet the highest standards. This crucial stage includes proactively detecting and mitigating potential problems while carefully confirming that the installed functionalities match the specified criteria. To guarantee smooth functioning and compliance to the established criteria, we thoroughly assess different system components using a combination of automated and manual testing approaches. In this project, software testing's main objective is to find and fix any inconsistencies, flaws, or unfulfilled expectations before to deployment, guaranteeing a faultless user experience for every user. Functional, performance, and usability testing belong to the many testing approaches used to maintain the platform's reliability and excellence.

6.1 Unit Testing

Every module of the GIGCCL Official Website project is carefully examined throughout the unit testing process to confirm its stability and functioning. Academic program development, updating, and deletion are handled by the Program Management Module with ease, guaranteeing accurate display throughout the website. The Event Management Module provides users with the most recent information about forthcoming events by effectively storing, updating, and displaying event details. Admins may easily create, edit, and remove job postings according to the Job Posting Module's perfect job posting management. Users are certain to be immediately notified of significant announcements due to the Notification System's accurate and timely updates. The uploaded schedules are shown accurately and are easily editable or retracted as needed thanks to the Timetable and Exam Schedule Modules. All interactive components, including the Programs, Admissions, and Co-Curricular Activities navigation buttons, are verified to function as intended by front-end testing, which effortlessly guides people to the relevant sites. Admin can locate relevant data fast thanks to the search function's accurate results. By ensuring that every module functions flawlessly when tested separately, this intensive procedure improves the user experience overall and offers a strong and reliable foundation for further integration testing.

6.2 Test Cases

6.2.1 Login with valid credentials

Project Name	GIGCCL Official Website
Module Name	Login
Test Case ID	ID-001
Test Case Description	Login with valid credentials
Test Steps	• Go to the login page. • Enter a valid username and password. • Click the "Login" button.
Test Data	Username: admin Password: 123
Pre-Condition	User should have valid login credentials.
Post-Condition	User is redirected to the admin or faculty dashboard.
Expected Result	User successfully logs in and accesses the dashboard.
Actual Result	Successfully Login and access the dashboard.
Status	Pass

Table 21: 6-1. Login with valid credential

6.2.2 Login with invalid credentials

Project Name	GIGCCL Official Website
Module Name	Login
Test Case ID	ID-002
Test Case Description	Login with invalid credentials
Test Steps	• Go to the login page. • Enter an invalid username and password. • Click the "Login" button.
Test Data	Username: invalid_username

	Password: invalid_password
Pre-Condition	None
Post-Condition	Error message appears, and user remains on the login page.
Expected Result	Invalid username or password" error message is displayed.
Actual Result	Error message displayed
Status	Pass

Table 22: 6-2. Login with invalid credential

6.2.3 Login with blank fields

Project Name	GIGCCL Official Website
Module Name	Login
Test Case ID	ID-003
Test Case Description	Login with blank fields
Test Steps	• Login with blank fields • Leave both fields blank. • Click the "Login" button.
Test Data	None
Pre-Condition	None
Post-Condition	Error message appears, prompting the user to fill in the required fields.
Expected Result	Please fill out this field" error message is displayed.
Actual Result	Please fill out this field
Status	Pass

Table 23: 6-3. Login with blank fields

6.2.4 Home page load properly

Project Name	GIGCCL Official Website
Module Name	Home Page

Test Case ID	ID-004
Test Case Description	Verify Home page loads properly
Test Steps	• Navigate to the Home page. • Check if all sections (banner, quick links, announcements, etc.) are visible.
Test Data	None
Pre-Condition	None
Post-Condition	All components load without errors.
Expected Result	Home page components display correctly.
Actual Result	Home page displayed successfully
Status	Pass

Table 24: 6-4. Home page load properly

6.2.5 Verify navigation to other modules

Project Name	GIGCCL Official Website
Module Name	Home Page
Test Case ID	ID-005
Test Case Description	Verify navigation to other modules
Test Steps	• Click on the links from the navigation menu.
Test Data	None
Pre-Condition	None
Post-Condition	User is redirected to the correct module pages.
Expected Result	Navigation works as expected, without errors.
Actual Result	Navigation links works properly
Status	Pass

Table 25: 6-5. Verify navigation to other modules

6.2.6 Verify the departments list loads

Project Name	GIGCCL Official Website
Module Name	Departments Page
Test Case ID	ID-006
Test Case Description	Verify the Departments list loads
Test Steps	• Navigate to the Departments page. • Check if all departments and their descriptions are displayed.
Test Data	None
Pre-Condition	Departments should be added in the database.
Post-Condition	All departments are visible along with their respective details.
Expected Result	Departments page displays all information correctly.
Actual Result	Departments are shown successfully
Status	Pass

Table 26: 6-6. Verify the departments list loads

6.2.7 Verify event list display

Project Name	GIGCCL Official Website
Module Name	Events Page
Test Case ID	ID-007
Test Case Description	Verify Events list displays
Test Steps	• Navigate to the Events page. • Check if all event titles, descriptions, and dates are displayed.
Test Data	None
Pre-Condition	Events should be added in the database.
Post-Condition	All events are visible with correct details and images.

Expected Result	Events page displays all information correctly.
Actual Result	Event page load and view successfully
Status	Pass

Table 27: 6-7. Verify events list display

6.2.8 Verify read more functionality

Project Name	GIGCCL Official Website
Module Name	Event page
Test Case ID	ID-008
Test Case Description	Verify "Read More" functionality
Test Steps	• Click on the "Read More" button for an event. • Verify that the event details are displayed.
Test Data	None
Pre-Condition	None
Post-Condition	Event details open on a separate page or a modal.
Expected Result	Event details load without errors.
Actual Result	Event detail shows successfully
Status	Pass

Table 28: 6-8. Verify read more functionality

6.2.9 Verify Programs list displays

Project Name	GIGCCL Official Website
Module Name	Programs Page
Test Case ID	ID-009
Test Case Description	Verify Programs list displays
Test Steps	• Navigate to the Programs page. • Check if all program names and sections (morning/evening) are displayed.

Test Data	None
Pre-Condition	Programs should be added in the database.
Post-Condition	All programs are visible along with their sections.
Expected Result	Programs page displays all information correctly.
Actual Result	Programs displayed successfully
Status	Pass

Table 29: 6-9. Verify programs list display

6.2.10 Verify contact us page loads

Project Name	GIGCCL Official Website
Module Name	Contact Us Page
Test Case ID	ID-010
Test Case Description	Verify Contact Us page loads
Test Steps	• Navigate to the Contact Us page. • Check if the page loads without errors and all elements are visible.
Test Data	None
Pre-Condition	None
Post-Condition	Contact Us page loads correctly with all elements visible.
Expected Result	Contact Us page displays all components properly.
Actual Result	Contact us page shows successfully
Status	Pass

Table 30: 6-10. Verify contact us page loads

6.2.11 Verify notifications display

Project Name	GIGCCL Official Website
Module Name	Notifications
Test Case ID	ID-011
Test Case Description	Verify notifications display.
Test Steps	• Navigate to the homepage. • Check if notifications are displayed properly in the notification section.
Test Data	Sample notification text
Pre-Condition	Notifications must exist in the database.
Post-Condition	Notifications should be visible and match the database.
Expected Result	Notifications are displayed correctly on the homepage.
Actual Result	Notification shown successfully
Status	Pass

Table 31: 6-11. Verify notifications display

6.2.12 Verify display job opportunities

Project Name	GIGCCL Official Website
Module Name	Job Opportunities
Test Case ID	ID-012
Test Case Description	Verify display of job opportunities
Test Steps	• Navigate to the "Job Opportunities" page. • Check if the list of jobs is displayed properly.
Test Data	Sample job postings
Pre-Condition	Jobs must exist in the database.
Post-Condition	Jobs should be visible on the page.

Expected Result	All job opportunities are displayed correctly.
Actual Result	Job page shows successfully
Status	Pass

Table 32: 6-12. Verify display jobs opportunities

6.2.13 Verify display of alumni profiles

Project Name	GIGCCL Official Website
Module Name	Alumni (Old Franians)
Test Case ID	ID-013
Test Case Description	Verify display of alumni profiles
Test Steps	• Navigate to the "Alumni" page. • Check if the list of alumni profiles is displayed correctly.
Test Data	Sample alumni profiles
Pre-Condition	Alumni profiles must exist in the database.
Post-Condition	Alumni profiles are visible on the page.
Expected Result	All alumni profiles are displayed properly.
Actual Result	Alumni profiles shows successfully
Status	Pass

Table 33: 6-13. Verify display of alumni profiles

6.2.14 Verify display of the exam schedule page

Project Name	GIGCCL Official Website
Module Name	Exam Schedule
Test Case ID	ID-014
Test Case Description	Verify display of the exam schedule page
Test Steps	• Navigate to the "Exam Schedule" page.

	• Check if the page loads with a list of exam schedules.
Test Data	Sample exam schedules
Pre-Condition	Exam schedule data must exist in the database.
Post-Condition	Exam schedules are visible on the page.
Expected Result	All exam schedules are displayed properly.
Actual Result	Exam schedule page display successfully.
Status	Pass

Table 34: 6-14. Verify display of exam schedule page

6.2.15 Verify the display of the timetable page

Project Name	GIGCCL Official Website
Module Name	Timetable
Test Case ID	ID-015
Test Case Description	Verify the display of the timetable page
Test Steps	• Navigate to the "Timetable" page. • Check if the page loads with a list of timetables.
Test Data	Sample timetable data
sPre-Condition	Timetable data must exist in the database
Post-Condition	Timetables are visible on the page.
Expected Result	All timetables are displayed properly.
Actual Result	Timetable page display successfully.
Status	Pass

Table 35: 6-15. Verify display of timetable page

6.3 Integration Testing

The **GIGCCL Official Website's** Departments, Programs, Events, Exam Schedule, and Timetable modules are all tested for integration to make sure they all function as a cohesive whole. By validating how modules connect, this testing makes sure that data flows correctly and that adjustments made in one module are reflected appropriately in related areas. A change to an academic program or event, for instance, is checked to make sure it fits within the appropriate category. Updates to the Exam Schedule and Timetable are also synchronized and shown consistently throughout the website. By ensuring that the website works as a unified platform, integration testing guarantees that users receive accurate, consistent, and trustworthy information. Additionally, it proves that error handling works well, providing a seamless and intuitive experience.

References

1) Abhijeet Kaldate. (2024). *Best University and College Website Designs To Use as Inspiration in 2024.*
   Retrieved from:
   https://wpastra.com/resources/college-websites/

2) Wikipedia. (2024). *Website.*
   Retrieved from:
   https://en.wikipedia.org/wiki/Website

3) Ankur. (2024). *How Push Notifications Are Transforming Higher Education.*
   Retrieved from:
   https://feedify.net/blog/how-push-notifications-are-transforming-higher-education/

4) Virtual University of Pakistan. (2024). *Virtual University of Pakistan*. Retrieved from:
   https://www.vu.edu.pk/

5) Massachusetts Institute of Technology. (2024). *Massachusetts Institute of Technology.*
   Retrieved from:
   https://www.mit.edu

6) Stella Power. (2024). *Enhancing the search experience on higher education websites.*
   Retrieved from:
   https://www.annertech.com/blog/enhancing-search-experience-higher-education-websites

7) Rayed Chaudhry. (2024). *How to Improve Web Accessibility in Higher Education.*
   Retrieved from:
   https://www.thinkorion.com/blog/how-to-improve-web-accessibility-in-higher-education

8) Guy Lawrence. (2024). *A Complete Guide to Cyber Security for Universities, Colleges, and Schools.*
   Retrieved from:
   https://www.stonegroup.co.uk/blog/a-complete-guide-to-cyber-security-for-universities-colleges-and-schools/

9) Cybercivics. (2024). *Cybersecurity Best Practices for Educational Institutions.*
Retrieved from:
https://www.cybercivics.com/single-post/cybersecurity-best-practices-for-educational-institutions

10) Campoverde-Molina, M., Luján-Mora, S. & Valverde, L. (2023). Accessibility of university websites worldwide: a systematic literature review. *Univ Access Inf Soc* **22**, 133–168.
Retrieved from:
https://doi.org/10.1007/s10209-021-00825-z

11) Kalpa Kalhara Sampath. (2023). *educational-institutions-how-to-mitigate-cybersecurity-threats.*
Retrieved from:
https://www.isc2.org/Insights/2023/07/educational-institutions-how-to-mitigate-cybersecurity-threats

12) Mayank Joshi. (2023). *The Need for Website speed: Exploring the Impact of Fast Load Times on User Experience.*
Retrieved from:
https://auditzy.com/blog/impact-of-fast-load-times-on-user-experience

13) Taylor Precure. (2023). *Details Every College Website Needs to Boost User Engagement.*
Retrieved from:
https://www.educationdynamics.com/increase-user-engagement-on-college-websites/

14) Gideon Harrington. (2023). *Comprehensive Guide to College and University Website Development and Design Strategies.*
Retrieved from:
https://designweblouisville.com/comprehensive-guide-to-college-and-university-website-development-and-design-strategies/

15) Andy Shen. (2023*). Reasons-to-use-SQLite.*
Retrieved from:

https://medium.com/@itsahaotian/5-reasons-to-use-sqlite-and-when-not-to-use-it-a8caeea5f183

16) Gkrimpizi, T., Peristeras, V., & Magnisalis, I. (2023). Classification of Barriers to Digital Transformation in Higher Education Institutions: Systematic Literature Review. *Education Sciences*, *13*(7), 746.
Retrieved from:
https://doi.org/10.3390/educsci13070746

17) Caglar, E. (2022). Importance and Usability of University Websites. In Y. Akgül (Ed.), *App and Website Accessibility Developments and Compliance Strategies* (pp. 1-37). IGI Global Scientific Publishing.
Retrieved from:
https://doi.org/10.4018/978-1-7998-7848-3.ch001

18) Chuck Bankoff. (2022). *How to Design and Build a School or University Website.*
Retrieved from:
https://www.constantcontact.com/blog/how-to-design-and-build-a-school-or-university-website/

19) IJEDICT. (2021). *Usability Evaluation of Web Portals.*
Retrieved from:
https://files.eric.ed.gov/fulltext/EJ1334654.pdf

20) Jeroen G. Mombarg. (2021). *USABILITY AND USER EXPERIENCE DESIGN ANALYSIS ON A UNIVERSITY WEBSITE.*
Retrieved from:
https://essay.utwente.nl/85946/1/Mombarg_MA_BMS.pdf

21) TIMIFY. (2020). *The benefits of an online scheduling system for the education sector.*
Retrieved from:
https://www.timify.com/en/blog/benefits-online-scheduling-schools-universities/

22) Fatemeh Bordbar. (2016). *The Effectiveness of Website Design in Higher Education Recruitment.*
Retrieved from:
https://www.suu.edu/hss/comm/masters/capstone/project/f-bordbar.pdf

23) Manzoor, Mirfa & Hussain, Walayat & Ahmed, Aftab & Iqbal, Mohammad. (2012). The importance of Higher Education Website and its Usability. *International Journal of Basic and Applied Sciences.* 1(2), 150-163.
Retrieved from:
https://doi.org/10.14419/ijbas.v1i2.73

24) Ramakrishnan, Thiagarajan & Hossain, Muhammad Muazzem & Prybutok, Victor. (2011). *Modeling the impact of college's website on the individual's impression of the college.*
Retrieved from:
https://roam.macewan.ca/items/b13a7d51-e62d-46ac-a9af-372afdb978fe

25) Beck, S. E. (2009). *The Good, The Bad y The Ugly: or, Why It's a Good Idea to Evaluate Web Sources. New Mexico State University Library [on line].*

26) Marzie Astani. (2003). *AN EMPIRICAL STUDY OF THE EFFECTIVENESS OF UNIVERSITIES' WEB SITES.*
Retrieved from:
https://www.researchgate.net/publication/253691024_AN_EMPIRICAL_STUDY_OF_THE_EFFECTIVENESS_OF_UNIVERSITIES'_WEB_SITES

27) Kenneth Mashinchi. (2024). *The Vital Role of High-Quality Websites in Higher Education.*
Retrieved from:
https://evolllution.com/the-vital-role-of-high-quality-websites-in-higher-education

28) SkoolBeep. (2024). *Education Portal – Benefits (SkoolBeep).*
Retrieved from:
https://www.skoolbeep.com/blog/education-portal

29) Manno Notermans. (2024). *Higher Education Website Design: 10 Tips With Examples.*
Retrieved from:
https://www.thinkorion.com/blog/higher-education-website-design-tips

30) Government Graduate College of Science Lahore. (2024). *Home.* Retrieved from: https://sites.google.com/gcslahore.edu.pk/ggcs/home

31) Government College University Lahore. (2024). *GCU Lahore.* Retrieved from: https://gcu.edu.pk/

32) Harvard University. (2024). *Harvard University.* Retrieved from: https://www.harvard.edu

33) Harvey Mudd College. (2024). *Harvey Mudd College.* Retrieved from: https://www.hmc.edu

34) University of Michigan. (2024). *University of Michigan.* Retrieved from: https://www.umich.edu

35) Stanford University. (2024). *Stanford University.* Retrieved from: https://www.stanford.edu

36) iPistis. (2024). *The Vital Importance of School Websites: Enhancing Communication and Education in the Digital Age*. Retrieved from: https://blog.ipistis.com/the-vital-importance-of-school-websites-enhancing-communication-and-education-in-the-digital-age

37) Jakob Nielsen. (2024). *Usability Principles – Jakob Nielsen's 10 Usability Heuristics for User Interface Design.* Retrieved from: https://ux247.com/usability-principles

38) W3C. (2024). *Web Content Accessibility Guidelines (WCAG).* Retrieved from: https://www.w3.org/WAI/standards-guidelines/wcag

39) Jenelle Kruse. (2024). *How to Build a Powerful College Website.*
Retrieved from:
https://phoscreative.com/articles/how-to-build-a-powerful-college-website

40) Gurpreet Kaur. (2024). *Top 10 Advantages Of Agile Methodology In Web Development.*
Retrieved from:
https://bigohtech.com/advantages-of-agile-methodology

41) Julia Korsun. (2024). *Why we use Django Framework.*
Retrieved from:
https://djangostars.com/blog/why-we-use-django-framework

42) Sahare, Rekha. (2024). *REFINEMENT OF COLLEGE WEBSITE - GCOEC. INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT. 08. 1-5.*
Retrieved from:
https://doi.org/10.55041/IJSREM30119

43) Manno Notermans. (2024). *Higher Education Website Design: 10 Tips With Examples.*
Retrieved from:
https://www.thinkorion.com/blog/higher-education-website-design-tips

44) Nick Babich. (2023). *Static vs. dynamic websites — what's the difference?*
Retrieved from:
https://webflow.com/blog/static-vs-dynamic-websites

45) Loveneet Singh. (2023). *Mobile First Design - Best Practices, Benefits & Challenges in 2024.*
Retrieved from:
https://redblink.com/mobile-first-design-benefits-challenges

46) Gustavo Goncalves. (2023). *Strategies for attracting students for higher education*.
Retrieved from:
https://www.mkt4edu.com/en/blog/strategies-for-attracting-students-for-higher-education

47) Varma, A., & Sharma, A. (2022). Proprietary software management system for educational institutes. *International Journal for Research in Applied Science and Engineering Technology.*

48) Webdesign-Inspiration. (2021*). Designing a University Website: 9 Must-Haves.* Retrieved from: https://www.webdesign-inspiration.com/article/designing-a-university-website-9-must-haves

49) Daniel Danielyan. (2021). *10 reasons for designers to use Figma.* Retrieved from: https://medium.com/design-bootcamp/10-reasons-for-ux-designers-to-use-figma-981e9efd9f13

50) Simon Dwight Keller. (2019). *Benefits of using Bootstrap.* Retrieved from: https://www.creative-tim.com/blog/web-design/8-reasons-to-use-bootstrap-for-web-development

51) Mahafzah, A. (2017). *Higher education in Jordan: History, present status, and future.* Retrieved from: https://qs-gen.com/higher-education-in-jordan-history-present-status-and-future/

52) Sara Jelen. (2024). *Education Sector Common Breaches and Cyber Threats.* Retrieved from:

https://www.offsec.com/blog/education-sector-common-breaches-and-cyber-threats/

53) Government College Shalimar Lahore. (2024). *Shalimar College Lahore.* Retrieved from:

https://shalimarcollege.edu.pk/

54) Karan Datwani. (2023). *Why build an admin panel for your website or web app?* Retrieved from: https://backpackforlaravel.com/articles/recommendations/why-build-an-admin-panel-for-your-website-or-web-app

55) Joswellahwasike. (2024). *The Role of HTML in Web Development: Conclusion.*
Retrieved from:
https://dev.to/joswellahwasike/the-role-of-html-in-web-development-3k0k

56) Priya Pedamkar. (2023). *Introduction to Uses of CSS.*
Retrieved from:
https://www.educba.com/uses-of-css/

57) Codeworks. (2024). *Why JavaScript is Used in Web Development.*
Retrieved from:
https://codeworks.me/blog/why-javascript-is-used-in-web-development/

58) Shruti Sharma. (2021). *Why to choose VS Code?*
Retrieved from:
https://dev.to/shruti2303/why-to-choose-vs-code-47ph

59) Fita. (2024). *What is FIGMA: Advantages and Disadvantages.*
Retrieved from:
https://www.fita.in/what-is-figma-advantages-and-disadvantages/

60) Chris Bunn. (2018*). network-security-in-universities-colleges-and-schools websites.*
Retrieved from:
https://www.isdecisions.com/en/blog/it-security/network-security-in-universities-colleges-and-schools